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Proceedings of the 11th International Symposium on Computer Science in Sport (IACSS 2017)

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Preface

The 11th International Symposium of Computer Science in Sports (IACSS 2017) took place September 6 – 9, 2017 at the University of Konstanz, Konstanz, Germany. The symposium continued a tradition of conferences starting in 1997 at Cologne, Germany, which were held every other year and traveled through many countries and continents since then.

Though the topics of the presentations have changed, the aims of the symposium are still the same. The symposium engages in building links between computer science and sports science, and showcases a wide variety of applications of computer science techniques to a wide number of problems in sports and exercise sciences. Moreover, it provides a platform for researchers in both computer science and sports science for mutual understanding, discussing the respective ideas, and promoting cross-disciplinary research.

This year, the symposium addressed the following topics:

Computer Science

- Modeling and Simulation
- Sports Data Acquisition Systems
- Image and Video Processing
- Sports Data Analysis
- Machine Learning and Data Mining
- Visualization and Visual Analytics
- Presentation, Communication
- Decision Support
- Robotics
- Virtual Reality
- Digital Games

Sports and Exercise Science

- Biomechanics and Neuromuscular Control
- Exercise Physiology and Sports Medicine
- Performance Development and Analysis

- Training, Coaching and Feedback
- Modeling of Adaptation, Fatigue, and Performance
- Optimization of Strategies for Best Performance
- Movement, Motor Control and Learning
- Sports Management

We received 27 submitted papers and all of them underwent strict and blind reviews by the Program Committee. At least two reviewers commented on each paper, resulting in an acceptance rate of 59%. Authors of the sixteen accepted papers were asked to revise their papers carefully according to the detailed comments so that they all meet the expected high quality of an international conference.

Four keynote speakers and authors of the accepted papers presented their contributions in the above topics during the three-day event.

A get-together reception, a guided tour to the city, and a boat trip to the famous island of Mainau with the conference dinner held there in historic venues were the highlights of the social program.

We thank Springer Publishers for providing the opportunity of continuing the tradition that started in Loughborough 2015, of publishing the conference proceedings in their series “Advances in Intelligent Systems and Computing.”

We thank the participants for coming to Konstanz and hope that it was an enjoyable and fruitful event for all participants. We also thank the Program Committee members, the Local Organization Committee members, the reviewers, the invited speakers, and the presenters for their contributions to make the event a success.

Finally, we thank the DFG Collaborative Research Center TRR 161 “Quantitative Methods of Visual Computing” for support two of the invited speakers and the University of Konstanz that has generously hosted the symposium in its lecture rooms.

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Peter Wolf	ETH Zürich, Switzerland
Hui Zhang	Shanghai University of Sport, China

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Betty Mohler	Max Planck Institute for Biological Cybernetics, Tübingen, Germany
Robert Schumaker	University of Texas, Tyler, USA
Manuel Stein	University of Konstanz, Germany

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Predicting Team Sports Results

An Empirical Analysis on European Odds of English Premier League

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Abstract. This study, taking the game results of 3800 matches of the English Premier League (2006/2007–2015/2016 seasons) and their odds offered by 347 lottery corporations worldwide as samples, built indices like Odds Adjustment Difference (OAD), Odds Difference Level, as well as Concordance Rate (CR), Draw Rate (DR), Reversal Rate (RR) between instant odds and game results, and applied those indices to the analysis of basic characteristics of European odds (also known as decimal odds). In addition, this study also put forward Odds Difference Coefficient (ODC), and compared it with the Competitive Balance Entropy (CBE) of the English Premier League, leading to the following conclusions: (1) The relationship between odds and the winning percentage followed a curve similar to “L” with an obvious inflection point; (2) In the English Premier League, the CR, DR and RR between instant odds and game results were 53.6, 25.8 and 20.6% respectively. The OAD between instant odds and low initial odds was the smallest, followed by that between instant odds and odds on draw, and that between instant odds and high initial odds; (3) The difference between low and high initial odds had a significant influence on the CR, DR and RR; to be more specific, when the odds difference increased, the CR would be significantly increased while the DR and RR were significantly reduced. The CR, DR and RR of the English Premier League were relatively stable in the recent 10 years; (4) The CBE of the English Premier League had a moderate negative correlation with its initial ODC; (5) Gamblers can refer to the OAD, and instant odds on CR, DR and RR, as well as ODC and CBE when betting on football matches, and some formulas and models are used by lottery corporations for analysis.

Keywords: European odds · English premier league · Odds difference level · Odds difference coefficient · Competitive balance entropy

1 Introduction

The English Premier League (known as “EPL”), consisting of 20 football teams, was established on February 20, 1992, and has become a league at the highest level in the English football league system. With plenty of rounds, fast pace and fierce competition, EPL has always been regarded as the best league in the world, making it a hot gambling object of lottery corporations worldwide.

Odds are a set of numerical values precisely calculated by lottery corporations on the basis of the recent performance of football teams (the number of goals, wins, draws and losses), and are able to reflect the gap in strength between the two sides of a game. Moreover, a regular adjustment to odds makes them an effective tool for lottery corporations (Yan and Li 2002). North American team sports lottery corporations operate on a spread betting system, whereby the lottery corporations quotes a points spread, reflecting an assessment of the expected point difference between the favorite and the underdog. Once a bet is placed the payoffs are fixed, but the bookmaker lottery corporations can adjust the spread as the time of the match approaches in order to equalize the volume of bets placed on either team (Goddard and Asimakopoulous 2004).

Yet to gamblers, as they can win a corresponding amount of bonus if they gamble on the outcome of a game correctly, the ratio of the bonus to the gamble is referred to as odds, which are consistent with odds set by lottery corporations (Wang and Huang 2002). The essence of odds can also be regarded as probability. Football matches have three possible results, namely, winning, draw or losing, and each result has its corresponding odds or probability, but it should be noted that a high or low probability doesn't necessarily turn out to be the actual result of a game. The practical significance of converting probability into odds is to satisfy the sales demand of lottery corporations and the needs of consumers, and thus, odds reflect the probability of each possible game result in price (Liu 2010). If the bet market can be considered to be effective and betting odds allow accurate predictions is not clear from the outset. This is due to the fact that the emergence of betting odds is a complex process and can be influenced by the opinion of professionals (i.e., lottery corporations) as well as laypeople (i.e., punters) with various degrees of knowledge. It can hardly be quantified which group has which influence on the betting odds (Wunderlich and Memmert 2016).

This study puts forward assumptions that the odds have certain regularity, as initial odds serve as a predictor of teams' strength, home-away effect and external environment, while instant odds are the second predictor based on the betting of gamblers in football matches. To some extent, the odds serve as an accurate predictor of game results. And the study uses some variables in its mathematical model in order to find out the objective regularity of OAD, characteristic of instant odds on CR, DR and RR, as well as the function of CBE. And these research questions are important for gamblers, lottery corporations and researchers.

Based on the game results of EPL in the recent 10 years and their corresponding European odds, the aim of this study is to conduct an empirical analysis with different kinds of indices and algorithms in order to explore the basic characteristics of European odds and their correlation with competitive balance of EPL. And these research questions are important for gamblers, lottery corporations and researchers.

2 Methods

2.1 Data Sample

This study collected the game results (win, draw or lose) of 3800 matches of the EPL (ten seasons in total from 2006/2007 to 2015/2016) as well as their instant odds and

initial odds offered by 347 lottery corporations worldwide as samples, the data source¹: <http://www.win007.com/>.

2.2 Initial Odds, Instant Odds and Odds Adjustment Difference

European odds are calculated on the basis of the probability of each possible game result, namely winning, draw or losing. Initial odds refer to odds in the earliest stage without any gamble or change; instant odds are those being adjusted in real time till a game starts; and Odds Adjustment Difference (OAD), as its name suggests, is the difference between instant odds and initial odds, reflecting the basic characteristics of instant odds' adjustments to low initial odds (odds on winning the game), draw odds (odds on draw), and high initial odds (odds on losing the game). By comparing the instant and initial odds, it can be concluded that OAD reflects the expectations of lottery corporations and gamblers on game results, and the higher the parameter is, the greater the uncertainty a game result has, as is shown in Formula (1):

$$OAD = O_{ins} - O_{ini} \quad (1)$$

In Formula (1), OAD stands for Odds Adjustment Difference, O_{ins} for instant odds and O_{ini} for initial odds.

2.3 Concordance Rate, Draw Rate and Reversal Rate Between Instant Odds and Game Results

This study calculated the mean of instant odds offered by all the lottery corporations of each game so as to figure out the Concordance Rate (CR), Draw Rate (DR) and Reversal Rate (RR) between the mean and the actual game result. When A team with low odds (lottery corporations and gamblers think that A team has a higher probability to win a match) wins, then this game result is considered to be in concordance with the odds; when A team with high odds (lottery corporations and gamblers think that A team has a lower probability to win a match) wins, it is considered to be a reversal. According to the above definitions, the calculation methods of CR, DR and RR between odds and game results are shown in Formula (2), (3) and (4):

$$CR = \frac{N_{LW}}{N} \times 100\% \quad (2)$$

$$DR = \frac{N_{DR}}{N} \times 100\% \quad (3)$$

$$RR = \frac{N_{HW}}{N} \times 100\% \quad (4)$$

¹ <http://zq.win007.com/cn/League/36.html>

In Formula (2), (3) and (4), CR, DR and RR stand for Concordance Rate, Draw Rate and Reversal Rate respectively, and N_{LW} refers to the number of games won by a team with low odds, N_{DR} the number of draws, N_{HW} the number of games won by a team with high odds, and N the total number of games.

2.4 Odds Difference and Its Classification Standards

Odds reflect not only values, but also the comprehensive strength of the both sides of a game. The lower the odds, the higher the probability of winning a game, while higher odds represent a lower probability of winning a game. Hence, the difference between low odds and high odds to some extent reflects the gap in winning percentage and comprehensive strength between the two sides. Of course, as final results of football games are influenced by various factors, a team with a higher winning percentage (with greater comprehensive strength) doesn't necessarily win the game. For instance, the term "dark horse" in football games refers to the phenomenon that a team with weaker comprehensive strength defeated a strong team unexpectedly.

In order to examine the CR, DR and RR between odds and results of games played by teams with different odds differences, this study applied the method of quartile and classified the differences between high instant odds and low instant odds of 3800 matches into four levels, as is shown in Table 1:

Table 1. Classification standards of odds difference levels

Odds difference level	Classification standards
Odds difference level I	Difference between low odds and high odds $(X) \leq 1.0$
Odds difference level II	$2.2 \geq$ Difference between low odds and high odds $(X) > 1.0$
Odds difference level III	$4.6 \geq$ Difference between low odds and high odds $(X) > 2.2$
Odds difference level IV	Difference between low odds and high odds $(X) > 4.6$

2.5 Odds Difference Coefficient

This study calculated the mean of instant odds and initial odds on winning, draw and losing of each game so as to figure out the Odds Difference Coefficient (ODC) of 380 matches of each season, which reflects lottery corporations' different expectations for teams in EPL, as is shown in Formula (5):

$$ODC = \frac{\sqrt{(w_i - d_i)^2 + (w_i - l_i)^2 + (d_i - l_i)^2}}{w_i + d_i + l_i} \quad (5)$$

In Formula (5), w_i , d_i and l_i represent the mean of odds on winning, draw and losing respectively, while $i \in X$ represents the mean of instant and initial odds offered by lottery corporations.

2.6 Competitive Balance Entropy (CBE)

The general formula to calculate balance is: $\text{balance} = \frac{D}{D_{\max}}$, in which D represents diversity index, and $D_{\max}$ represents the maximum diversity index when there are S species and N total individuals. This study calculated Competitive Balance Entropy (CBE) based on Shannon-Wiener Index, the formula of which is:

$$J = \frac{H'}{H'_{\max}} = \frac{H'}{\ln S} \quad (6)$$

In Formula (6), which is a general formula, J represents CBE; H' represents Shannon-Wiener Index; $H'_{\max}$ is the maximum of H' , equaling $\ln S$. The formula J' has different meanings in different sports, it represents CBE of a certain season in football matches and S equals 3 (win, draw or lose). So, the CBE (formula J') of a certain season can be calculated by the following formula:

$$J' = \frac{\sum_{i=1}^n H'_i}{n} = \frac{-\sum_{i=1}^n \sum_{k=1}^3 P_{ik} \ln P_{ik}}{n \ln 3} \quad (7)$$

In Formula (7), n is the total number of teams in a certain season; P_{i1} = the number of games won by team i /the total number of matches of team i ; P_{i2} = the number of draws of team i /the total number of matches of team i ; and P_{i3} = the number of games lost by team i /the total number of matches of team i .

According to Formula (7), CBE (J') lies between 0 and 1. The closer the CBE (J') gets to 1, the better competitive balance the teams enjoy in that certain season (Zhang and Zhao 2017).

3 Results

3.1 The Relationship between Odds and the Winning Percentage

The winning percentage and profits are the two important factors directly related to odds, and the winning percentage can be calculated by subtracting profits from the reciprocal of odds. Although profits of lottery corporations differ from one to another and are unknown to outsiders, the overall relationship between odds and the winning percentage is the same. Figure 1 shows their relationship under the assumption that profit margin was of 10%, from which we can see that the relationship between odds and the winning percentage followed a curve similar to “L” with its inflection point at around 10. To be more specific, when the odds were between 1 and 10, an increase in the odds would lead to a significant drop in the winning percentage; but when the odds were above 10, there was merely an insignificant drop in the winning percentage if there was an increase in the odds.

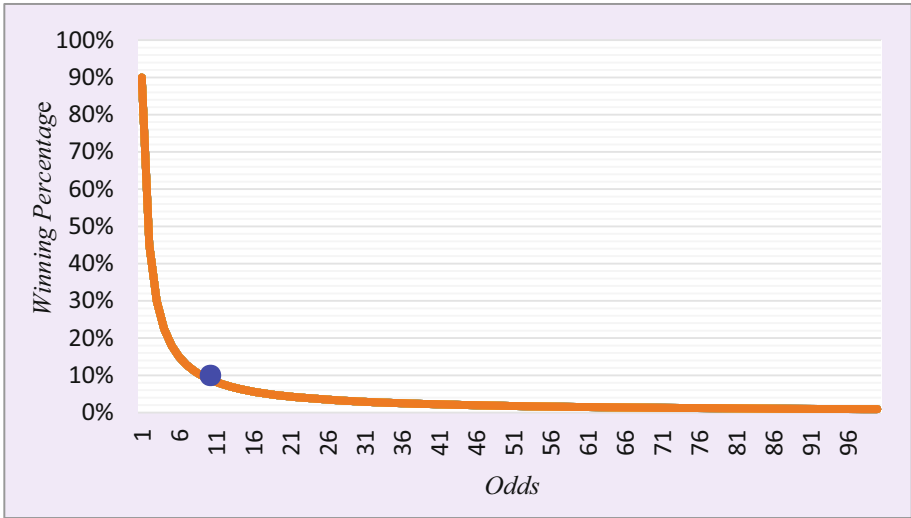


Fig. 1. Relationship between odds and the winning percentage

3.2 Odds Adjustment Difference

Table 2 can lead to the conclusion that the OAD between instant odds and low initial odds was the smallest, followed by that between instant odds and odds on draw, and that between instant odds and high initial odds. Besides, all of the OADs have a very significant difference ($p<0.01$), and the effect size of 2006/2007–2009/2010 are moderate, which $\eta_p^2 = 0.070, 0.125, 0.109$ and 0.113 , respectively.

Table 2. Odds adjustment difference

EPL	N	Mean and standard deviation of OAD			F	P	η_p^2
		Low odds	Draw odds	High odds			
06–07	380	−0.003 ± 0.046 ^c	0.019 ± 0.055 ^c	0.148 ± 0.416 ^a	42.552	<0.01	0.070
07–08	380	−0.001 ± 0.054 ^c	0.030 ± 0.075 ^c	0.179 ± 0.347 ^a	81.385	<0.01	0.125
08–09	380	−0.002 ± 0.069 ^c	0.035 ± 0.095 ^c	0.207 ± 0.453 ^a	69.742	<0.01	0.109
09–10	380	−0.007 ± 0.067 ^c	0.071 ± 0.140 ^c	0.257 ± 0.516 ^a	72.145	<0.01	0.113
10–11	380	−0.004 ± 0.073 ^c	0.043 ± 0.129 ^c	0.182 ± 0.568 ^a	30.905	<0.01	0.052
11–12	380	−0.000 ± 0.072 ^c	0.072 ± 0.154 ^c	0.162 ± 0.500 ^a	26.986	<0.01	0.045
12–13	380	0.001 ± 0.076 ^c	0.054 ± 0.119 ^c	0.128 ± 0.377 ^a	28.564	<0.01	0.048
13–14	380	0.005 ± 0.057 ^c	0.042 ± 0.134 ^c	0.152 ± 0.430 ^a	32.531	<0.01	0.054
14–15	380	0.011 ± 0.073 ^c	0.020 ± 0.119 ^c	0.112 ± 0.366 ^a	23.521	<0.01	0.040
15–16	380	0.007 ± 0.131 ^c	0.053 ± 0.238 ^c	0.139 ± 0.621 ^a	11.287	<0.01	0.019
Total	3800	0.001 ± 0.075 ^c	0.043 ± 0.135 ^c	0.167 ± 0.467 ^a	349.657	<0.01	0.058

In addition, Table 2 also shows that only in the four seasons from 2012/2013 to 2015/2016, the means of OAD to low initial odds were positive (meaning adjusting instant odds upward); while in the other six seasons, the means of OAD to low initial odds were negative, indicating that in the most cases, instant odds were more likely to be adjusted downward from low initial odds. This shows that gamblers always bet on the lowest odds in matches, so the lottery corporations adjust downward the instant odds.

3.3 CR, DR and RR Between Instant Odds and Game Results

Figure 2 shows the CR, DR and RR between instant odds and game results, in which 2037 games were won by the one side with low odds (the CR being 53.6%), 981 games on draw (the DR being 25.8%), and 782 games were won by the one side with high odds (the RR being 20.6%).

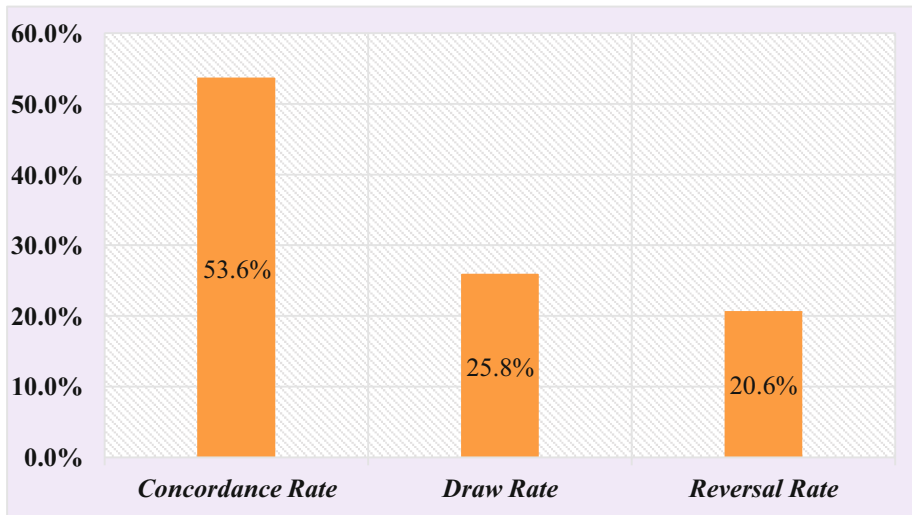


Fig. 2. CR, DR and RR between odds and game results

3.4 CR, DR and RR of Different Odds Difference Levels

Figure 3 shows that when the odds difference between low and high instant odds increased, the CR between instant odds and game results also increased significantly, while the DR and RR dropped significantly. At Odds Difference Level I, the CR, DR and RR between instant odds and game results were quite close to one another, as they were at 38.2, 32.3 and 29.5% respectively; while at Odds Difference Level IV, the CR, DR and RR between instant odds and game results differed from one another significantly, as the CR reached 72.4%, DR was at 17.7% while RR was merely at 9.9%.

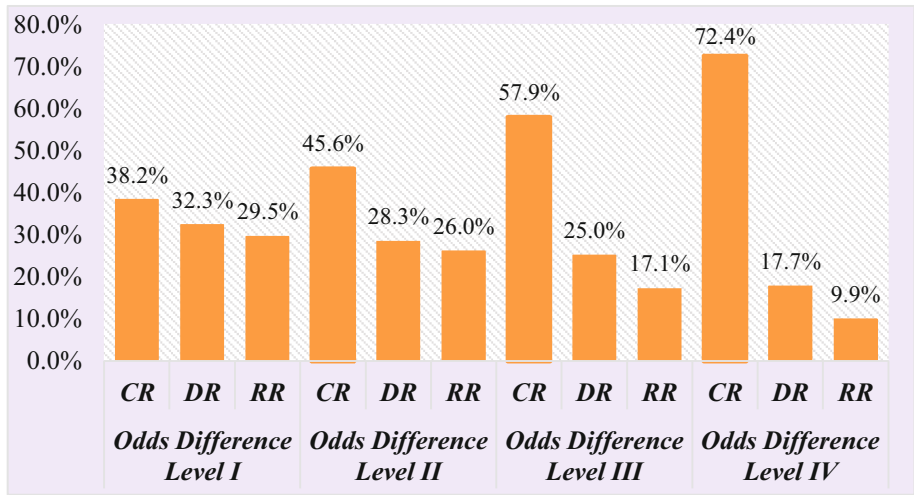


Fig. 3. CR, DR and RR of different odds difference levels

3.5 A Longitudinal Analysis of Changes in CR, DR and RR Between Instant Odds and Game Results

From Fig. 4, we can see that the CR, DR and RR between instant odds and game results of EPL in the recent 10 years remained stable on the whole despite a few fluctuations. The maximum of CR reached 60.3% (season 2013/2014), and its minimum was at 47.6% (season 2015/2016); the DR peaked at 29.2% (season 2010/2011), and hit its lowest level

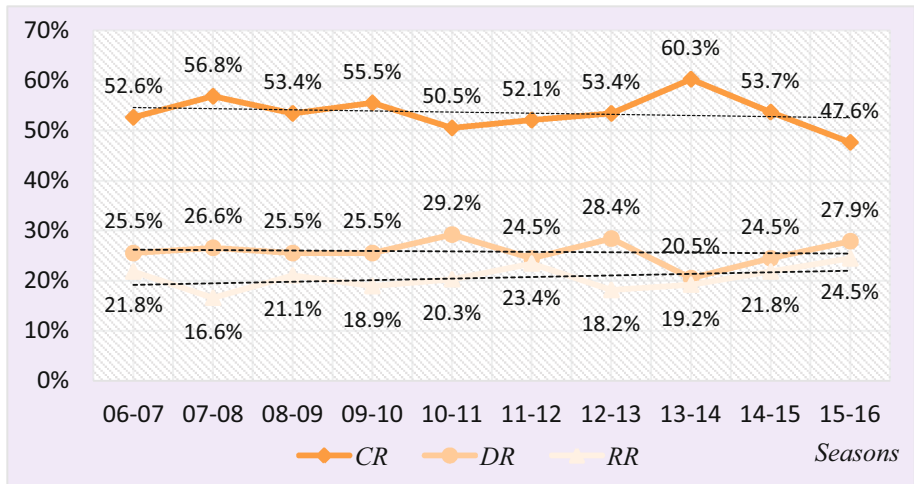


Fig. 4. Longitudinal changes in CR, DR and RR between instant odds and game results of EPL in the recent 10 seasons

at 20.5% (season 2013/2014); and the peak value of RR reached 24.5% (season 2015/2016), and its minimum was at 16.6% (season 2007/2008).

3.6 The Correlation Between CBE of EPL and Its Initial and Instant ODC

Figure 5 is a comparison of CBE of EPL as well as its initial ODC and instant ODC in the recent 9 seasons. From season 2006/2007 to season 2014/2015, the CBE of EPL peaked at 0.954 in the season 2010/2011, and hit its lowest level in the season 2013/2014 (0.890), with its mean value at 0.920. Figure 5 also demonstrates that instant ODC was always higher than initial ODC, indicating that ODC increased continuously after gamblers had their gambles on initial odds offered by lottery corporations; in other words, in the most cases, gamblers believed that the gap in strength between the two sides was widening.

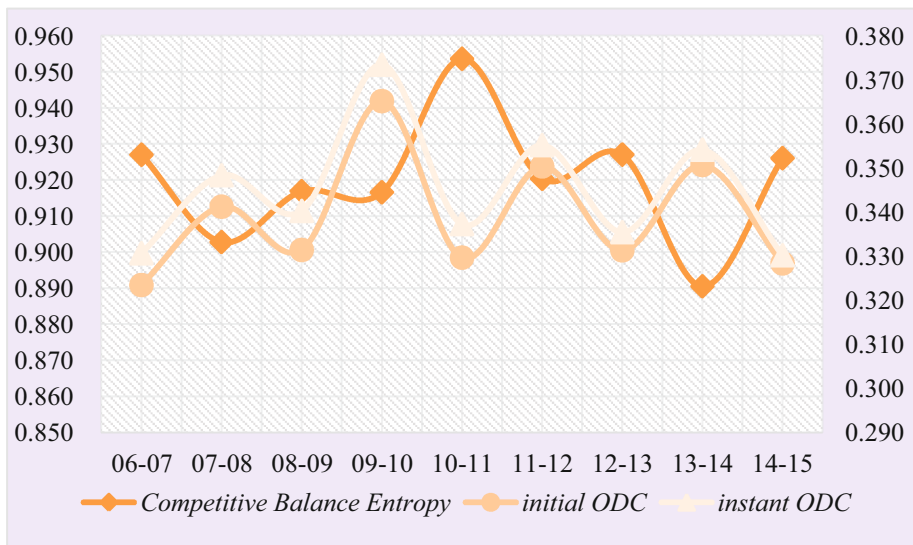


Fig. 5. Correlation between CBE and initial and instant ODC

A further analysis shows that the CBE of EPL had a strong negative correlation with the initial ODC ($r = -0.717$, $P < 0.05$), but had no significant correlation with instant ODC ($r = -0.630$, $P > 0.05$). Initial ODC reflects expectations of lottery corporations on the gap in comprehensive strength between the two teams in a match, including the home-away effect, coaches, referees, injuries of players and so on. As initial ODC is calculated by certain models and formulas, it enjoys a certain regularity, while CBE measures the excitement and uncertainty of football games in a league, but the two parameters have one thing in common: they are not affected by gamblers. The instant ODC represents the gap in comprehensive strength between the two teams after gamblers have made their bets and lottery corporations have adjusted the odds in order to earn bonuses or profits. In other words, the instant ODC cannot completely represent

expectations of lottery corporations on game results. Hence, the initial odds offered by lottery corporations could reflect the gap in strength among teams in EPL to a certain extent. To be more specific, the lower the initial ODC is, the more competitive balanced the two teams' strengths are (the higher the CBE is) and the more uncertain the game result will be. On the contrary, the higher the initial ODC is, the wider their gap in strength is (the lower the CBE is) and the more certain the game result will be.

4 Discussion

The analysis of OAD shows its regularity that the higher the parameter is, the greater the uncertainty a game result has, as OAD reflects the expectations of lottery corporations and gamblers on game results.

In Fig. 2, the CR is 53.6%, the DR is 25.8% and the RR is 20.6%. These illustrate that the team with relatively lower odds has an over 50 percent probability of winning the game, while the team with higher odds has an about 20% probability of winning the game; and probability of the game ending in a draw is about 25%.

In Table 2, in the ANOVA, numbers marked with the same letter indicate $P > 0.05$, those marked with adjacent letters indicate $P < 0.05$, and those marked with non-adjacent letters indicate $P < 0.01$ (Hao and He 2008).

In Fig. 5, the CBE of EPL had no significant correlation with instant ODC ($r = -0.630$, $P > 0.05$), which caused by some reasons as follow: (1) the instant ODC only had 9 samples, as same as the CBE of EPL and the initial ODC; (2) the standard deviation of instant ODC is bigger than initial ODC.

5 Conclusion

- (1) The relationship between odds and the winning percentage followed a curve similar to "L" with an obvious inflection point. Before the inflection point, an increase in odds would lead to a significant drop in the winning percentage; while after the inflection point, an increase in odds would merely lead to an unobvious decrease in the winning percentage.
- (2) In terms of the EPL, the CR, DR and RR between instant odds and game results were 53.6, 25.8 and 20.6% respectively. The OAD between instant odds and low initial odds was the smallest, followed by that between instant odds and odds on draw, and that between instant odds and high initial odds.
- (3) The difference between low and high initial odds would have a significant influence on the CR, DR and RR; to be more specific, when the odds difference increased, the CR would be significantly increased while the DR and RR were significantly reduced. The CR, DR and RR of the EPL in the recent 10 years were relatively stable.
- (4) The CBE of the EPL had a moderate negative correlation with its initial ODC. In other words, the lower the initial ODC was, the more balanced the two teams' strengths were (the higher the CBE was) and the more uncertain the game result would be, and vice versa.

- (5) Gamblers can refer to the OAD, and instant odds on CR, DR and RR, as well as ODC and CBE when betting on football matches, and these indicators should be put together to figure out their meanings and their correlation with the winning probability. Some formulas and models are used by lottery corporations for analysis, which could make odds more reasonable in football matches.

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Study on the “Hot Match” Effect in Professional Football Leagues

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Abstract. This study takes match outcomes of European eight leagues, China Super League (CSL) and Japan J1 League (J1) in 20 seasons (1996–2015) as samples, and proposes UNIANOVA (multi-factors: home or away game, results of the previous games, combination of the two) to explore the Hot Match Effect in professional football leagues. The results show that (1) “Home advantage” phenomenon is significant in professional football leagues; (2) Hot Match Effect exist in professional football leagues, and is obvious in Home matches; (3) the interaction effect between the two factors (home or away, results of the previous games) is significant in terms of winning percentage.

Keywords: Professional football leagues · Hot match effect · Hot hand

1 Introduction

In sports a widespread belief exists that success breeds success and failure breeds failure. One major example is the “hot hand” or “streak shooting”—terms referring to the belief that the performance of a player during a particular period is significantly better than could be expected on the basis of the player’s overall record (Gilovich et al. 1985). The related research on “hot hand” or “cold hand” has mainly focused on basketball, volleyball, baseball, tennis and golf. Those sports have two common characteristics: 1. Two possible outcomes (Hit or Miss, Win or Lose); 2. A large number of shoots or scores in one game. Compared with those, football is a sport with three possible outcomes (win, draw or lose) and fewer shoots or scores in one game. As far as we are concerned, no study has yet been conducted on Momentum effect in football matches.

In professional football leagues, each team competes with its opponents in succession according to the fixtures, and the outcomes form a natural time series. In the light of principles concerning Momentum Effect, the study propose UNIANOVA (multi-factors: home or away, result of the previous games, combination of the two) to explore the Hot Match Effect in professional football leagues.

2 Method

In this paper, the “Hot Match” Effect is defined as the phenomenon that as the winning streak increases, the winning percentage of the next game grows as well.

2.1 Subjects

This study takes match outcomes of European eight leagues, China Super League (CSL) and Japan J1 League (J1) in 20 seasons (1996–2015) as samples. Information about the leagues is shown in the following Table 1 in detail.

Table 1. Schedule of 10 leagues.

Abbreviation	Name of the League	Begin	End
England1	English Premier League	1996–1997	2015–2016
Spain1	Spanish la Liga	1996–1997	2015–2016
German1	German Bundesliga	1996–1997	2015–2016
Italy1	Italian Serie A	1996–1997	2015–2016
French1	French Ligue 1	1996–1997	2015–2016
Holland1	Netherlands Eredivisie	1996–1997	2015–2016
Portugal1	Portugal Primeira Divisão	1996–1997	1998–1999
	Portugal Primeira Liga	1999–2000	2015–2016
England2	English First Division	1996–1997	2003–2004
	English League Championship	2004–2005	2015–2016
Japan1	Japan J1 League	1997	2015
China1	C League	1996	2003
	Chinese Super League	2004	2016

Data are collected from <http://wildstat.com/> and <http://data.j-league.or.jp/>. The study only focuses on the regular season’s data (Holland1, England2). The matches don’t include the over-time and penalties (Japan1). As Wuhan Guang-gu’s withdrawal from CSL (China Super League) tournament in 2008, CFA judged its outcomes in 2008 season to be loses. For this reason, the data concerning the CSL2008 are not included in the study. The study conducts UNIANOVA (multi-factors: home or away, results of the previous games, combination of the two) with the help of SPSS19.0.

2.2 Procedures

As our statistics are defined in terms of streaks, defining the term “streak” should be the first step. Let G be a set of games indices and $\{X_g\}_{g \in G}$ be the sequence of match outcomes, where $X_g = 3$ if the outcome is win, $X_g = 1$ if the outcome is draw, $X_g = 0$ if the outcome is lose. A streak occurs in game g if a team has just completed k or more games with the same outcome, i.e. if $X_{g-1} = X_{g-2} = \dots = X_{g-k} = 3$. For the analysis we report below, we set $k = 3, 2, 1$ based on evidence that the intuitive sense of the emergence of a streak generally begins at the third successive event (Carlson and Shu 2007),

and previous studies have focused on streaks of at least three hits as indicative of hot hand shooting (Koehler and Conley 2003; Rao 2003).

$$W_k^i = \frac{|G_w^k|}{\|G_w^{k-1}\|} G_w^k := \{g \in G : X_{g-1} = X_{g-2} = \dots = X_{g-k} = 3\} \quad (1)$$

In the above formula, $|\cdot|$ counts the number of wins in a set of games, $\|\cdot\|$ counts the number of elements in a set of games and G_w^k for the subset of games g immediately follow a streak of k wins, $X_{g-1} = X_{g-2} = \dots = X_{g-k} = 3$. W_k^i indicates the team i 's conditional probability (to win game immediately after streak of k wins).

$$P_W^k = \sum_{i=1}^n (W_k^i)/n \quad (2)$$

P_W^k is the mean value among n team in one season. The calculation only takes the non-zero value in W_k^i into account. Using those indicators, the study conducted UNIANOVA (multi-factors: home or away game, results of the previous games, combination of the two) with the help of SPSS19.0.

3 Results

3.1 Analysis of the Winning Percentage of Home and Away

Table 2 shows that the mean of 10 leagues' home winning percentage is 0.427, and the mean of away winning percentage is 0.261. In term of home winning percentage, German1 has the highest (0.486), followed by Holland1 (0.485), Japan1 has the lowest (0.425); and with regard to away winning percentage, Japan1 has the highest (0.347), followed by Holland1 (0.284), China1 hits the lowest (0.213). The overall mean of winning percentage was 0.366. Japan1 hits the highest (0.386), with the second being Holland1 (0.385) and the lowest being China1 (0.344).

3.2 Analysis of Winning Percentage with Conditional Probability (Previous Match Outcomes)

From Table 3, as the winning streak increases, the winning percentage of the next game grows as well. Under the condition that previous game was won, the mean of winning percentage of the next game is 0.356; under the condition of two games winning streak, the mean of winning percentage of the next game is 0.461; under the condition that three games winning streak, the mean of winning percentage of the next game is 0.543.

Table 2. Winning percentage of home and away games

	N	Home		Away		Total	
		$\bar{X}$	SD	$\bar{X}$	SD	$\bar{X}$	SD
England1	20	0.463	0.006	0.268	0.005	0.366	0.004
Spain1	20	0.484	0.004	0.254	0.009	0.369	0.004
German1	20	0.486	0.015	0.264	0.01	0.375	0.008
Italy1	20	0.472	0.008	0.231	0.014	0.351	0.007
French1	20	0.483	0.01	0.233	0.006	0.358	0.003
Holland1	20	0.485	0.003	0.284	0.005	0.385	0.003
Portugal1	20	0.483	0.013	0.249	0.013	0.366	0.002
England2	20	0.46	0.014	0.267	0.007	0.364	0.006
Japan1	20	0.425	0.013	0.347	0.026	0.386	0.008
China1	20	0.475	0.009	0.213	0.014	0.344	0.009
Total	200	0.472	0.021	0.261	0.03	0.366	0.014

Table 3. Winning percentage with conditional probability (pervious match outcomes)

	N	$P(W 1W)$		$P(W 2W)$		$P(W 3W)$	
		$\bar{X}$	SD	$\bar{X}$	SD	$\bar{X}$	SD
England1	20	0.364	0.035	0.464	0.036	0.522	0.07
Spain1	20	0.347	0.038	0.448	0.048	0.575	0.055
German1	20	0.354	0.035	0.463	0.054	0.532	0.074
Italy1	20	0.345	0.034	0.460	0.042	0.540	0.075
French1	20	0.335	0.034	0.446	0.051	0.510	0.089
Holland1	20	0.385	0.031	0.487	0.051	0.583	0.06
Portugal1	20	0.350	0.036	0.499	0.059	0.588	0.079
England2	20	0.351	0.023	0.405	0.031	0.494	0.05
Japan1	20	0.387	0.036	0.449	0.06	0.518	0.071
China1	20	0.341	0.046	0.484	0.072	0.564	0.105
Total	200	0.356	0.038	0.461	0.057	0.543	0.079

3.3 Analysis of the Winning Percentage with Conditional Probability (Home or Away Games and Previous Match Outcomes)

From Table 4, under the condition of the same winning streak, the winning percentage of the next home game is higher than that of the next away game. Winning percentage with conditional probability of home games increases monotonically with the number of pervious wins. But the same phenomenon doesn't exist in the away games of German1, French1, England2 or China1.

Table 4. Analysis of winning percentage

	N	Home			Away		
		P(W 1W)	P(W 2W)	P(W 3W)	P(W 1W)	P(W 2W)	P(W 3W)
		$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$
England1	20	0.463 $\pm$ 0.043	0.550 $\pm$ 0.067	0.611 $\pm$ 0.064	0.381 $\pm$ 0.065	0.508 $\pm$ 0.094	0.530 $\pm$ 0.215
Spain1	20	0.484 $\pm$ 0.040	0.565 $\pm$ 0.060	0.622 $\pm$ 0.067	0.372 $\pm$ 0.046	0.523 $\pm$ 0.090	0.584 $\pm$ 0.247
German1	20	0.492 $\pm$ 0.037	0.575 $\pm$ 0.061	0.658 $\pm$ 0.090	0.396 $\pm$ 0.035	0.498 $\pm$ 0.087	0.478 $\pm$ 0.308
Italy1	20	0.478 $\pm$ 0.050	0.576 $\pm$ 0.058	0.634 $\pm$ 0.061	0.389 $\pm$ 0.041	0.532 $\pm$ 0.098	0.581 $\pm$ 0.242
French1	20	0.471 $\pm$ 0.040	0.536 $\pm$ 0.044	0.585 $\pm$ 0.066	0.364 $\pm$ 0.035	0.548 $\pm$ 0.089	0.525 $\pm$ 0.231
Holland1	20	0.502 $\pm$ 0.028	0.590 $\pm$ 0.061	0.631 $\pm$ 0.076	0.425 $\pm$ 0.043	0.552 $\pm$ 0.078	0.620 $\pm$ 0.200
Portugal1	20	0.489 $\pm$ 0.040	0.584 $\pm$ 0.068	0.646 $\pm$ 0.102	0.420 $\pm$ 0.051	0.561 $\pm$ 0.095	0.590 $\pm$ 0.178
England2	20	0.440 $\pm$ 0.023	0.508 $\pm$ 0.037	0.561 $\pm$ 0.066	0.338 $\pm$ 0.035	0.464 $\pm$ 0.069	0.446 $\pm$ 0.172
Japan1	20	0.461 $\pm$ 0.049	0.531 $\pm$ 0.072	0.586 $\pm$ 0.091	0.392 $\pm$ 0.032	0.516 $\pm$ 0.058	0.528 $\pm$ 0.210
China1	20	0.498 $\pm$ 0.052	0.572 $\pm$ 0.079	0.638 $\pm$ 0.151	0.440 $\pm$ 0.077	0.575 $\pm$ 0.196	0.396 $\pm$ 0.350
Total	200	0.478 $\pm$ 0.044	0.559 $\pm$ 0.065	0.617 $\pm$ 0.090	0.392 $\pm$ 0.055	0.528 $\pm$ 0.104	0.528 $\pm$ 0.245

3.4 UNIANOVA of Winning Percentage

In Table 5, there is a very significant difference in the winning percentage between home and away games ($P < 0.001$), and the effect size is great, which imply that the phenomenon of Home Game Advantage is widespread and obvious in professional football leagues.

There is a very significant difference in the winning percentage among different winning streak groups ($P < 0.01$). It indicates that there is “Hot Match” Effect in professional football leagues, especially in home games. Winning percentage with conditional probability of home games increases monotonically with the number of pervious wins.

In the analysis of the factor (home or away and winning streak), there is a very significant difference in Italy1 ($F = 8.341$, $P < 0.001$, $\eta_p^2 = 0.141$), French1 ($F = 13.977$, $P < 0.001$, $\eta_p^2 = 0.216$), Holland1 ($F = 9.726$, $P < 0.001$, $\eta_p^2 = 0.161$) and Portugal1 ($F = 11.789$, $P < 0.001$, $\eta_p^2 = 0.189$), and all the effect sizes are great. There is also a very significant difference in England1 ($F = 3.165$, $P < 0.01$, $\eta_p^2 = 0.089$), Spain1 ($F = 7.923$, $P < 0.001$, $\eta_p^2 = 0.135$), England2 ($F = 7.119$, $P < 0.001$, $\eta_p^2 = 0.123$) and China1 ($F = 7.018$, $P < 0.001$, $\eta_p^2 = 0.123$), while all the effect sizes are medium. There is a significant difference in the German1 ($F = 3.206$, $P < 0.05$, $\eta_p^2 = 0.06$), and the effect size is medium, but there is no significant difference in Japan1. All the results suggest that there is an interactive effect between home or away games and the number of previous wins.

Table 5. The UNIANOVA of winning percentage

	England1	Spain1	German1	Italy1	French1	Holland1	Portugal1	England2	Japan1	China1
Home or Away F	23.032	44.059	56.608	45.800	46.796	37.064	48.801	97.952	15.106	31.283
Home or Away P	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**
Home or Away η_p^2	0.234	0.225	0.271	0.232	0.235	0.196	0.243	0.392	0.090	0.171
Streak of wins F	40.104	45.004	21.086	53.846	42.115	64.127	69.399	36.779	31.634	15.294
Streak of wins P	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**
Streak of wins η_p^2	0.443	0.470	0.294	0.515	0.454	0.559	0.578	0.421	0.384	0.232
Home or Away $\times$ Streak of wins F	4.905	7.923	3.206	8.341	13.977	9.726	11.789	7.119	0.945	7.018
Home or Away $\times$ Streak of wins P	<0.01 **	<0.001**	0.025 *	<0.001**	<0.001**	<0.001**	<0.001**	<0.001**	0.421	<0.001**
Home or Away $\times$ Streak of wins η_p^2	0.089	0.135	0.060	0.141	0.216	0.161	0.189	0.123	0.018	0.122

3.5 UNIANOVA of Winning Percentage (Overall Data)

Taken together, the 10 leagues’ overall data paint a picture of how “League” as a factor was introduced in the UNIANOVA.

In Table 6, there is a very significant difference in the winning percentage among different leagues ($P < 0.001$), but the effect size is small.

A very significant difference exists both in the winning percentage between home or away games ($P < 0.001$) and in the winning percentage among different winning streak groups ($P < 0.01$), which implies that the phenomenon of Home Game Advantage and “Hot Match” Effect is widespread and obvious in professional football leagues.

Table 6. The UNIANOVA of winning percentage (overall data)

	F	P	η_p^2
League	7.010	<0.001**	.040
Home or Away	403.214	<0.001**	.210
Streak of wins	374.965	<0.001**	.425
League * Home or Away	1.856	0.055	.011
League * Streak of wins	2.010	0.002*	.034
Home or Away * Streak of wins	65.576	<0.001**	.115
League * Home or Away * Streak of wins	2.003	0.002*	.034

4 Conclusion

Hot Match Effect exists in professional football leagues, and is particularly obvious in home games. The Hot Match Effect is more significance with the increase of the winning streak in home games. The interaction effect between the two factors (home or away, results of the previous games) is significant in terms of winning percentage.

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Modeling and Prediction

Artificial Neural Networks Predicting the Outcome of a Throwing Task – Effects of Input Quantity and Quality

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Abstract. Internal *forward models* are used to explain motor prediction processes in motor control and learning e.g. predicting an upcoming miss in a throwing task before the knowledge of results is available. In this study we used artificial neural networks (ANN) to model such movement outcome prediction processes. Additionally, we varied the inputs of four different multilayer perceptrons (MLP) with respect to the quantity and the reliability (quality) of input parameters to account for perceptual noise. The results show that ANNs are able to learn the non-linear input-output mapping underlying the throwing task even with few input variables (velocity and angle at ball release). Results improve when providing additional information about the ball flight (prediction accuracy increases from $RMSE = 7.9$ mm to $RMSE = 3.9$ mm). However, when a model is provided with noisy inputs only, model training and prediction suffers substantially ($RMSE = 53.8$ mm). Yet, additional reliable information about the ball flight (in addition to noisy velocity and angle) leads to very high model prediction accuracy again ($RMSE = 4.1$ mm). In a nutshell, ANNs can be used to model internal *forward model* predictions, but the availability of reliable input information is essential at least to some extent.

Keywords: Skittles · Multilayer perceptron · Task performance prediction · Forward model · Motor control

1 Introduction

Predicting the outcome of a movement is one major prerequisite to act successfully in daily life. This ability is essential in various situations like walking or grabbing a cup of coffee. The benefit of such a prediction is that it provides the sensory-motor system with information about possible upcoming errors. This error prediction can occur even before a person receives a feedback or knowledge of results (i.e. feeling and seeing the cup of coffee in the hand). This prediction can be used to correct an ongoing movement in order to achieve the intended movement aim. Such movement related processes are called feedforward motor control [1–3]. However, studies investigating event-related potentials (ERP) in the electroencephalogram (EEG) were able to show that such forward model predictions about the movement outcome are not only generated when executing closed-loop movements, but also during open-loop (ballistic) movements like throws [4].

The theory of internal models is a prominent concept contributing to the question of how predictions about movement outcomes can be achieved [5–7]. Within this concept of internal models, the *forward model* is thought to be capable of learning input-output mappings of a movement and thus predicting movement related outcomes based on different inputs. These inputs can be derived from different sensory channels including vision and proprioception and also the efference copy. It is also important to note that the inputs to the internal *forward model* are not perfectly reliable due to perceptual noise [9].

Support for the concept of an internal *forward model* comes from different research areas, such as from neurophysiological [4, 10, 11] and behavioral studies [12]. However, it is still not finally clear how an internal *forward model* integrates multiple inputs to generate a prediction of a movement outcome and how such a model can be trained.

In this study, we will contribute to this question by using a computational model that is able to make outcome predictions of a virtual pendulum-like throwing task (a ball has to be thrown towards a target) based on different input quantities. As motor related *forward model* predictions are supposed to be represented in the brain, artificial neural networks (ANN) seem a reasonable approach to combine both physiological and computational models [13]. ANNs are inspired by physiological neural networks within the human brain and they are often used as a research tool for the interpretation of neurobiological phenomena. In addition, ANNs are widely used in sports science to model specific skills or sports motions [14]. Multilayer perceptrons (MLP), as one kind of ANN, are able to learn a desired input-output-mapping based on a set of examples (training data) and may, thus, be used to predict a movement outcome based on movement specific input data. However, although MLP seem to be appropriate to learn the movement related input-output mapping in this study, one general disadvantage is that the mapping is encoded in its connection weights and it is often difficult to understand how such a black box model produces the desired output [15].

Based on MLPs as a tool to analyze the outcome prediction in a virtual throwing task, the aims of this study are threefold: (I) We model the outcome prediction in a virtual throwing task by an MLP (MLP-1) and assess whether it can learn the mapping between sensory inputs (velocity and angle at ball release) and the movement outcome (minimum distance between a thrown ball and a target). (II) We compare the results of this MLP-1 with the prediction accuracy of another MLP (MLP-2) which uses information about the early part of the ball flight in addition to the velocities and angles. (III) We analyze two additional MLPs (MLP-1b, MLP-2b) based on MLP-1 and MLP-2 which are trained on modified sensory inputs. This modification means that “perceptual”-noise (stochastic errors) is added to the inputs of the nets and we test the influence of these noisy inputs on the prediction accuracy.

2 Methods

2.1 The Motor Task: Virtual Skittles

Virtual skittles is an adaptation of the original British pub game (related to the American game tetherball) in which a ball is tethered to a center post by a string and has to be thrown to the opposite side of the post in order to hit a target skittle [16] (Fig. 1).

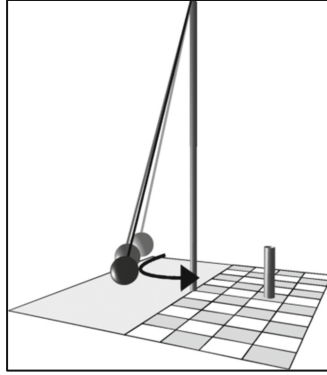


Fig. 1. Illustration of the real skittles task. A ball is attached to the top of a post and has to be thrown in order to hit a target skittle on the other side.

Virtual skittles has the same goal of the task, but uses a virtual ball and a virtual target. To be precise, a test person uses a real manipulandum in form of a metal lever. While executing the task, the test person sits on a stool and rests the arm on the foam padded metal lever. A contact sensor is placed at the tip of the lever and whenever the test person's finger gets in contact with the sensor, the virtual ball is picked up. Subsequently, the lever can be rotated within the horizontal plane around a vertical rotation axis that is located approximately under the test person's elbow joint. To throw, the players pick up the ball by placing the index finger on the contact sensor. Then the lever is rotated clockwise in order to release the ball at any time during the movement by taking off the finger from the sensor. The resulting elliptic flight trajectory of the ball can be precisely derived using the underlying physical model (parameters: velocity and angle at ball release) introduced by Müller and Sternad [17]. Furthermore, by using the physical model a solution manifold can be calculated containing all possible velocity-angle combinations and their produced outcome when the location of the target is given. Relevant geometric parameters of this model in our study were: center post (radius = 0.25 m; position: $x = 0.0$ m, $y = 0.0$ m), target (radius = 0.05 m; position: $x = 0.35$ m, $y = 1.0$ m), ball (radius = 0.05 m), lever (position: $x = 0.0$ m, $y = -1.5$ m, length = 0.4 m). The relevant movement outcome – i.e. the minimum distance between trajectory of the center of the ball and the center of the target (DBT) – can be precisely derived through the execution variables velocity (m/s) of the ball and the angle (degree) measured at the moment of release from the lever. All DBTs < 0.1 m were categorized as target hits (the ball touches the target when the distance between centers is less than the sum of both radii, i.e. 0.05 m + 0.05 m).

In our experimental setup, a right handed test person trained on the virtual skittles task in order to hit the target as frequently as possible for a total of 1500 trials at four consecutive days. The person received visual feedback after a fixed time of 850 ms. This static feedback (see Fig. 2, left part) depicted ball positions (B1, ..., B15) at 15 temporally equidistant moments of ball flight. The position of B1 represents the location of the ball at the time of ball release and B15 at the time of minimum distance to the target. Data from these trials were used for further analyses.

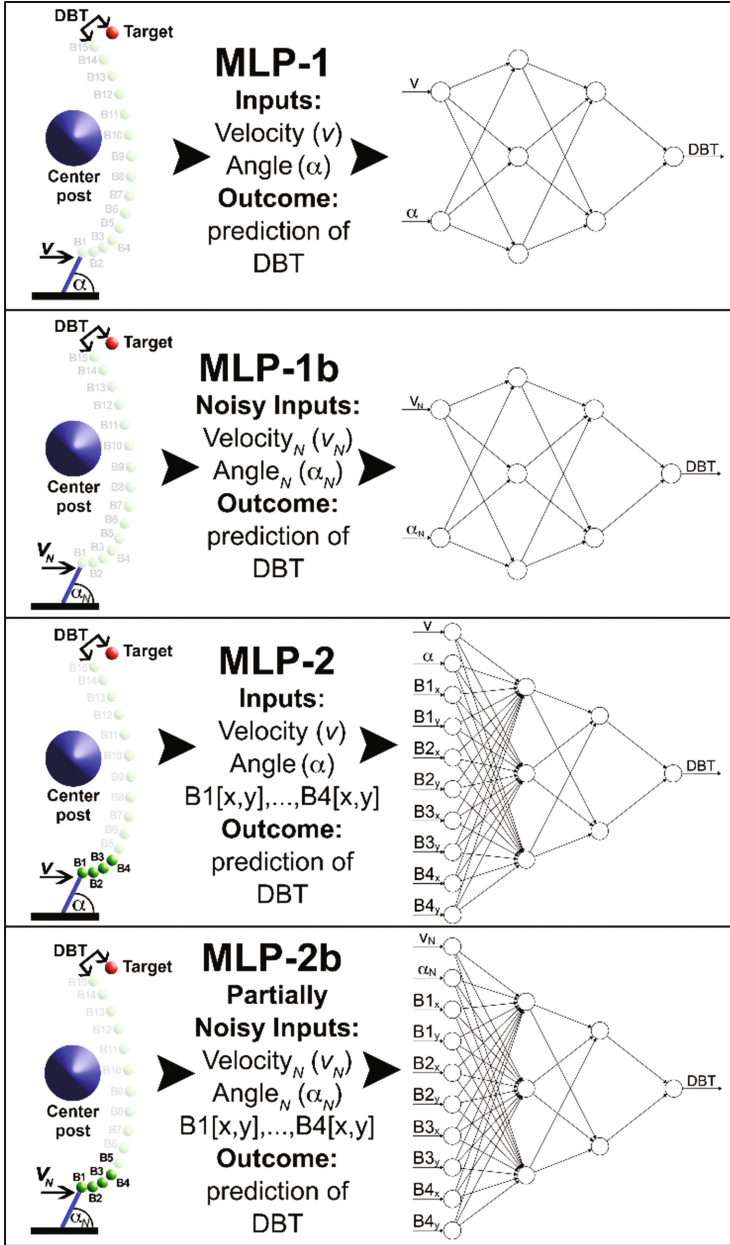


Fig. 2. Construction of four different multilayer perceptrons to predict the outcomes of the skittles throwing task with different amounts of input information and input noise. Each row illustrates the inputs/outputs and the architecture of one MLP (MLP-1, MLP-1b, MLP-2, MLP-2b). The left part of each row illustrates the skittles task with the input parameters used for model training and validation. The architecture of each model can be seen on the right. Parameters with noise are marked with N .

2.2 Modeling the Task: Multilayer Perceptrons with Different Amounts of Input Information

Type and architecture. We used two different architectures of multilayer perceptrons (MLP) to predict the minimum distance between the ball and the target object (DBT). Furthermore, each architecture was trained and evaluated in two different ways: firstly, with original input data of our test person trials (MLP-1, MLP-2) and, secondly, with random noise added to velocity and angle of the performed trials (MLP-1b, MLP-2b). We designed MLP-1 and MLP-1b with two inputs, two hidden layers (three and two neurons), and one neuron in the output layer predicting the DBT (measured in m; see Fig. 2). Velocity (measured in m/s) and angle (measured in degree) at the time of ball release served as inputs to the input layer of MLP-1 (and MLP-1b after adding random noise). MLP-2 and MLP-2b had a similar architecture like MLP-1 and MLP-1b, but used 10 input neurons according to velocity and angle as well as the information about the x-y-coordinates of the first four ball positions ($B1_x, B1_y, \dots, B4_x, B4_y$; see Fig. 2). The position of the first four balls were chosen for two reasons. Firstly, the ball's position at these time points carry information that is most closely linked to the release parameters, i.e. velocity and angle. Secondly, we assume that due to inevitable conduction and processing delays, an internal *forward model* can only use information that is available up to a certain time before the real outcome will be present, thus, we assume that only the initial part of the complete ball flight trajectory can be integrated to generate a prediction. Each MLP used sigmoid activation functions in the hidden layers and a linear activation function in the output layer.

Generating noisy input data. To generate noisy input data (to account for perceptual noise) we used the original data and added one error value to each of the velocity and angle input values assuming the overall error distribution is Gaussian shaped with a mean at zero. To generate error terms that are reasonable in size for release velocity and angle, we used data of a previous study in which we determined the detection threshold for manipulations of angle and velocity [18]. Subjects were not able to reliably detect (detection rate >50%) manipulations of the release angle smaller than approx. 30° . Velocity manipulations were only noticed if they were larger than approx. 30% of the average release velocity. We are convinced that the detection threshold reflects the accuracy of perception. Therefore, we set the standard deviation of the applied noise-error distribution to $SD_{v\text{-Noise}} = 0.8568$ m/s for the velocity input and to $SD_{\alpha\text{-Noise}} = 41.8^\circ$ for the angles. Noise was only added to the input values of velocity and angle because this information is primarily available through proprioception. Information transported through vision, e.g. the ball flight, seems to be more reliable [9, 19] and thus no random errors were added to these variables in our modeling approach.

Data preprocessing, network training, and postprocessing. Data preprocessing included scaling the input variables (velocity, velocity_{noisy}, angle, angle_{noisy}, x-y-coordinates of ball positions B1 to B4) to a range of [0; 1]. Thus, each variable had a minimum of zero, a maximum of one, and a range of one. Before network training, we assigned random weights to each connection between neurons. These random weights were then adjusted using a resilient backpropagation algorithm, which has proven to be

significantly better and faster than other network training methods [15]. Postprocessing included descaling the outputs to the original range of DBT data. Due to the fact that the network’s training data was recorded from 1500 trials of a test person, it did not cover all theoretically possible velocity-angle combinations. However, the resulting combinations were congruent with those usually executed in this task.

Model Validation. After training, the four MLPs were validated and (a) the correlation between the measured real and the model-predicted DBT was derived (b) the root mean squared error (RMSE) between the real and the model-predicted DBT was calculated, (c) a 10-fold cross validation was run for each MLP as well as (d) a calculation of a compliance-rate (*c-rate*) representing the relative number of correctly predicted movement outcomes with regard to the possible outcomes *hit* and *miss*. For cross validation, the data was randomly split into 10 folds of 150 trials each. Then, one fold was excluded and each model was trained on the data of the other nine folds. The trained models were tested in predicting DBTs based on inputs of the excluded fold. This procedure was repeated 10 times for each model (MLP-1, MLP-1b, MLP-2 and MLP-2b) yielding an average cross validation RMSE (cv-RMSE) for every single model.

All calculations were done with R (3.2.4) (R Core Team, 2016) – especially the neuralnet package [20].

3 Results

3.1 Descriptive Data of the Test Person’s Task Performance

The test person’s data included a total amount of 1500 trials. The average measured velocity at the time of ball release was 3.06 m/s ($SD_v = 0.98$ m/s; $range_v = 4.48$ m/s) whereas the average release angle was 87.5° ($SD_\alpha = 14.5^\circ$; $range_\alpha = 125.6^\circ$). As a result of each throw, DBT was calculated. On average, the test person’s DBT was 0.09 m ($SD_{DBT} = 0.07$ m; $range_{DBT} = 0.42$ m)¹. In addition, Fig. 3 shows the test person’s improvement in the task over the practice trials (one practice block equals 100 consecutive trials of the virtual skittles task). The test person started with a hit rate of 44% in the first block and improved its task performance up to 82% hits in the last block (block 15). Furthermore, in blocks 5, 9 and 13 the hit rate decreased. The appearance of these performance drops corresponds with the beginning of each practice day (first 100 trials) and can be explained by a well-known effect in the literature on motor learning and training science, the *warm-up decrement* [21].

¹ As the test person was instructed to hit the target as frequently as possible, the relevant outcome categories of the virtual skittles task were binary, i.e. *hit* and *miss*. Please note that a *hit* was achieved whenever $DBT < 0.10$ m. Every $DBT \geq 0.10$ m was categorized as a *miss*.

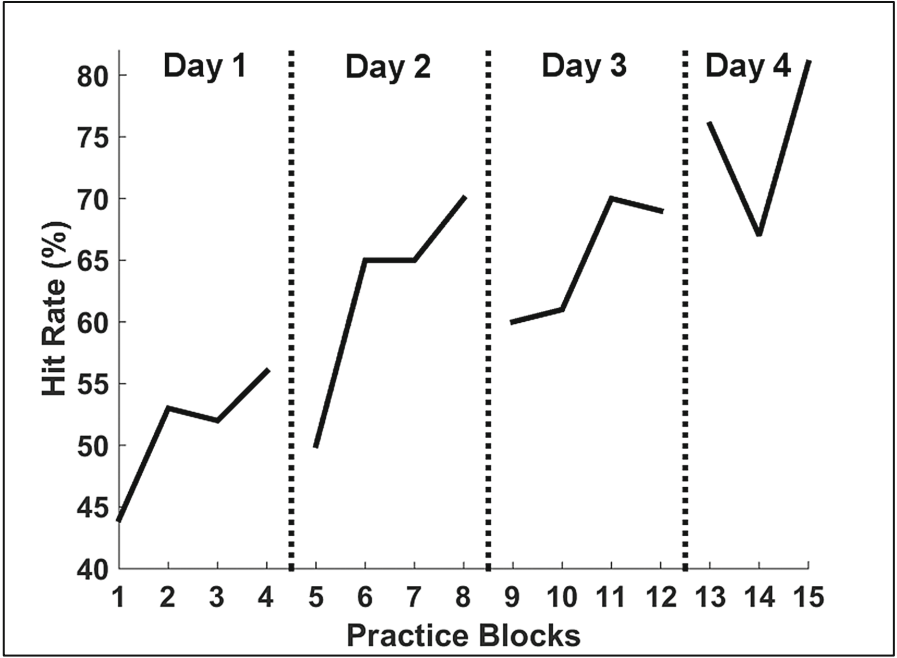


Fig. 3. Development of the test person's hit rate over time (practice blocks). Each practice block consists of 100 consecutive trials, summing up to a total amount of 1500 practice trials.

3.2 Prediction Accuracy Based on Original Input Data (MLP-1 & MLP-2)

MLP-1. The first prediction model used only information about velocity and angle (measured at ball release) for the prediction of DBTs. Comparing the predicted and the measured DBTs, we found a high correlation of $r_{MLP-1} = 0.993$. A model's average error of predicting the measured DBTs is expressed with the root mean square error (RMSE). For MLP-1, this average prediction error was $RMSE_{MLP-1} = 7.9$ mm. The 10-fold cross validation yielded a $cv\text{-}RMSE_{MLP-1} = 9.2$ mm. Furthermore, the actual outcomes (hit or miss) matched the model predicted outcomes in $c\text{-}rate_{MLP-1} = 97.9\%$.

MLP-2. In the second model, information about the ball's trajectory ($[B1_x, B1_y], \dots, [B4_x, B4_y]$) was provided additionally to the release angle and velocity information already used in MLP-1. This additional information led to an increase of prediction performance with respect to its accuracy. The correlation between predicted and measured DBTs was $r_{MLP-2} = 0.998$. The average prediction error of MLP-2 decreased compared to MLP-1 ($RMSE_{MLP-2} = 3.9$ mm; $cv\text{-}RMSE_{MLP-2} = 6.4$ mm). Furthermore, in accordance to the other parameters, the $c\text{-}rate$ of MLP-2 also slightly improved to $c\text{-}rate_{MLP-2} = 98.6\%$.

3.3 Prediction Accuracy Based on Noisy Input Data (MLP-1b and MLP-2b)

MLP-1b. This MLP received the same input information as MLP-1 (v , α), but with normally distributed random errors added to the values in order to simulate perceptual noise. Compared to the models using the original data (not noisy), we found a clear drop in prediction performance in MLP-1b. The transition from the original to the noisy velocity and angle data led to a correlation of $r_{MLP-1b} = 0.564$ (compared to $r_{MLP-1} = 0.993$) between predicted and measured DBTs. The average prediction error also increased clearly when using noisy input data ($RMSE_{MLP-1b} = 53.8$ mm and $cv-RMSE_{MLP-1b} = 54.3$ mm). In addition, the regression plot of MLP-1b in Fig. 4 shows that the linear relationship between predicted and measured DBTs is very weak and assumptions about linearity may be violated. The c -rate of MLP-1b was $c\text{-rate}_{MLP-1b} = 72.1\%$, which is clearly lower than in the models using the original input data (MLP-1 and MLP-2).

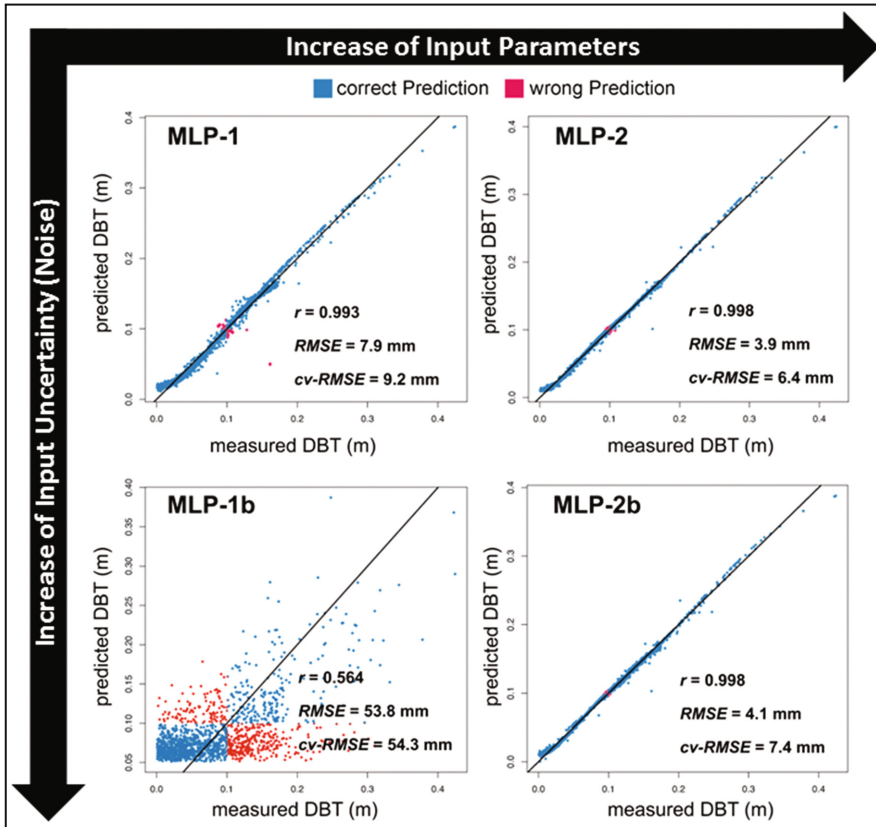


Fig. 4. Overview of prediction accuracies of the four models (MLP-1, MLP-1b, MLP-2, and MLP-2b). The blue points illustrate the trials in which the measured and model predicted movement outcomes (hit or miss) are conform.

MLP-2b. This model was trained with noisy velocity and angle inputs and additional ball flight information (not noisy). Testing the prediction accuracy, we found similar results in MLP-2b relative to MLP-1 and MLP-2, i.e. a very high positive correlation ($r_{MLP-2b} = 0.998$) between the model predictions and the actually measured DBTs. Finally, MLP-2b's average prediction error was also comparable to the errors found for MLP-1 and MLP-2 ($RMSE_{MLP-2b} = 4.1$ mm; $cv-RMSE_{MLP-2b} = 7.4$ mm; $c-rate_{MLP-2b} = 98.5\%$).

For an overview, Fig. 4 shows regression plots of measured vs. model predicted DBTs for each MLP.

4 Discussion

In this article we aimed to model the internal *forward model* predictions in a virtual throwing task (skittles). To this end, multilayer perceptrons (MLP) were trained with input data (i.e. velocity, angle and partly information about the ball flight) that would also be available to the internal *forward model* in humans. As output values we used the minimum distance between ball and target (DBT). In detail, we constructed four MLPs that differed with respect to the amount of input information (MLP-1 & MLP-2) and input uncertainty to account for perceptual noise that typically inflicts human sensory information (MLP-1b & MLP-2b).

Our findings address two different aspects: firstly, the effect of input quantity and, secondly, the influence of input quality (with respect to noise or random error). Regarding the amount of input information used for training and validation of the models, we found that information about the release velocity and angle are already sufficient to generate a relatively precise prediction of the error in the skittles task (see results of MLP-1 in Fig. 4). The integration of additional information about the ball flight slightly improved the prediction accuracy comparing the results of MLP-1 ($r_{MLP-1} = 0.993$; $RMSE_{MLP-1} = 7.9$ mm; $cv-RMSE_{MLP-1} = 9.2$ mm; $c-rate_{MLP-1} = 97.9\%$) and MLP-2 ($r_{MLP-2} = 0.998$; $RMSE_{MLP-2} = 3.9$ mm; $cv-RMSE_{MLP-2} = 6.4$ mm; $c-rate_{MLP-2} = 98.6\%$). However, considering that the goal of the task was to hit the target and hits were achieved if $DBT < 0.1$ m, the benefit of having additional information of the ball flight (B1,..., B4) is rather small.

According to the second analyzed aspect, when looking at the influence of input reliability, we find that the prediction accuracy decreases dramatically comparing MLP-1 and MLP-1b ($r_{MLP-1b} = 0.564$; $RMSE_{MLP-1b} = 53.8$ mm; $cv-RMSE_{MLP-1b} = 54.3$ mm; $c-rate_{MLP-1b} = 72.1\%$), i.e. perceptual noise in input data has the expected detrimental effect on error prediction. Although the correlation between model predicted and measured DBT is moderate, we argue that this model would not be appropriate for predicting hits and misses in the skittles throwing task. This is because considering the c-rate of MLP-1b, the model would predict a wrong outcome in 27.9% of cases, whereas the other three models predict the actual movement result with much higher c-rates of about 98%. However, in MLP-2b we provided information of the ball flight (with no random error) in addition to the noisy velocity and angle inputs also used in MLP-1b. The results show that this model is able to make accurate predictions of the outcome ($r_{MLP-2b} = 0.998$; $RMSE_{MLP-2b} = 4.1$ mm; $cv-RMSE_{MLP-2b} = 7.4$ mm; $c-rate_{MLP-2b} = 98.5\%$), which are about the same size or at least

as accurate as the predictions of MLP-1 and MLP-2 that had no noisy data. One possible explanation for this finding could be that the noisy inputs of MLP-2b are downscaled before passing them from the input layer to the three neurons in the first hidden layer. There are some indications that this might hold true for some of the weights, but there is no consistent picture for a general downscaling of noisy inputs. With regard to internal *forward models* in the human motor system, our results suggest that for the development of an internal *forward model* for a certain motor task, the availability of at least one reliable input information has to be given in order to generate an accurate outcome prediction. Nevertheless, the ball positions were not noisy in all of our four models, so further analysis should address this issue in more detail in the future.

Considering input quantity and input quality, we assume that both input characteristics interact with each other and are somehow connected in *forward model* learning and prediction. As the results of MLP-1 show, it is possible to successfully train a model on an input-output mapping of a throwing task with a small amount of input information if this information is reliable (no noise overlay). However, inputs of a human *forward model* are always superimposed with a certain amount of noise or random errors. In this case, increasing the amount of input information and especially providing information from more reliable sensory channels could help to develop a *forward model* that is capable of accurately predicting a movement's outcome.

Future investigations with this modeling approach can go in different directions. As we used an arbitrary number 1500 trials of only one test person in this study, it would be reasonable to test how the model is able to generalize and predict on data of other test persons. It would especially be interesting having two test persons using different strategies (i.e. different velocity-angle combinations at ball release) to achieve a target hit. Another possible direction would be to test the impact of different inputs on the generalizability of the trained input-output mapping. With our modeling approach we are able to test the influence of different input parameters on the model's training and performance. In detail, we are able to tackle questions like: Are there input parameters that contribute to the ability of a model predicting outcomes outside of the subarea of velocity-angle combinations they were trained on? For instance, additional parameters for these analyses might be information based on muscle activity (measured by electromyography – EMG).

5 Conclusion

Multilayer perceptrons can be used to model human internal *forward models*, i.e. they show similar behavior like the ones assumed in theories of motor control and learning. They are able to model processes that relate to the predictions of a movement outcome based on different inputs that are available through different sensory channels (e.g. velocity and angle through proprioception and information about the ball flight through vision). However, perceptual noise can be an issue when it comes to model training and prediction while only uncertain (noisy) inputs are provided to the system. As reliable information, e.g. from vision, helps a computational model to significantly increase its prediction accuracy, this supports the assumption that reliable *forward model* input

information are the key for an accurate prediction of a movement outcome. Therefore, our findings might help to establish a basis for supporting athletes and coaches when it comes to focus on relevant information in motor learning and control.

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Activity Recognition of Local Muscular Endurance (LME) Exercises Using an Inertial Sensor

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Abstract. In this paper, we propose an algorithmic approach for a motion analysis framework to automatically recognize local muscular endurance (LME) exercises and to count their repetitions using a wrist-worn inertial sensor. LME exercises are prescribed for cardiovascular disease rehabilitation. As a technical solution, we propose activity recognition based on machine learning. We developed an algorithm to automatically segment the captured data from all participants. Relevant time and frequency domain features were extracted using a sliding window technique. Principal component analysis (PCA) was applied for dimensionality reduction of the extracted features. We trained 15 binary classifiers using support vector machine (SVM) to recognize individual LME exercises, achieving overall accuracy of more than 98%. We applied grid search technique to obtain the optimal SVM hyperplane parameters. The learning curves (mean $\pm$ stdev) for each model is investigated to verify that the models were not over-fitted and performed well on any new test data. Also, we devised a method to count the repetitions of the upper body exercises.

Keywords: Local muscular endurance · Human activity recognition · Cardiovascular disease · Principle component analysis · Support vector machine

1 Introduction

Regular and appropriate exercise facilitates rehabilitation from chronic diseases such as cardiovascular disease (CVD). However, adherence to community based exercise and rehabilitation programmes is extremely low. Uptake and adherence to such programmes are very less [1]. Delivering a home-based programme via a mobile-phone can address some of these adherence issues (e.g., access to a targeted programme, travel time). However, the motivation to complete the exercise may be reduced because it is not being measured or seen by others (instructor

or fellow exercisers). One solution is to use a wearable sensor to recognize when a person is exercising and to record specifically how many repetitions they are performing. As a scientific solution, activity recognition and repetition counting during exercises can be studied under the broader category of human activity recognition (HAR) with the help of miniaturized and accurate wearable multi-modal wireless sensors [2, 5, 6, 8]. The specific exercises associated with cardiac rehabilitation are LME exercises (Table 1).

The use of wearable 3D accelerometer and gyroscope can provide accurate translational and rotational data [3, 5, 6, 8]. Use of variable sensors for ambulatory motions and HAR is discussed in [6] and specific to the detection of asymmetric running using time and frequency features is discussed in [3]. The scope of this paper is to design and implement a novel automated system for the recognition of specific LME exercises and counting the number of repetitions using a wearable inertial sensor.

In this paper, a Shimmer3 sensor unit (Shimmer, Ireland) is utilized to capture fifteen different LME exercises from six different participants. Segmentation is performed and statistical time-frequency domain features [4, 9] are extracted from the segmented data. PCA is then used to reduce the dimensionality of the feature vectors [10]. Activity recognition is studied using SVM. We applied a grid search algorithm [14] to obtain optimized hyperplane parameters and kernel options to overcome the overfitting of the trained classifier. A suitable counting mechanism, using peak-to-peak detection or threshold crossing, is examined to determine the number of repetitions for the upper body exercises only (future work will apply these to the lower limb exercises). The final results, including the training and validation accuracy, f-score, precision, recall for all 15 exercises, are presented and discussed. For illustration purposes, we chose one LME exercise (i.e. Bicep Curl) to demonstrate all the intermediate results and fully describe the entire proposed framework.

2 Proposed Framework

The major components of our framework are illustrated in Fig. 1. This end-to-end pipeline structure consists of five steps to recognize and count the repetitions of each exercise. Each component is discussed in detail in Sect. 3.

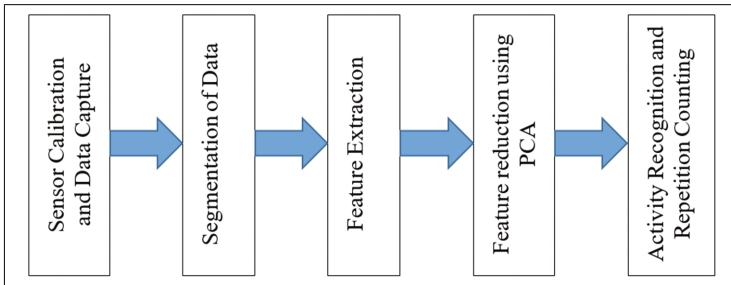


Fig. 1. End-to-end pipeline of the proposed framework

3 Methodology

3.1 Sensor Calibration and Data Capture

Data collection is carried out using Shimmer3, a lightweight and miniaturized wearable sensor encompassing a 3D accelerometer, 3D gyroscope, a 3 MHz MSP430 CPU with Bluetooth connectivity for remote access and a microSD card for local storage. The Shimmer unit is calibrated to obtain consistent and accurate data. The 3D accelerometer ($\pm 2g$) data is used to recognize the activities and data from the accelerometer or gyroscope are used for determining the repetition count. Data is collected from six healthy participants performing the exercises at a sampling rate of 512 Hz. The Shimmer sensor is securely placed on the right wrist of the participant using an elastic wrist strap. The sensor placement and orientation are shown in Fig. 2.

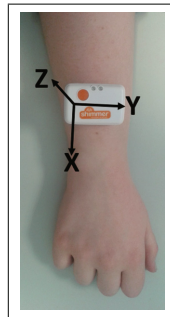


Fig. 2. Illustration of sensor placement and sensor orientation on right wrist

An experimental protocol is followed during data capture. Once the sensor is turned on, each participant claps three times which indicates the beginning of each recording. Subsequently, participants remain stationary for 5 s to ensure no random noise is introduced to the main signal before performing the exercise. Each participant then has to perform the exercise for 40 s. The trial is concluded with a further 5 s pause followed by three claps indicating the conclusion of the trial. Data collected with this approach for the Bicep curl is shown in Fig. 3. An identical protocol is used for all 15 exercises listed in Table 1. In addition the activity classification model needs to learn/identify when the person is not performing an exercise. Data capture is carried out for quasi static arbitrary (random) movements, such as participants standing still or actions which can be referred as non-performing an exercise.

3.2 Segmentation of Data

A segmentation algorithm is designed and implemented to automatically annotate the entire dataset (i.e. for each of 15 LME exercises and random movements

Table 1. List of LME exercises

Upper body LME exercises			
Ex 1	Bicep Curls	Ex 6	Pec Dec
Ex 2	Triceps extension (right arm)	Ex 7	Trunk twist
Ex 3	Upright row	Ex 8	Side Bends - alternating sides
Ex 4	Lateral raise (arms up)	Ex 9	Bent Over Row (right arm)
Ex 5	Frontal raise (arms up)	Ex 10	Press up against wall
Lower body LME exercises			
Ex 11	Squats	Ex 14	Standing bicycle
Ex 12	Lunges - alternating sides	Ex 15	Leg lateral raise (right)
Ex 13	Calf raises		

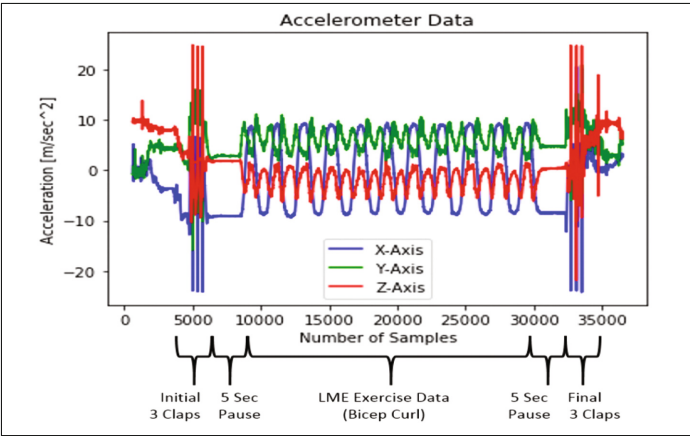


Fig. 3. Illustration of data capture protocol with accelerometer data for the Bicep Curls

from all 6 participants) for further analysis. The algorithm automatically detects the initial three claps from the accelerometer data and captures the activity signal for a period of 30 s in between the initial and final pause periods. The 30 s of segmented data is annotated as (**Class1**). Similarly, segmentation is carried out on the random movement accelerometer data and labeled as (**Class0**). The annotated signals are then passed on to the next part of the pipeline to extract more meaningful features from the accelerometer data. The segmented activity signal of the Bicep Curls with a period of 30 s is shown in Fig. 4.

Segmented data from four participants are used for training the model and for cross validation using SVM (discussed in classification section). Data from two participants are used for testing the model. For each exercise, segmented data

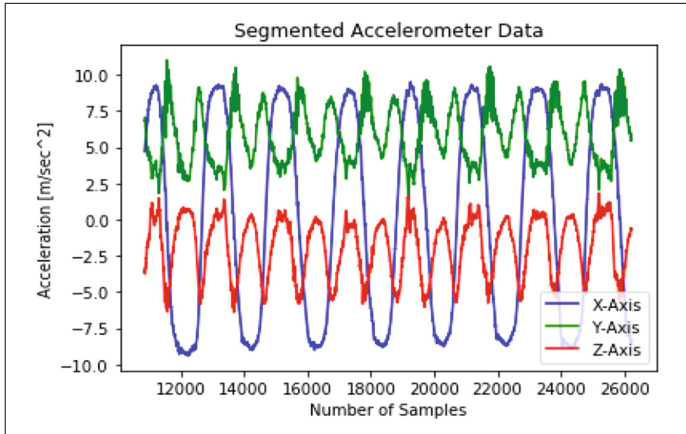


Fig. 4. Segmented Bicep Curls data for 30 s

(**Class1**) are concatenated with random movement data (**Class0**) from four participants. The concatenated data set is then subjected to feature extraction.

3.3 Feature Extraction

As each repetition lasts approximately for 4 s, a sliding window of 4 s with 50% (2 s) overlap is used for feature extraction from segmented 3D accelerometer data. Statistical features, which are widely used in the field of HAR (i.e., mean, standard deviation, min and max values, RMS values, Pearson correlation coefficients, FFT coefficients and entropy values), were computed from each window [4,7,9–11]. A feature vector comprising of $n=24$ features, is computed and extracted from each 4 second window. Extracted time and frequency domain features are listed in Table 2. These feature vectors are input to the next phase of the pipeline for dimensionality reduction.

Table 2. List of time and frequency domain features

Number of features	Feature description
6	Mean and Std Deviation from accelerometer, gyroscope
6	Minium and Maximum on each axis
3	RMS values on accelerometer on each axis
3	Pearson correlation coefficients between the axis
3	Energy from FFT coefficient on each axis of accelerometer
3	Entropy value computed from each axis of accelerometer

3.4 Feature Reduction Using PCA

The total $\mathbf{n}$ extracted features give an $\mathbf{n}$ -dimensional perception classification problem. The basic understanding is to have the minimum $\mathbf{k}$ correlated features that can completely define the characteristics of the underlying classification problem [4, 10]. PCA is a linear transformation of a number of correlated features into a smaller set of uncorrelated features. The significance of PCA is to obtain the reduced dimensionality from the feature set and to have fewer significant, meaningful relationships among the features. Normalizing and computing the correlation matrix among the extracted time-frequency features to obtain the Eigenvectors and associated values is the primary stage of PCA. Eigenvectors are sorted in descending order based on their corresponding Eigenvalues. The first principle component corresponds to the maximum variability of the original data and succeeding components add up to the remaining variability. In our analysis, we performed PCA on the data set and retained Eigenvectors that explain the variance of 98%, as shown in Fig. 5, which is discussed in Sect. 4. The computed PCA components are used as input to the activity recognition model.

3.5 Activity Recognition and Repetition Counting

SVM classifiers (one SVM model for each exercise) with optimum hyperplane parameters are used to recognize each LME exercise from the random movement [5, 12]. SVM is a supervised learning method that tries to minimize the cost function while enabling the identification of different classes (**Class1**) and (**Class0**), while maximizing the margin between margin vectors. In developing our approach for exercise classification, the model needs to learn from the input feature set $\mathbf{X}$ and the output target set $\mathbf{Y}$. Learning occurs with the mapping $\mathbf{X} \mapsto \mathbf{Y}$ with some object, $\mathbf{x} \in \mathbf{X}$ and with class label $\mathbf{y} \in \mathbf{Y}$. To recognize whether a specific exercise is being performed or not can be thought of as a binary classification problem with model input values $\mathbf{x} \in \mathbb{R}^n$ and the output target value $\mathbf{y} \in \{\pm 1\}$. The SVM model is then trained with the training set $(x_1, y_1), (x_2, y_2) \cdots (x_m, y_m)$. The basic linear SVM classifier $\mathbf{y} = f(\mathbf{x}, \alpha)$ with α the hyperplane parameters, weight (w) and y-interceptor (b), of the function is shown in Eq. (1) [12, 13].

$$f(x, (w, b)) = w \cdot x + b \quad (1)$$

The objective of the developed model is to minimize the cost function $I[f_m]$ or the associated empirical error with the linear fit over m training set (see (2)).

$$I[f_m] = \frac{1}{m} \sum_{i=1}^m l(f(x_i, \alpha), y_i) \quad (2)$$

where l is the zero-one loss function. Optimized fit can be obtained by minimizing the error through minimizing $\|w\|^2$ subject to the satisfaction of relations (3).

$$\begin{aligned} (w \cdot x_i + b) &\geq +1, & \text{if } y_i &= +1 \\ (w \cdot x_i + b) &\leq -1, & \text{if } y_i &= -1 \end{aligned} \quad (3)$$

A statistical model is considered best fit if it approximates the target function with a smaller error. When the model learns from the underlying data set and the effect is negative on the new data set, then the model is termed over-fitted or overlearned. If the relationship between the target function and training sets provided are more nonlinear, the model fails to learn the relationship. Typically, when the new data is added to the model, the model fails to learn from the data and it yields an output with the memorized behaviour from the training data. This results in a poor predictive performance of the model and even small noises or fluctuations can be classified as an activity and therefore efficiency of the classification model reduces on unseen data.

Regularization and cross validations are used to overcome the overfitting problem. An additional parameter $\lambda R(f)$ can be added (see (4)) to smooth the cost function to minimize the overall error and the term $\lambda R(f)$ is the regularization parameter [12]. The value of regularization parameter λ emphasizes the significance of misclassification. In kernelized SVM higher the value of regularization may lead to overfitting and lower the value leading to under-fitting.

$$I[f_m] = \frac{1}{m} \sum_{i=1}^m l(f(x_i, \alpha), y_i) + \lambda R(f) \quad (4)$$

We also used 10-fold cross validation where we randomly split the training dataset into 10 folds without replacement; with 9 folds used for the model training and one fold used for testing. This procedure is repeated 10 times so that we obtain 10 models and performance estimates. The described technique is applied to all developed models (one for each exercise) and the precision, recall and F-Score parameters are computed to validate the performance of each model.

SVMs can be easily kernelized to solve nonlinear classification problems as well. The basic idea behind the kernel methods to deal with such linearly inseparable data is to create nonlinear combinations of the original features to project them onto a higher dimensional space via a mapping function where it becomes linearly separable. We implemented a grid search hyperplane parameter optimization technique to improve the performance of the designed models by finding the optimal combination of hyperplane parameter values, including the regularization parameters and kernel options. [4, 12]

Reporting repetition count after each exercise is an important feedback for the patient. A best suitable axis from the accelerometer or gyroscope sensor is selected for this purpose and a Savitzky - Golay filter [15] of order 4 with a 2 second window is used for smoothing the data. Two algorithms are tested: one using peak-to-peak (**PP**) detection and another using threshold crossing (**ThC**). A threshold value of 75% of the difference between max and min peaks is used. Results obtained for Bicep Curl is shown in Fig. 8. The algorithms are studied on all upper body LME exercises. A similar approach will be used subsequently for lower body LME exercises.

4 Results

The study involves fifteen LME exercises (Table 1), with six participants and 30 s of segmented and annotated data. We developed a python script to extract 24 time-frequency features per a single window of 4 s duration for each exercise from each participant. PCA is performed on the training data set and 10 principle components which explain the variance of more than 98% are retained from the 24 features. A plot of principle components with descending significance and cumulative variance ratio is shown in Fig. 5. A grid search algorithm is used to find the optimum hyperplane parameters such as gamma, C and the kernel to

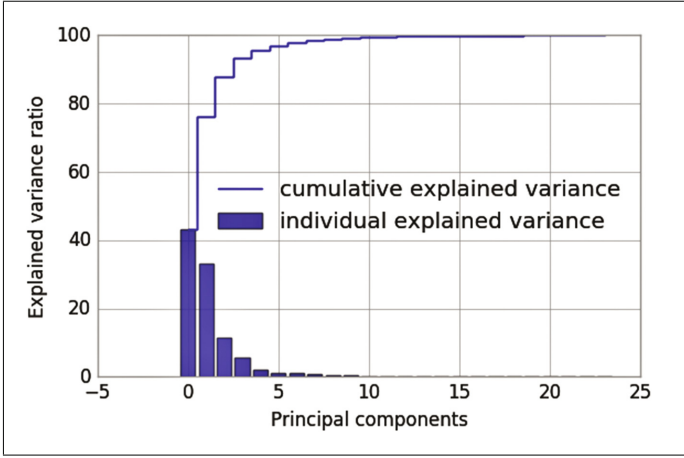


Fig. 5. Plot of principle components vs explained variance ratio and plot of cumulative explained variance for Bicep Curls.

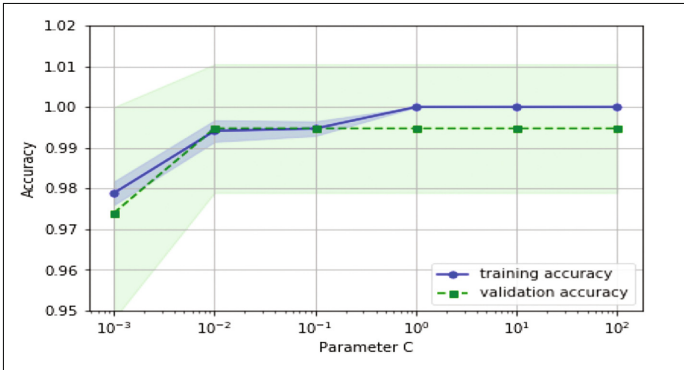


Fig. 6. Plot of calculation of C value required to attain desired Accuracy for Bicep Curls.

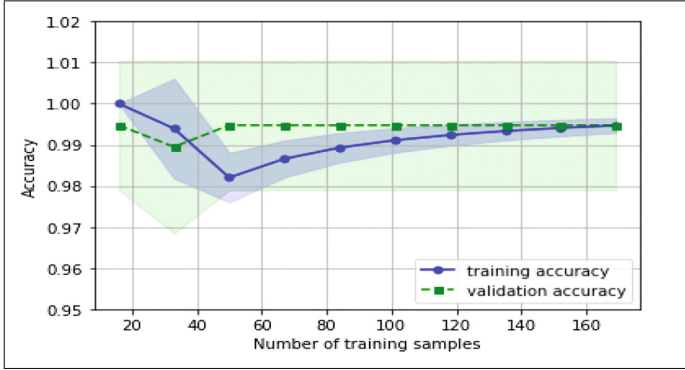


Fig. 7. Plot of number of training samples vs Accuracy for the Bicep Curls exercise.

be used with SVM for each exercise. The low C parameter value indicates the smooth decision surface. Both γ and C parameters are usually well within the range of 10^{-3} to 10^3 . The cross-validation curve is shown in Fig. 6. The γ parameter in SVM represents the influence of a single training example with values varying from low, indicating far reach, to high, indicating close reach. The number of training samples required to attain accuracy of $\geq 98\%$ is shown in Fig. 7 for the Bicep Curls. From Fig. 7, it can be seen that with a total number of approximate 100 samples, validation accuracy is close to the training accuracy.

The gray shade surrounding the learning curves indicate the tolerance level (mean $\pm$ stdev) in Figs. 6 and 7. From these plots, it is evident that validation accuracy almost matches the training accuracy, indicating our model is not suffering from overfitting. It also illustrates that the number of subjects used in this study is sufficient. The optimum hyperplane parameters and the type of kernel are selected using a grid search algorithm for the SVM model of each exercise (see Table 3).

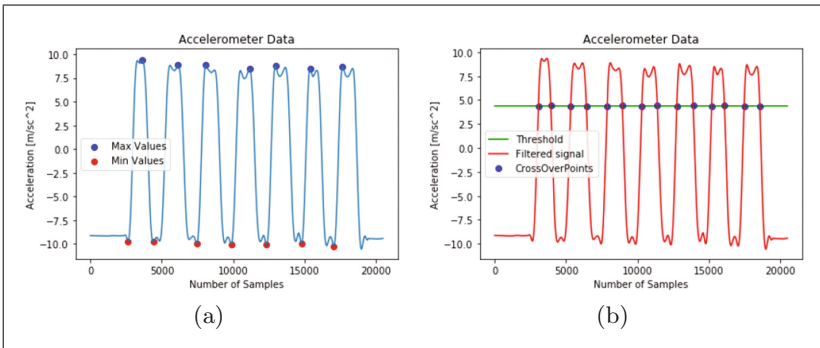


Fig. 8. Demonstration of repetition counting for the Bicep Curls: (a) Peak-to-peak method. (b) Threshold Cross method.

Table 3. Optimum hyperplane parameter and kernel selection for each LME exercise

Exercise	Best score	Gamma value	C value	Preferred kernel
Ex 1	0.9935	0.1	0.1	linear
Ex 2	1.0000	0.1	1.0	rbf
Ex 3	0.9869	0.1	0.1	linear
Ex 4	1.0000	0.1	1.0	rbf
Ex 5	0.9864	0.01	10.0	rbf
Ex 6	0.9935	0.01	100.0	rbf
Ex 7	1.0000	0.01	100.0	rbf
Ex 8	1.0000	0.1	0.1	rbf
Ex 9	1.0000	0.1	0.01	linear
Ex 10	1.0000	0.1	0.001	linear
Ex 11	0.9804	0.1	10.0	linear
Ex 12	1.0000	0.1	0.001	linear
Ex 13	1.0000	0.1	1.0	rbf
Ex 14	0.9935	0.1	0.01	linear
Ex 15	1.0000	0.1	0.1	rbf

Table 4. Performance measures for each LME exercise

Exercise	Precision	Recall	F1 score
Ex 1	1.000	1.000	1.000
Ex 2	1.000	1.000	1.000
Ex 3	1.000	1.000	1.000
Ex 4	1.000	1.000	1.000
Ex 5	1.000	1.000	1.000
Ex 6	1.000	1.000	1.000
Ex 7	1.000	1.000	1.000
Ex 8	1.000	0.963	0.981
Ex 9	1.000	1.000	1.000
Ex 10	1.000	1.000	1.000
Ex 11	1.000	1.000	1.000
Ex 12	0.963	0.963	0.963
Ex 13	1.000	0.963	0.981
Ex 14	1.000	1.000	1.000
Ex 15	1.000	0.963	0.981

Statistical performance measures of the SVM for recognizing each exercise are listed in Table 4. The proposed framework recognizes LME exercises (Table 1) with an overall accuracy $\geq 98\%$.

Figure 8(a) represents the peak-to-peak method used to count the repetition for bicep curls where the user has performed seven repetitions and the algorithmic count are marked on the graph. A count for repetition is considered when a pair of max and min is calculated. A threshold value of 75% of the max to min difference is computed. A repetition is counted when the threshold value is crossed twice. Figure 8(b) represents the threshold-cross method used to count the repetitions for the Bicep Curls where the user has performed seven repetitions with a computed threshold of 4.3766 m/s^2 . The sensor and axis, and associated method that determines the number of repetition for each upper body LME exercise is listed in Table 5.

Table 5. Sensor and Axis, and associated method for correctly determining the number of repetitions

Exercise	Sensor & Axis that correctly identified the repetition count	Accurate method
Ex 1	Accelerometer: X - axis Gyroscope: Z - axis	Accelerometer: PP, ThC Gyroscope: ThC
Ex 2	Accelerometer: X - axis	Accelerometer: PP, ThC
Ex 3	Accelerometer: X - axis	Accelerometer: PP, ThC
Ex 4	Accelerometer: X - axis	Accelerometer: PP, ThC
Ex 5	Accelerometer: X - axis Gyroscope: Y - axis	Accelerometer: PP, ThC Gyroscope: PP, ThC
Ex 6	Gyroscope: X - axis	Gyroscope: PP, ThC
Ex 7	Gyroscope: Y - axis	Gyroscope: PP
Ex 8	Accelerometer: Z - axis	Accelerometer: PP
Ex 9	Accelerometer: X - axis	Accelerometer: PP
Ex 10	Accelerometer: X - axis	Accelerometer: PP, ThC

5 Conclusions

In this paper, we described a novel framework, with a wrist worn inertial sensor, capable of automatically segmenting and recognizing whether individual LME exercise that were prescribed for cardiovascular rehabilitation were performed or not. The proposed framework employs optimized SVM training algorithm in conjunction with the selected time-frequency feature extraction technique to effectively recognize the LME exercises with more than 98% accuracy. Using the PCA technique, the dimensionality of the computed feature vectors are reduced while more than 98% of the variance is retained. All hyperplane parameters are tuned and different kernels are investigated to optimize the performance of the

training models. The learning curves are also plotted to ensure that the model is not over-fitted. Finally, the number of times each exercise is performed was calculated by detecting the peaks of the signals along with utilizing the pre-defined threshold values.

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Gait Stability During Shod and Barefoot Walking and Running on a Treadmill Assessed by Correlation Entropy

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Abstract. This study tests correlation entropy, K_2 , as a measure of stability for gait analysis. An average of 13 strides from 10 participants in each combination of one footwear (barefoot vs shod) condition and one gait mode (walking vs running) were collected during treadmill walking and running. Sagittal plane ankle, knee and hip angular displacement and velocity data were used for analysis. Two-way repeated measures ANOVA showed a main effect for gait mode ($p = .03$) – running had lower K_2 than walking, indicating higher stability. Although the sample of strides and participants was small, we speculate that the greater inertia for running helped stabilize movement control, making the running coordination pattern more resilient against small stride-to-stride perturbations.

Keywords: Correlation entropy · Gait · Biomechanics · Dynamical systems

1 Introduction

Human gait is a complex process involving the coordinated action of many muscles to mobilize the body in such a way as to satisfy the demands of the task. According to the concept of *motor abundance*, a vast number of configurations at the musculoskeletal level can be assembled to accomplish the same task [15]. This abundance is said to provide the movement system with flexibility to operate in the presence of numerous combinations of varying constraints, e.g., locomotion speed, obstacle avoidance. The interaction between the ground and the foot is one of the most important constraints as the dynamics between the two determine how forces are transmitted through the body, braking and propulsive forces, vertical acceleration of the center of mass and the stability of the movement system. With the term *stability* here, we refer to the resistance of the neuromuscular system to perturbations [9].

Barefoot gait has received an increasing amount of attention in the biomechanics literature in recent years. Barefoot gait presents an opportunity to investigate how the runner adapts to the changing foot-ground interface; however,

its effects have mostly been measured in terms of extracted, discrete measures such as loading rates [3, 5], joint moments [22], angles and angular velocities [11] and spatio-temporal parameters [4]. Moreover, gait variability is typically characterized by scalar values such as standard deviation [18]. Such measures of variability have been shown not to reflect the control of the movement system in response to varying constraints [8].

The emergence of coordinated movement in gait has been treated as a dynamical system [2]; therefore, measures that analyse the development of the system over time, so-called dynamic invariants, should be applied. In particular, the greatest Lyapunov exponent (λ^*) provides information on whether the trajectories of a dynamical system converge ($\lambda^* < 0$) or diverge ($\lambda^* > 0$) with respect to its starting conditions. Thus, λ^* assesses whether the investigated dynamical system is chaotic [7]. However, estimating the Lyapunov exponents usually requires thorough phase space reconstruction when scalar measurements are used [7].

Different types of entropy measures also assess the predictability of an investigated dynamical system. One such measure is the Kolmogorov-Sinai (K) entropy, which is, however, difficult to calculate from experimental data. Therefore, in practice, different approaches are used to approximate K . A frequently used approach is to estimate K through the correlation entropy, K_2 . There are several methods for computing K_2 , including the recurrence plot approach. K_2 can be derived from the structure of diagonal lines in a recurrence plot [17]. An advantage of this approach is that K_2 can be determined without phase space reconstruction [23]. Bypassing the phase space reconstruction also affords the analysis of multivariate datasets – an attractive prospect for biomechanics.

The following interpretations can be drawn from K_2 : $K_2 \approx 0$ means that the system under investigation is nearly regular or periodic, $K_2 > 0$ means the system is chaotic and if K_2 is infinite the system is random.

The purpose of this study is to test K_2 as a measure for determining differences in dynamical stability between barefoot versus shod gait and for walking versus running. We expected that, since the study participants were new to barefoot running, shod gait would be more stable, and also that the greater momentum associated with the comfortable running speed would stabilize the dynamics of the system compared to walking and result in greater K_2 for walking.

2 Material and Method

2.1 Procedures

After giving written consent and following a seven minute warm-up and familiarization trial, ten participants (five male, five female; age 26.8 ± 7.7 years) walked and ran for 30 s at two speeds normalized to leg length. The dimensionless Froude number was used to set the treadmill speed to limit the effect of body size on gait parameters. Based on pilot data [1], the Froude numbers 0.28 and 1.08 were used, which roughly corresponded to comfortable walking and

running, respectively. Treadmill speeds, v , were determined by

$$v = \sqrt{F l g}, \quad (1)$$

where F is the Froude number, l is the leg length of the participant, without shoes, measured from the greater trochanter of the femur to the ground, and g is the acceleration due to gravity [6]. Mean speeds for walking and running were 1.43 m/s and 2.79 m/s, respectively. Each of the speeds were performed in two conditions: barefoot and shod. In the shod condition, participants wore their own running shoes. One of the participants' running shoes were considered minimalist [?]. None of the participants were experienced in barefoot running. All procedures in the study were approved by the Human Ethics Committee of the University of Otago.

Data were collected at 100 Hz using a ten-camera Vicon system (Oxford Metrics Ltd., UK). Local coordinate systems were established for the lower limbs according to the procedures in Robertson et al. [19]. Joint centers and subsequent ankle, knee and hip Cardan angles and angular velocities for the left leg were calculated in Visual3D (C-Motion, Germantown, MD). Stride movement times varied such that the number of strides available for analysis differed between participants; therefore, between 12 and 14 strides in each condition were selected for analysis. Toe-off and touchdown events were determined by finding local minima in vertical spatial position of the foot markers at the calcaneus and the head of the second metatarsal. Trials were then trimmed to individual strides from right toe-off to the next toe-off. A dual-pass 2nd order Butterworth filter with a cutoff of 10 Hz was applied to remove noise from the joint angular data [20]. All processing and analysis procedures were performed in MATLAB (The Mathworks, Natick, MA).

2.2 Method

The data analysis involved several steps and was conducted for each participant and condition individually. The analysis is based on the multivariate time series $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, where $\mathbf{x}_i \in \mathbb{R}^{6 \times 1}$ consisting of the six kinematic variables described in the previous section. Since the value ranges of the variables were different, each variable was normalized using the z-score transformation. Further, each variable was resampled to 400 Hz using a cubic spline interpolation since a proper approximation of K_2 requires a certain length of the time series [17]. Additionally, to reduce correlation effects between the variables of the multivariate dataset, a principle component analysis was conducted to the respective multivariate dataset $D \in \mathbb{R}^{n \times 6}$ with sampling rate τ . Based on the resulting principal component coefficients $C \in \mathbb{R}^{6 \times p}$, p is the number of principal components, the new feature set N was prepared through

$$N = D \times C \quad (2)$$

The next step is the calculation of the recurrence matrix on which recurrence plots are based. A recurrence plot is a method to visualize recurring states

of a dynamical system. It is computed based on a trajectory containing measurement(s) of the system under investigation. Determining a recurrence plot usually requires phase space reconstruction which, however, is not necessary for the determination of the invariant K_2 [23]. In our case, the recurrence matrices were calculated with respect to the trajectory represented by N . According to Marwan et al. [17] the recurrence matrix is

$$\mathbf{R}_{i,j} = \Theta(\epsilon - \|N_{i,\cdot} - N_{j,\cdot}\|) \quad (3)$$

where ϵ is a predefined threshold, $\Theta(\cdot)$ is the Heaviside function, and $N_{i,\cdot}$ denotes the i -th row of N .

Finally, the correlation entropy K_2 can be estimated from a set of recurrence plots calculated for a range of ϵ [17]. For each $\mathbf{R}(\epsilon)$ the cumulative distribution of diagonal lines $P_\epsilon^c(l)$ must be determined. The probability of finding a diagonal line of at least length l in a $\mathbf{R}$ is given by

$$P_\epsilon^c(l) \propto \exp(-K_2(\epsilon)l) \quad (4)$$

For large l 's and independent of ϵ , $P_\epsilon^c(l)$, represented on a logarithmic scale, should become a straight line with slope $-K_2 \tau$ [23]. In this study, $P_\epsilon^c(l)$ was determined for 101 different $\epsilon \in [1, 3]$. The final K_2 values were determined as the mean of the 101 values of $K_2(\epsilon)$.

2.3 Statistical Analysis

A two-way repeated measures ANOVA was conducted to compare the effects of footwear condition (barefoot versus shod) and gait mode (walk versus run) on the K_2 values of stability. Significance was set at $\alpha = 0.05$. All statistical procedures were conducted in MATLAB.

3 Results

Exemplar kinematics for Participant 1 walking at a comfortable speed are shown in Fig. 1, and running at a comfortable speed in Fig. 2. The ankle angle was defined so that neutral standing position is 0° , plantarflexion is negative and dorsiflexion is positive. Generally, throughout most of the stride the foot was more dorsiflexed when barefoot. Knee and hip angles were similar between barefoot and shod conditions (0° is extended, positive is flexed and negative hyperextended). Angular velocities were also similar between conditions, but statistical analysis of range of motion and angular velocities was not included in this paper.

Figure 3 illustrates for Participant 3 how K_2 values were determined. Panel A) shows the log of the cumulative line distributions of the respective recurrence plots for each ϵ . To derive the K_2 values from such a histogram one identifies a line length l_t , the longer the better, where the distribution lines are roughly parallel for all ϵ . Subsequently, panel B) shows the $K_2(\epsilon)$ which are related to the respective absolute value of the slope at l_t times the sampling frequency for ϵ .

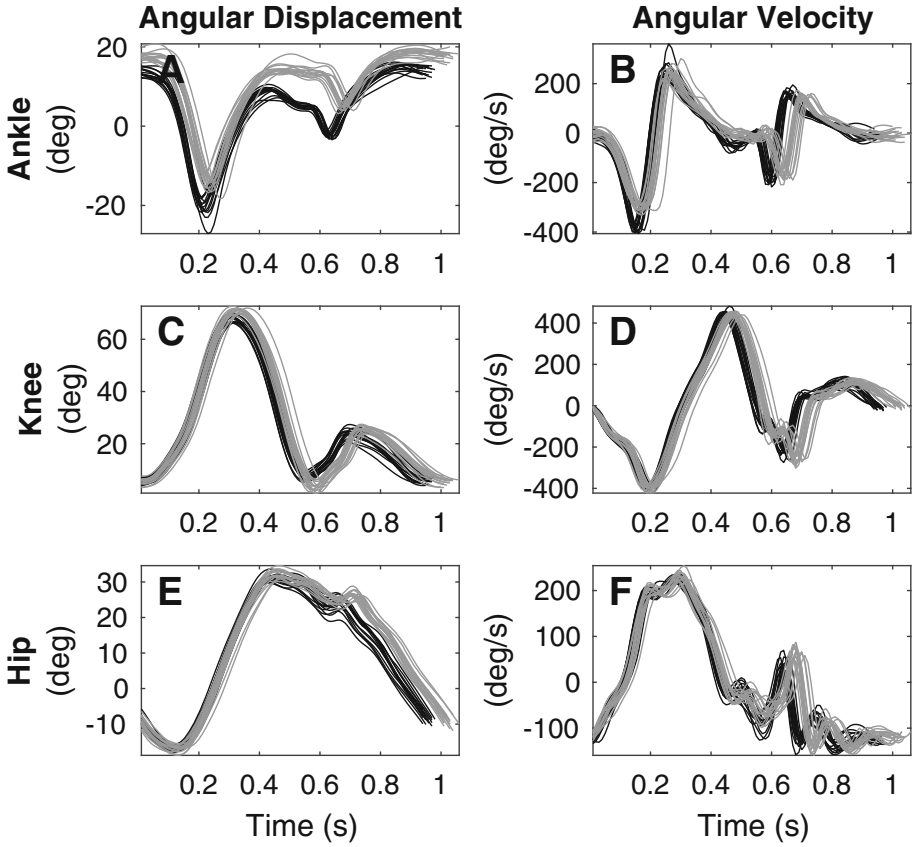


Fig. 1. Kinematic data for one example participant at the respective walking speed. (A) ankle angle, (B) ankle angular velocity, (C) knee angle, (D) knee angular velocity, (E) hip angle and (F) hip angular velocity. Barefoot strides are shown in black and shod in grey. Strides begin and end with right foot toe-off.

The resulting K_2 that was used for further analyses was determined as the mean of the $K_2(\epsilon)$. For the exemplar participant in Fig. 3, l_t was 500 and $K_2 = 0.70 \pm 0.07$.

K_2 values were significantly higher for the walking strides compared to running, $F_{(1,9)} = 6.90, p = .03$ (Fig. 4). K_2 values did not differ between barefoot compared to shod, $F_{(1,9)} = 0.01, p = .92$, and there was no significant interaction between gait mode and footwear, $F_{(1,9)} = 0.02, p = .73$. We also note that the variability in K_2 was high for all conditions (Fig. 4).

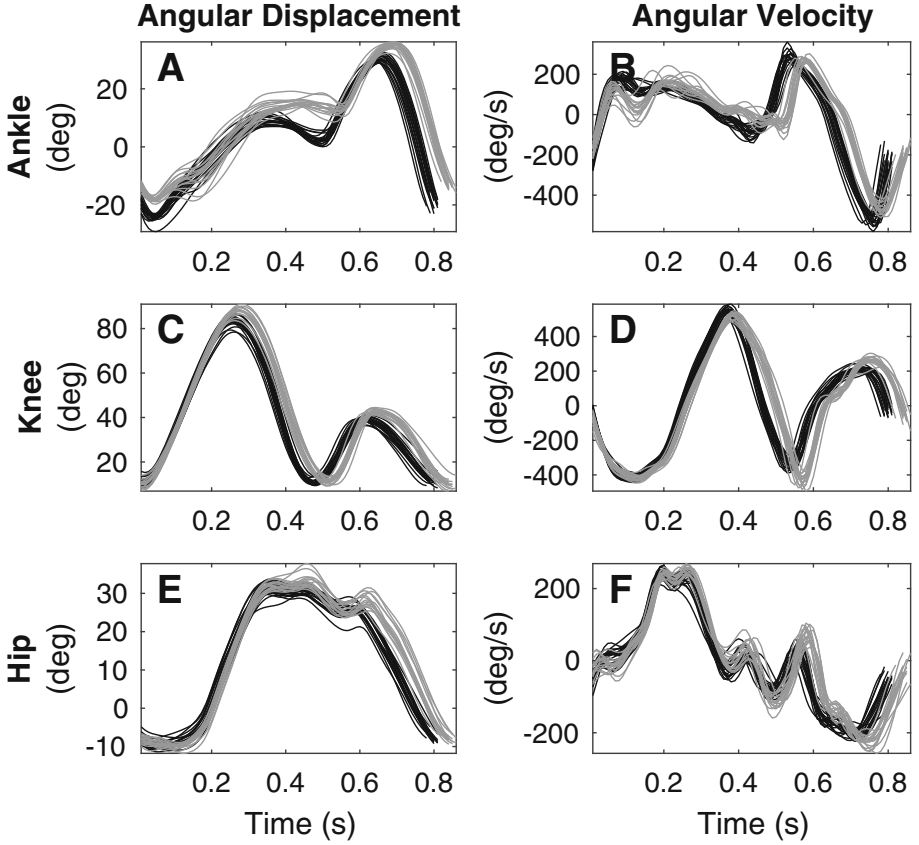


Fig. 2. Kinematic data for one example participant at the respective running speed. (A) ankle angle, (B) ankle angular velocity, (C) knee angle, (D) knee angular velocity, (E) hip angle and (F) hip angular velocity. Barefoot strides are shown in black and shod in grey. Strides begin and end with right foot toe-off.

4 Discussion

This study tested the correlation entropy value K_2 as a measure of stability in gait when representing strides by unilateral sagittal plane kinematics. K_2 is a measurement to assess the predictability of the investigated dynamical system and, thus, describes the (chaotic) variability of the system's behaviour; the more variable a system is, the less predictable it is. We expected shod running to be the most stable combination of gait mode and footwear condition, but only the main effect of running achieved significance. If the main effect of gait mode on stability is real, we suggest that increased momentum or inertia of the runner limits the complexity of the movement. When considering compensations along the kinematic chain of the lower extremity, running at comfortable pace is less complex than comfortable walking. In other words, the greater inertia associated with

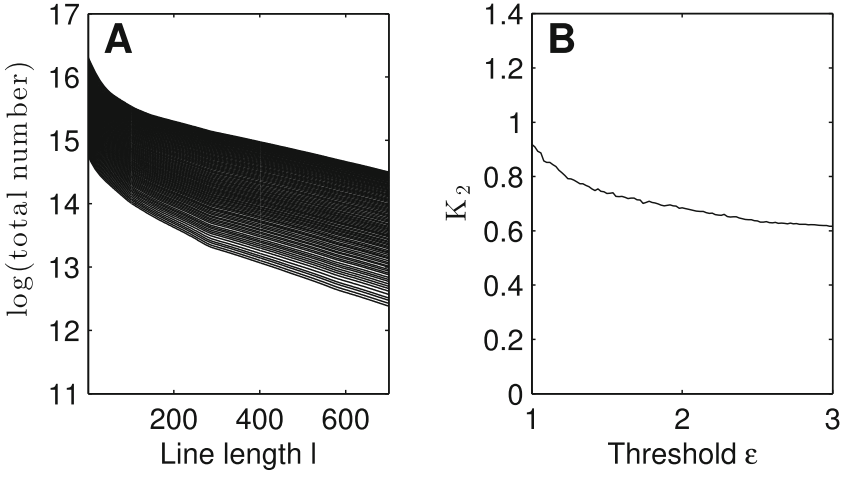


Fig. 3. A) Distribution of line length l of one participant at one condition, B) resulting $K_2(\epsilon)$ for $\epsilon \in [1, 3]$ at line length 500.

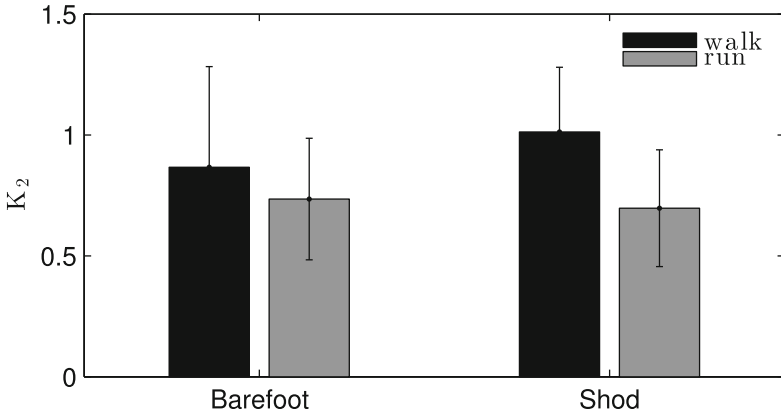


Fig. 4. Mean K_2 values for both walking and running in barefoot and shod conditions for all participants. Error bars represent one standard deviation.

running simplified movement control, making the running coordination pattern more resilient against small stride-to-stride perturbations. However, we would need to use data from more participants to substantiate these interpretations.

Jordan and colleagues showed that stability increased at the preferred walking speed [14] as well as at the preferred running speed [13]. Furthermore, they showed greater ankle displacement stability during running compared to walking [12], although head displacement was more stable for walking. Given that Latt et al. [16] showed that preferred walking speeds stabilize head displacement and correlation entropy allows the analysis of stability using multivariate data,

in future work we may look to combine lower extremity kinematics, as in the current study, with upper body kinematics to compare our work with that of Jordan et al. Conversely, England and Granata [10] showed greater stability at the slowest walking speeds assessed by λ^* . We did not look at speeds slower than the preferred walking speed, but future work may focus on these speeds as well as speeds around the transition speed to gain a better comparison between the multidimensional K_2 and λ^* used by [10].

Correlation entropy might be a promising measure of stability for biomechanical data. According to [21], K_2 is a lower bound for the sum of the positive Lyapunov exponents, which enables the stability of the system to be investigated. K_2 and λ^* , however, are different measures, which makes comparisons between studies difficult. Moreover, there are several different algorithms for calculating the phase space reconstruction as well as for approximating λ^* , which further complicates comparisons between e.g. [13] and [10]. K_2 , calculated using the recurrence plot method alleviates the necessity to reconstruct the phase space; furthermore, it allows multi-dimensional time-series to be used – two features that make K_2 attractive for studying stability in biomechanical systems.

A major limitation of this analysis is the small sample of participants and the number of strides per condition. The approximation of K_2 values is based on time series of length $\sim 5,500$ in this research. According to [17] more than 10,000 data points are needed to achieve good K_2 approximations through the proposed recurrence plot based method. However, the measurements did not contain enough strides to achieve such long time series. Further resampling would not have added more information to the time series; therefore, we suggest for future research either to collect data for more strides, or to use bootstrapping techniques based on the measured strides to extend the respective time series for K_2 approximations using the proposed recurrence plot approach.

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Statistical Models for Predicting Short-Term HR Responses to Submaximal Interval Exercise

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Abstract. Aim of the study was to identify possible predictors influencing the variability of individual short-term heart rate (HR) responses to submaximal interval exercise using a probabilistic model. Short-term HR responses to the change of load bouts obtained in a twelve-week training intervention were analyzed. Questionnaires gathered preceding sport activity, sleep, nutrition, health and mood prior to each training session. Additionally, time of the day and number of interval was included in calculation. Multiple regression method was used to identify predictors for start heart rate (HR), steady state HR, and for the slope of the HR curve. Especially the number of the interval, physical and mental health, and negative mood were influencing these responses. The start heart rate was identified as predictor in five of eight response parameter. Time was a factor highly varying between participants. Future research need to validate the results in a wider sample and integrate more parameters in the analysis.

Keywords: Short-term HR responses · Modeling · Individual

1 Introduction

All over the world, the level of physical inactivity is rising [19] although the positive influence of physical activity on health has been empirically verified and widely propagated. However, significant changes on health require an individually optimized training to induce the desired positive effects. As insufficient load may result in sub-optimal or missing adaptations, overload may cause risks like injuries.

In endurance training, the heart rate (HR) is considered a valid and reliable indicator for measuring individual strain. HR is a non-invasive and instant marker and the measurement does not interfere with the trainee's movements.

For optimal adaptations, the individual optimal strain needs to be evoked in the training person. However, predicting this optimal strain is challenging as identical load can induce completely different responses in different individuals. These responses depend on a high variety of influencing factors, e.g., age, gender, or training status [12]. In order to enable an immediate adjustment of induced load, the individual responses have to be modeled and predicted during or prior to training.

Numerous studies can be found focusing on modeling individual strain [15]. Most models require the estimation of individual parameters derived from the training data to

fit the model individually to the subject. In general, the parameters characterize time constants [8] as well as constant or varying model parameters [2, 17]. Machine learning approaches such as artificial networks [20] require a data set for learning. Whereas some authors integrate time varying parameters [2, 6] expecting an increase of HR over the training time, most models predict individual HR presupposing similar or equal levels at a given stress. The HR response to the change of load bouts is a varying signal that seems to be additionally influenced by rather temporarily and fluctuating factors. Several studies show an influence of current psychophysical states of the individual HR. Detailed analyses of multiple factors influencing short-term HR response in different individuals are (still) missing.

The aim of this paper is to identify the influence of temporary factors on HR and develop individual probabilistic prediction models. Therefore, HR data obtained in a 12-week endurance training intervention on a bike ergometer was analyzed. Based on literature analysis, questionnaires were used to gather information about the psychophysical state of the individual.

2 Materials and Methods

The study presented here was approved by the Ethics Committee of the Technische Universität Darmstadt in 2016.

2.1 Participants and Apparatus

Four healthy and active adults volunteered to participate in the study after having signed an informed consent. The participants' characteristics are displayed in Table 1.

Table 1. Demographic and anthropometric description of the participants.

	Participant 1 (P1)	Participant 2 (P2)	Participant 3 (P3)	Participant 4 (P4)
Age [years]	31	32	32	31
Height [m]	168	175	182	185
Weight [kg]	79.2	69.5	88.2	96.1
Sex	Female	Female	Male	Male
BMI [kg/m ²]	28.1	23.0	26.7	27.6
Activity time per Week [h]	3.5	2	7	3

All tests were performed using a cycle ergometer with a flywheel (Daum Ergometer 8008 TRS 3; Fürth, Germany). The power was controlled by the resistance at the flywheel and measured in Watts by the ergometer. HR data was successively recorded beat by beat using a Polar V800 sport watch (Polar Electro, Kempele, Finland) and a corresponding Polar chest belt (T31). Data recording started at the beginning of the training protocol. Respiratory parameters were recorded using the spiroergometry device K5 (COSMED, Rome, Italy) during the exhaustion and the sub tests using a mixing chamber. First anaerobic threshold (ANT1) and second anaerobic threshold

(ANT2) were automatically calculated from the respiratory parameters using the OMNIA Software (COSMED, Rome, Italy).

2.2 Protocol

All data was obtained during a twelve-week endurance training intervention on a bike ergometer. This duration was chosen as adaptations to the training can be reliably observed after this training period [3]. Prior to and after completion of the intervention, the participants performed an all-out exhaustion test to estimate the individual maximal HR, maximal Oxygen Uptake and the individual anaerobic thresholds of the participants. Additionally, two subtests were performed in week 4 and 8 to adapt the training intensity to the training status of the participants during the training process.

The protocol of the exhaustion test started with a resting period with the participants sitting still on the ergometer. After 3 min the participants started pedaling for 2 min at 25 W, followed by 3 min at 50 W. After this warm-up period the load at the ergometer was successively increased by 50 W every 3 min until exhaustion. In the subtests, the warm-up period of the exhaustion tests was repeated. Subsequently, the participants had to pass an exercise phase of 3 increasing load levels for three minutes each. The load for these load levels was calculated to induce responses corresponding to the individual's anaerobic thresholds. Therefore, the first load was calculated to evoke responses below ANT1, the second load was calculated to evoke responses between ANT1 and ANT2 and the third load was calculated to evoke responses above ANT2. After this 9 min exercise period a resting period of 5 min active recovery at 25 W was applied. Subsequently, another 9 min exercise phase and 5 min recovery phase were successively added to validate the results.

According to the guidelines of the WHO [18] the training volume for the intervention was set to 25 min of intensive training three times a week. Three different training methods were applied: the intensive continuous method (ICM), the extensive interval method (EIM) and the intensive interval method (IIM) [13]. The load for each protocol was calculated to evoke the target HR using the HR responses and the corresponding load levels from the first exhaustion test and the subtests. The training protocols are displayed in Table 2 in detail.

Table 2. Protocols for the training intervention

	Intensive Endurance Method (ICM)	Extensive Interval Method (EIM)	Intensive Interval Method (IIM)
Intensity	75% HR _{max}	80% HR _{max}	95%HR _{max}
Load period	25 min	3:30 min	1:00 min
Recovery Time between load intervals	0 min	1:30 min	1:30 min
Repetitions	1	5	10

All protocols were automatically applied at the ergometer. The participants were advised to keep the pedal rate (PR) constant at 80 revolutions per minutes (RPM) [7].

2.3 Data Processing

HR data for the exercise phases of the EIM training protocol was analyzed. In this training protocols, the HR of the participant is expected to stay constant within a submaximal range. Therefore, short term HR responses to the change of load bouts are expected to increase linearly in the beginning, followed by an exponential increase into a steady state. Steady state is reached when the variation of HR does not exceed a corridor of 5 bpm for the rest of the exercise interval [14]. The time of the load phases in the EIM is expected to be sufficient that HR can reach this steady state.

HR data of the participants was automatically divided into exercise and recovery phases for each training protocol. The phases started with the change of the load at the ergometer. The warm-up period and the recovery phases were not part of the computations. Thus, we obtained 60 h curves for each participant.

In a first step, specific values of each HR response were calculated. HR_{start} was calculated as minimum HR value in the first 10 s of each curve to compensate possible measuring errors caused by the variability of HR. The mean HR over the last 30 s of each curve was calculated as HR_{steady} for all given data. In total, 240 data sets were processed for calculating HR_{start} and HR_{steady} .

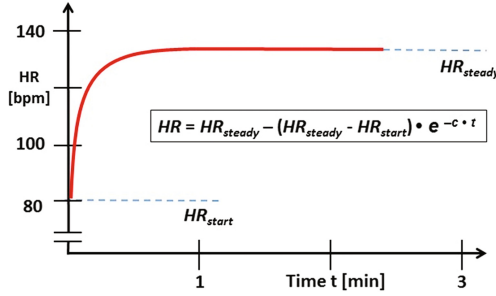


Fig. 1. Prototypical time course of HR according to the Bunc equation [5].

Additionally, the slope of each curve (c) was approximated using the monoexponential formula proposed by Bunc, Heller, & Leso [5] (see Fig. 1). According to Bunc et al. [5] the submaximal HR response to the change of load bouts is described by the following equation:

$$HR_{current} = a - b \cdot e^{-c \cdot t} \quad (1)$$

Legend:

- a – steady state HR level elicited by the change of load (HR_{steady}) [bpm]
- b – DeltaHR, deviation of HR_{steady} and HR at the start of exercise (HR_{start}) [bpm]
- c - slope of HR curve
- t - time [min]

For approximation, parameters a and b representing HR_{steady} and the deviation of HR_{steady} and HR_{start} were directly estimated from the data. Subsequently, the HR data was linearized and parameter c was estimated using the linear regression analysis. Due to incomplete HR courses caused by measuring errors, six curves were excluded for calculating value c . In total, 234 data sets were processed calculating value c .

2.4 Recording of Influencing Factors

Prior to the training, the participants completed questionnaires to identify possible influencing factors. The first questionnaire was a self-developed instruments based on literature reporting factors that might influence HR responses. Sport activity 24 h prior to the training was recorded regarding type, duration, and intensity. Additionally, the participants declared if they felt any pain, soreness or fatigue induced by this prior sport activity. The restfulness of the sleep [9] in the night prior to the training was assessed with a five point Likert scale (1: very restful, 5 not at all restful). The nutrition two hours before the training [11] was assessed by a three point scale (“1” - nothing, “2” - light meal, “3” - full meal). The current health status [18] was measured by three items regarding physical health, psychic health and social health on a 5 point scale ranging from 1 “very bad” to 5 “very high”.

The influence of mood on the HR and the HR response has been widely verified [4]. Therefore, the current mood of the participants was measured with the “Stimmungs- und Befindens-Skalen” (Trait and State of Mood Skale, SBS) [10]. The SBS consists of 8 items regarding “Activation”, “Contact Motivation”, “Self Confidence”, “Cheerfulness”, “Nervousness”, “Sleepiness”, “Irritability”, and “Dejectedness”. Whereas the first four items have a positive valence, the last four items have a negative valence. Each item is described by three words each (e.g. “Self Confidence”: confident, ascendant, experienced). All items are scored on a Likert Scale ranging from 0 (not applicable) to 10 (fully applicable). Additionally, the time of the day and the number of the interval was included in the analysis.

In total, 19 variables were measured. To integrate only variables that are independent of each other, some variables were grouped together. The type, duration and intensity of the prior sport activity was converted into the metabolic equivalent (MET) according to widely accepted compendium [1]. Mood was aggregated in positive mood (SBS +) and negative mood (SBS-). The mean value of the four corresponding items was used for calculation.

The constant parameters age, gender, and training status were excluded from calculation. Additionally, the load applied at the ergometer was also excluded from calculation as this load was expected to evoke a reproducible constant HR response to the defined target HR.

In total, 11 predictors were integrated in the analysis. Figure 2 displays the workflow.

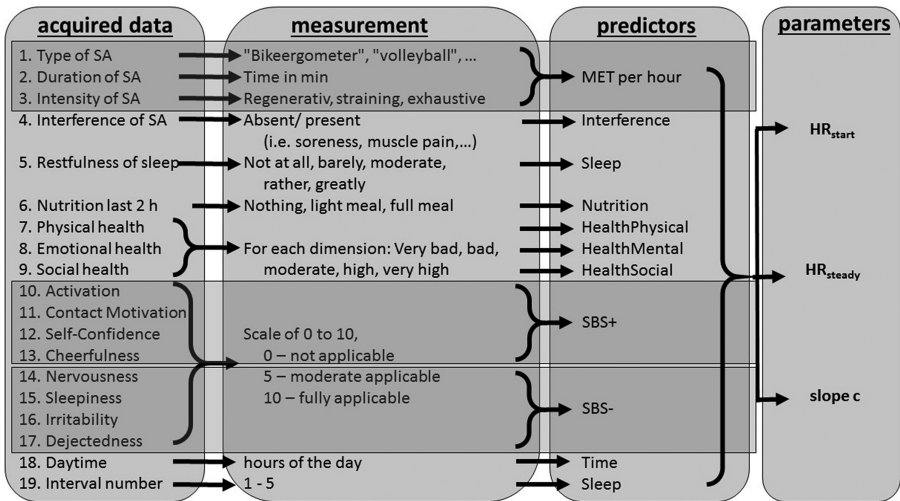


Fig. 2. Process of data acquisition, aggregation and predictors; SA: sports activity

2.5 Statistical Analysis

All statistical analyses were performed with IBM SPSS Statistics 24.

In order to confirm submaximal strain in all participants, we tested if steady state was reached in all HR curves. HR was preprocessed using the moving average method. Steady state was reached when the difference of the measured HR and HR_{steady} was smaller than 5 bpm and the increase of the averaged HR was smaller than 5 bpm. The deviation of HR_{steady} from the target HR was calculated for each participant to identify a varying response in each participant.

Multiple regression methods were used to evaluate the overall contribution of the influencing factors on the dependent variables HR_{start}, HR_{steady}, and c. As HR adaptations and perception of trait is very individual the tests were performed for each participant. At first, a regression model with stepwise integration was applied to identify significant influencing factors. Durbin-Watson values were calculated to indicate an absence of autocorrelation of the dependent values. Additionally, another stepwise multiple regression was calculated for HR_{steady}, and c integrating HR_{start} as possible influencing factor. This method was chosen after examining the raw HR data to analyze the prediction feasibility of HR_{start}. Factors that were retained in the stepwise model were then analyzed with a multiple regression model using forward selection.

3 Results

Steady State was reached in all HR curves (*Mean* = 117.5 s, *SD* = 25.8 s, *Max* = 190 s, *Min* = 48 s). In all participants, HR_{start} and HR_{steady} increased during the training session. The highest increase was found in participant 4 (see Table 3).

Table 3. HR_{start} and HR_{steady} for all participants

	Participant 1		Participant 2		Participant 3		Participant4	
	HR _{start}	HR _{steady}	HR _{start}	HR _{steady}	HR _{start}	HR _{steady}	HR _{start}	HR _{steady}
<i>Mean</i>	118.7	148.0	108.4	147.9	97.8	144.9	106.7	146.3
<i>SD</i>	8.86	8.75	7.84	4.57	7.91	7.53	11.16	7.98
<i>Max</i>	140	168	123	156	119	165	135	168
<i>Min</i>	97	132	82	131	81	128	86	127

No difference of HR_{start} or HR_{end} were observed depending on the week of the training. Statistical data of HR_{start} and HR_{steady} for all exercise phases is displayed in Table 3.

In all participants a high variation of the deviation of HR_{steady} from the target HR was observed (*Mean* = 3.83, *SD* = 7.28, *Min* = -14 bpm, *Max* = 27 bpm) The lowest mean deviation was found in participant 2 (*Mean* = 0.08 bpm, *SD* = 4.56 bpm, *Min* = -8 bpm, *Max* = 17 bpm).

The multiple regression method revealed significant predictors for all values except for c in participant 1, participant 3 and participant 4. If HR_{start} was included in calculation, predictors for all values were found. All displayed predictors meet the requirements for applying the regression method. The Durbin-Watson values varied but showed a higher acceptability only in participant 2 if HR_{start} was included. Only the values for HR_{end} in participant 2 and 3 require further attention. If the multiple regression did not reveal any predictors, no values were calculated. All Durbin-Watson values are displayed in Table 4.

Table 4. Durbin-Watson test for all calculated regressions

	Participant 1		Participant 2		Participant 3		Participant 4	
	-	+	-	+	-	+	-	+
HR _{start}	1.955	x	1.792	x	1.680	x	1.636	x
HR _{end}	2.100	1.812	1.206	1.384	1.682	1.137	2.183	2.183
c	x	1.627	2.009	2.009	x	2.064	x	1.866

Legend:

-: HR_{start} was not included in calculation

+: HR_{start} was included in calculation

x: no Durbin-Watson-Value calculated

3.1 Participant 1

The responses HR_{start} and HR_{end} of participant 1 showed a high influence of multiple predictors. If HR_{start} is not included in calculation no predictors were identified for value c. Time of the day and nutrition were predictors that were found in all HR values. HR_{start} was a main predictor for HR_{end} and the only predictor for value c. All data is displayed in Table 5.

Table 5. Revealed predictors in participant 1, left side: results for predictors without inclusion of HR_{start} in calculation, right side: results for inclusion of HR_{start} in calculation.

HR _{start}							
	R ²	F	Sig.		R ²	F	Sig.
HealthMental	.201	15.80	.000				
Time	.329	15.433	.000				
SBS–	.558	25.84	.000				
Interval	.640	27.17	.000				
Sleep	.686	26.78	.000				
HealthPhysical	.730	27.55	.000				
SBS+	.793	33.24	.000				
Nutrition	.822	35.03	.000				
HR _{end}				HR _{end} , with HR _{start} included			
HealthMental	.256	21.28	.000	HRstart	.732	162.43	.000
Time	.396	18.71	.000	MET	.754	91.39	.000
SBS–	.647	34.16	.000	Time	.789	74.54	.000
Interval	.723	35.83	.000	Nutrition	.812	64.68	.000
MET	.783	39.00	.000				
HealthSocial	.856	52.59	.000				
Interference	.876	52.50	.000				
c				c, with HR _{start} included			
no variables found				HFstart	.091	6.92	.011

3.2 Participant 2

The responses in participant 2 also showed multiple predictors for the HR responses. Especially nutrition and the physical health were predictors that were found in HR_{start} and HR_{end}. The slope of the curve represented by value c was only influenced by the sport activity prior to the training and by possible interference effects. If HR_{start} was included as predictor only HR_{start}, interval and the time of the day influenced HR_{end}. No change was observed for value c. All data is displayed in Table 6.

3.3 Participant 3

The main predictor for participant 3 was the number of the interval. No variables were found without inclusion of HR_{start} as predictor for value c. HR_{start} was the only predictor for HF_{end} and c if included in the calculation. Although we received low Durbin-Watson values, the F test of the developed models show a high significance. All data is displayed in Table 7.

Table 6. Revealed predictors in participant 2, left side: results for predictors without inclusion of HRstart in calculation, right side: results for inclusion of HRstart in calculation

HR_{start}							
	R ²	F	Sig.		R ²	F	Sig.
Interval	.142	10.75	0.002				
Nutrition	.269	11.85	0.000				
HealthPhysical	.531	23.28	0.000				
SBS+	.561	20.33	0.000				
HealthMental	.654	23.34	0.000				
HR_{end}				HR_{end}, with HR_{start} included			
Interval	.410	40.38	0.002	HFstart	.497	57.39	0.000
Nutrition	.475	25.75	0.000	Interval	.653	53.52	0.000
HealthPhysical	.576	25.35	0.000	Time	.689	41.29	0.000
SBS-	.699	31.90	0.000				
Sleep	.747	31.90	0.000				
c				c, HR_{start} included			
MET	.061	4.84	0.032	MET	.061	4.84	0.032
Interference	.195	6.92	0.002	Interference	.195	6.92	0.002

Table 7. Revealed predictors in participant 3, left side: results for predictors without inclusion of HRstart in calculation, right side: results for inclusion of HRstart in calculation

HR_{start}							
	R ²	F	Sig.		R ²	F	Sig.
Interval	.368	33.80	0.00				
MET	.588	40.61	0.00				
SBS-	.724	48.90	0.00				
HR_{end}				HR_{end}, HR_{start} included			
Interval	.254	21.14	0.000	HFstart	.583	83.59	0.000
Interference	.371	18.37	0.000				
HealthPhysical	.414	14.89	0.000				
HealthSocial	.572	20.74	0.000				
Nutrition	.701	28.72	0.000				
c				c, HR_{start} included			
No variables found				HFstart	.141	9.84	0.003

3.4 Participant 4

The predictors of HR_{end} and c in participant 4 were independent of inclusion of HR_{start}. Especially the number of interval explains 47% of the variance of HR_{start} and 42% of the variance of HR_{end}. All data is displayed in Table 8.

Table 8. Revealed predictors in participant 4, left side: results for predictors without inclusion of HR_{start} in calculation, right side: results for inclusion of HR_{start} in calculation

HR_{start}							
	R ²	F	Sig.		R ²	F	Sig.
Interval	.476	54.57	0.00				
Time	.571	40.22	0.00				
HR_{end}				HR_{end}, HR_{start} included			
Interval	.424	44.40	0.00	Interval	.424	44.40	0.00
MET	.641	53.62	0.00	MET	.641	53.62	0.00
SBS–	.702	47.43	0.00	SBS–	.702	47.43	0.00
Sleep	.751	45.45	0.00	Sleep	.751	45.45	0.00
SBS+	.791	45.62	0.00	SBS+	.791	45.62	0.00
HealthPhysical	.834	50.50	0.00	HealthPhysical	.834	50.50	0.00
Time	.867	56.12	0.00	Time	.867	56.12	0.00
HealthSocial	.878	53.93	0.00	HealthSocial	.878	53.93	0.00
c				c, HR_{start} included			
MET	.068	5.20	0.026	MET	.068	5.20	0.026
Time	.119	4.92	0.011	Time	.119	4.92	0.011
Interference	.209	6.12	0.001	Interference	.209	6.12	0.001

4 Discussion

The aim of the study was to identify individual predictors for short-term HR responses. All models including the proposed predictors the individual HR-responses show a high significance.

Especially, the number of the interval was a strong predictor that was found in all participants for HR_{start} and HF_{end}. This corresponds to results found in other studies, that HR is increasing during interval training [16]. Interestingly, negative mood effected the HR responses more often than positive mood. The predictor was found especially in HR_{end} except for P3. Additionally, physical health was found as strong predictor for HR_{end}, except for P1. This might be caused by the participant’s health status. The participant stated only to be either moderately or highly physically healthy, whereas in the other participants a greater heterogeneity was observed. Additionally, mental health was a predictor for HR_{end} in all participants except for P2. In general, the slope of the curve represented by value c revealed only weak predictors. The sport activity represented by MET and possible interference by previous sport activity explains as 19% of the variances of value c in participants 2 and 4.

Time of the day was a factor highly varying in the participants. It was revealed as predictor for HR_{start} and HR_{end} in participant 1 and for all values in P4. In contrast, neither P2 nor P3 showed an influence of Time.

HF_{start} was found to be a strong predictor in most participants except for P4. Especially in P3, HR_{start} was the only predictor for HR_{end} and the slope of the course. HR_{start} even explained 58% of the variability of HR_{end} in this participant. No specific gender-related predictors were found. All predictors are summarized in Table 9.

Table 9. Revealed predictors for all participants, HR_{start} was rated separately

	Participant 1			Participant 2			Participant 3			Participant 4			
	HR start	HR end	c	HR start	HR end	c	HR start	HR end	c	HR start	HR end	c	sum
MET		x				x	x				x		4
Interf		x				x		x					3
Sleep	x				x						x		3
Nutr	x			x	x			x					4
HPhys	x			x	x			x			x		5
HMent	x	x		x									3
HSoc		x						x			x		3
SBS+	x			x							x		3
SBS-	x	x			x		x				x		5
Time	x	x								x	x	x	5
Interv	x	x		x	x		x	x		x	x		8
HR _{start}		x	x		x			x	x				5

With the present data, the developed models show a reasonable approach for modeling individual short-term HR responses. Especially HR_{start} requires further attention. Regarding the literature, inducing the optimal training strain requires only the application of the correct training load. This study shows that there are more influencing factors ranging from physiological to mental that also needs to be considered.

In future research, the influence of additional predictors (e.g. weather, temperature or drinking behavior) need to be further investigated. Additionally, the assessment of a physiological profile might be useful to further analyze the resilience of the individuals.

5 Conclusion

In conclusion, possible predictors influencing the short-term HR responses are varying between individuals. Nevertheless, the number of interval, physical health, Time and negative mood are influencing these responses. The HR at the change of load bout is an important predictor even in the submaximal range. Future research needs to validate the results in a larger sample and integrate more parameters in the analysis.

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Sport Games Analysis and Management

Information Systems for Top-Level Football with Focus on Performance Analysis and Healthy Reference Patterns

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Abstract. In professional football clubs, information from many different sources has to be joined for informed decisions on team management. A central club management information system (CMIS) could help to support the employees at the club with information that they need for their daily decisions. This paper analyses the information demand and lists requirements for an optimal solution for the problem. With a beta version of a CMIS-software, field tests are conducted to improve the understanding of the needs and habits of employees at the club. This information will run into the development of the next version of this software. This iterative process aims at a final solution that meets the needs of football clubs.

Keywords: Soccer · Management information systems · Healthy reference pattern · Complex strength diagnostics · Software development

1 Introduction

In top-level football, clubs are made up of many individual sub-organisations with different sub-systems. There are many sources of information (such as club management, public relations, team management, medical, athletics... etc.), and every field has its own IT-systems to generate and store this information. Although there are many products promising solutions for these problems offered in the field (SAP Sports One, Easy2Coach...), practical experience shows that efficient use of computerised aids in this area is scarce. On the other hand, in the economic field of Business Intelligence (BI), there are powerful systems available that could be a model for football clubs. Subareas of BI are management information systems (MIS). This paper presents the idea of developing a MIS for professional football clubs allowing all employees in a club to access data relevant for them. Research cooperations with three clubs of the German Bundesliga have been started. Based on experiences with these clubs, a concept for such a system will be developed. The main target of this work will be the development of deeper analyses across different sub-systems of the clubs information bases. The above mentioned scarcity of usage despite the existence of systems promising the functionality of a CMIS gave rise to the idea to combine knowledge from sports science, sports

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practice and computer science. At a first stage, this paper deals with the potential and a solution of combining data from performance analysis and medical data.

2 Problems with CMIS in Football Clubs

Within the last years, many different systems for football clubs have been developed and offered. From the perspective of computer technology, developing a CMIS is no longer a problem. But still, practical experiences with - an admittedly low number of - three first league football clubs show that most of the times different data of several sub-systems are stored decentralized at the computers of each employee. Most of these sub-systems follow a strict proprietary logic and need a technical background to understand it. There are no interfaces to get the data of a subsystem into a central system automatically. As a consequence, drawing cross connection between data of different sources is hardly possible, which means, that existing information cannot unfold its full benefit.

Another big problem we detected is not technical, but more organizational. Each department at a football club is more or less working on its own. There is no efficient central administration of all systems and data of all departments. The same holds for a IT-management capable to implement something like a CMIS. Staff turnover leads to sudden changes in procedures and sometimes to a partial or complete shutdown of already existing systems.

Only in some instances, commercial CMIS are already implemented. But in general, they are still limited to special fields and cannot be used in the whole club. To make matters worse, individual suppliers sell their systems to the clubs, without knowing the specific needs and meeting the needs of the employees. They ignore the fact that practice needs to be a part of the development process. As a consequence, the club has then invested into an expensive system and is frequently prohibited by contract to work with other partners. This is of course counteracting against any progress and prevents IT from unfolding its benefits.

As a result of our experiences one may claim that the challenge of building a CMIS is still to find out how such a system should look like, which data should be included and how to convince and teach a club to implement and work with such a system.

3 Concept of a CMIS and Its Implementation in a Club

Figure 1 shows the idea of a CMIS. Different data of different departments and their sub-systems should be stored in a central system. Within this system, the staff members have access to their data or data of other departments that are relevant to their work.

Digital right management secures the unauthorized access to data. Cross-connections are possible and the barriers of data access are lowered. A uniform but user specific User Interface (UI) supports the users in conducting new and deeper analyses of all relevant data available in the club.

The concept of a CMIS should focus on different criteria (see Fig. 2). First, there should be an analyzing and consulting process. During this process, the different systems and data should be documented. In personal interviews, the needs of the employees should be found out and a general concept could be developed. Fundamental for the success of such a CMIS is the support and authority of the management. An authorized and responsible project manager with a mandate of the management should be chosen.

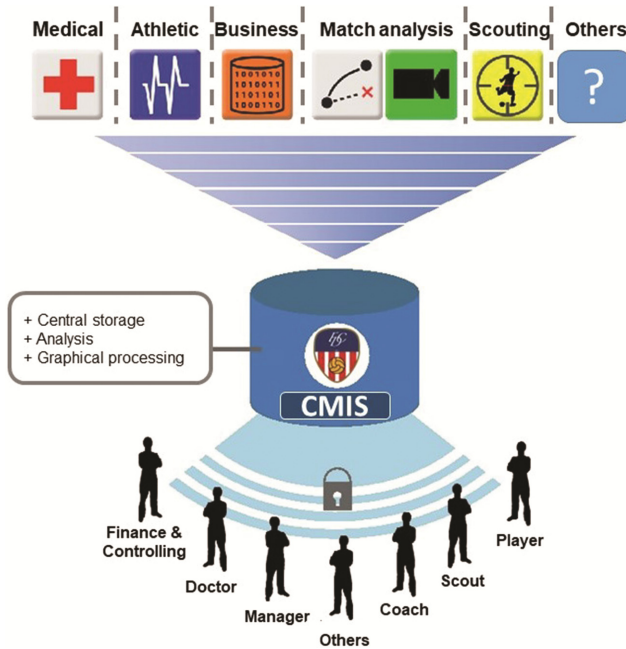


Fig. 1. The concept of a CMIS: A central system collects all data of sub-systems and provides it to the staff members.

This person must be well skilled in understanding the processes in football clubs, data collection, software development and project management to implement all the tools of software development into the very special field of football clubs. The development of a completely new system may not be recommended because each club is different and the requirements are uncertain in advance. As a consequence, a modular development

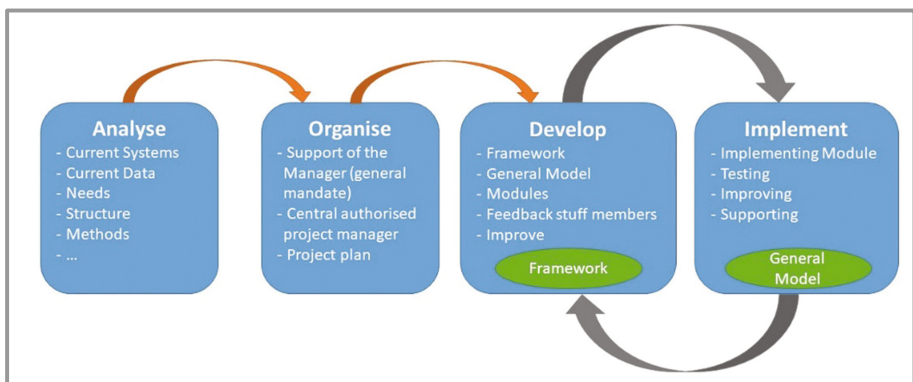


Fig. 2. Steps in developing and implementing a CMIS

is recommended. Based on a general template, modules for each department should be developed, that could be put together at the end.

Existing tools could be used as a starting point and adapted to the special needs of football clubs. First, with the help of these tools a demonstration model is developed, that is used to show the functionality of such a system to staff members and get their feedback. It should be as close as possible to the future system. This framework should be very flexible to change design and analysis very quickly and develop iteratively a system, that fits the needs of the employees and is very likely to be used by them in the future CMIS. Based on the refined demonstration model, the implementation of the module can be realized. This process is repeated for each module of each department.

4 Concept for Developing a CMIS

The complete development of an entire club management information system (CMIS) is an extensive task and not realized at the current stage of this project. What can be presented as a result of our work and experiences so far, is a concept of a specific software development model and the exemplary implementation of some sub-systems as prototypes. This corresponds to the concept of a modular development, that helps to develop step by step specific solutions considering the needs of the departments. After finishing each module, a usable solution could be integrated into the CMIS. For each individual module this general procedure should be followed to make sure to arrive at an integrated system.

4.1 General Model for Development

Figure 3 shows the general development process for individual applications. The first step is to select different tests and to choose the most appropriate measurements to obtain the desired information. After that, the system design starts containing two steps. The data

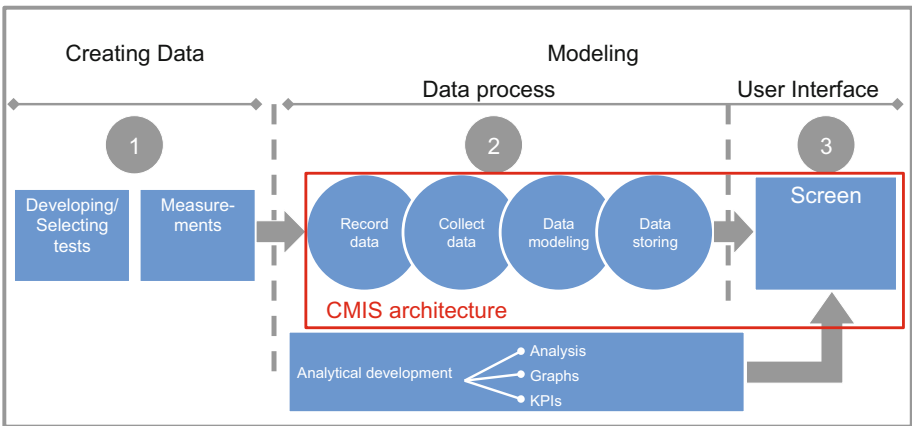


Fig. 3. Model of the general developing process.

process covers the chain from data recording to storing it in a database. At this point, it is important to adhere to the general architecture of the CMIS, because at the end, each individual application should fit into the whole system.

As shown in Fig. 4, two separate data storages are recommended. Media data typically require large storage, which might affect the performance of the system. Since media data are not needed in each use case working with two separated storages is expected to be beneficial for data handling and performance. These measures of internal organization are transparent for the user.

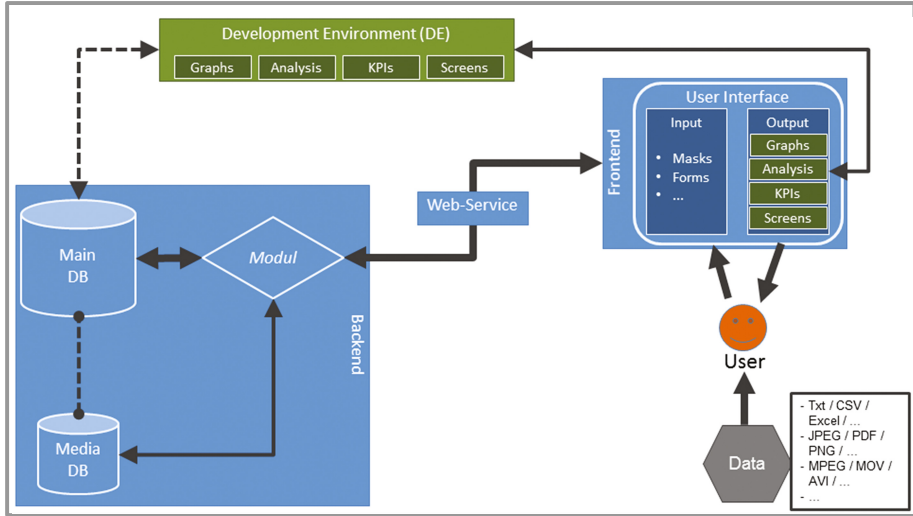


Fig. 4. The structure of the Software as Web Application.

Separately from data management the development of the analytical procedures is done (see Figs. 3 and 4). Based on the stored data, individual and user specific presentations, analyses, graphs and key performance indicators (KPIs) are designed typically with a close cooperation with the responsible employees. Feedback obtained by the employees after a test phase shows, whether these analyses are helpful, should be more sophisticated or skipped. The design of the user interface adheres to general principles of the CMIS as well. This is not contradictory to giving space for the user to create his individually designed screen meeting all information requirements desired.

4.2 Analysing the Information Demands of a Club

The first step of the development process must be the analysis of the current situation, structure and needs of a football club and its employees. Three different methods are used. Individual and personal contacts are important for that. With interviews, questionnaires and participant observations, the relevant information is acquired. With this information, different concepts and individual profiles are generated. One method is a “persona” (Cooper 2003). These are fictional profiles based on the job profile and work

flow of different future users to consider the needs and requirements of different users. Based on these persona so-called scenarios could be determined, to visualise the work of the users and understand their workflow better.

4.3 Software Selection

First, a software was chosen, that served as the basis for the development of a prototype (Perl et al. 2002). At the sector of Business Intelligence (BI) in economy, there are various software solutions with different specifications available. The following criteria were important for the software selection: Fast and flexible analyses (in memory technology), quick response, flexible user interface (UI) design, quick changes in design and analyses, online applications available and compatible with different data formats. One software that solves these requirements, was QlikView.

4.4 User Interface (UI)

The UI is the main part of an intuitive user guidance and important for the success of such a system. For that, it is important, to develop a design that is attractive to the user, generates interest and helps also to understand the analyses leading through the information (Few 2006). For the success of such a software tool in the practical environment, the design of the UI and its consistency is very important (Lames and Perl 1997). First, there was an overall design concept that is the basis for the whole software model. After that, buttons, boxes and different screens for each topic were developed.

At the UI, there are two different views for data presentation. First, the single player view: It depicts all relevant data of one player, depending on the department. Second, the team view: The whole team (or specific groups) may be analyzed and a single player could be compared against the average or against another player(s).

5 Focusing Healthy Reference Pattern (HRP) and Complex Strength Diagnostics (CSD)

According to the concept developed above, practical implementation started with two approaches to show how individual applications could be developed and integrated into a CMIS. The process of software development, described earlier, has been applied to each field to develop a model that will fit the needs of each department and could also be integrated into the whole CMIS. The selection was done because it meets the demands of the cooperating clubs and data were already existing or the club supported the collection of the relevant data. As mentioned in chapter 2, it is not sufficient to show the clubs an abstract concept containing just dummy data. Instead the chosen procedure made sure that the clubs were provided with data, they really needed. Performance diagnostics, injury prevention and rehabilitation are of great importance for clubs. Because of this, the two areas of Healthy Reference Patterns and Complex Strength Diagnostics have been chosen to start with the realization of two modules within the general concept of a CMIS.

5.1 Healthy Reference Pattern (HRP)

In general, there is a lack of long-term medical information about football players. This is not only due to a weak IT organization but must acknowledge that players and staff are typically employed on a short-term base in the clubs. This frequently leads to a situation where, after an injury, the exact health status of the player before the injury is unknown and being so, an essential information for medical treatment is missing. It is widely acknowledged that duration and quality of rehabilitation processes are of outstanding relevance in top level sports.

The issue is to develop a model for HRP that makes it possible to analyze the health status of each player at one specific point to determine the medical status. Different tests have been chosen to describe relevant aspects of the health status of players. This information could be used for injury prevention and additionally for rehabilitation after injuries have happened. The CMIS should provide a tool that can help to analyse the data and to provide access to crucial information for the relevant employees in their daily work. A similar approach has been followed by pre-injury screening (Chaitow et al. 2006, p. 329).

Interviews with the employees of the medical department of one German Bundesliga club had been done, leading to a test battery: 3D motion capturing for gait analysis (VICON) and electromyography (Myon), maximum strength of Quadriceps and Hamstring (IsoMed 2000), drop jump (Contemplas), 3D body scanning (VITUS 3D) and pressure sensor insoles (Moticon).

Sample and procedure 22 professional football players of a team of the German Bundesliga were tested at one day. For the gait analysis a walkway with a force plate, a Vicon 3D motion capturing system with 12 cameras ("Plug-in Gait" model for the lower body with 16 markers) and electromyography (4 sensors each leg) have been used. The test persons had to walk/run at two speed levels. For each side and each speed level, six trials were measured. At IsoMed maximum strength measurement, they had done isokinetic concentric knee extension and flexion of each leg in two speeds (60°/sec with 6 repetitions or 180°/sec with 20 repetitions). The range of motion of the knee was from 10° to 100°. For drop jumps (20 cm height, one-legged, three trials per leg) a force plate was used to measure the ground contact time. For the body scan, a Vitus 3D Body Scanner (300 dots per cm²) was used. With the pressure sensor soles, different tests had been done. At the soles, 16 sensors could measure pressure, the center of pressure and ground contact time on the pitch. For each player, a report was produced with all HRP information.

Example The software development is still in progress. Not all HRP data are already integrated. Figure 5 shows the status quo. All HRP data of one player (single player view) are presented within one screen. The user can select or deselect additional information. Coloured numbers show whether the player is above or below team average or whether values are critical. The depth of analysis is the choice of the user. Within the software only aggregated information on the tests will be presented. All details may be assessed by opening the report via link.

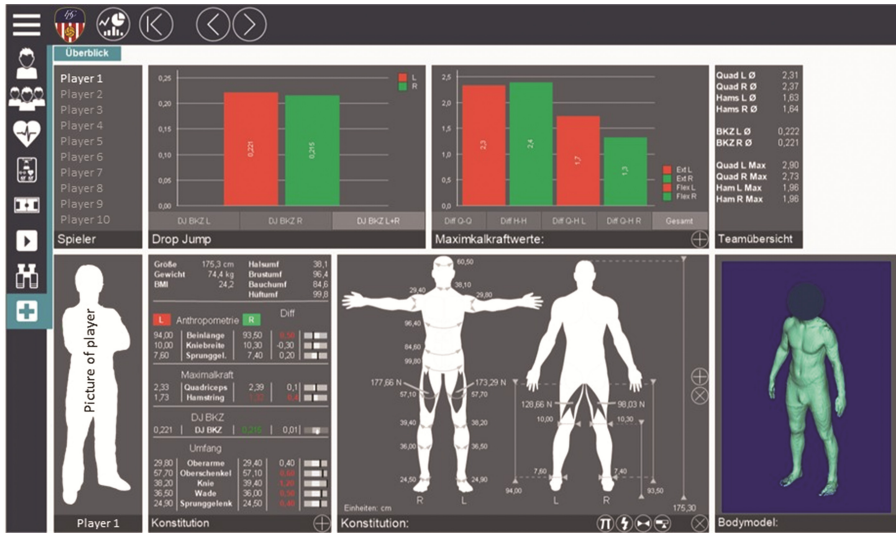


Fig. 5. Screenshot of the current status of the HRP UI with all additional information selected.

5.2 Complex Strength Diagnostics (CSD)

Within the HRP diagnostics, maximum strength tests are included to see the hamstring-quadriceps ratio (HQR). These tests are commonly used as injury risk prediction. But the maximum values of strength ratio appear to be poor predictors of non-contact injury (Pinto et al. 2017). So the first step of development was to use the data of the central database and visualize them in graphs, tables and charts for the needs of this specific field. After that, further analyses will be developed, to get better insights and use these data as predictors for injuries and add these to the application.

Data Sources In addition to the isokinetic strength measurement of the HRP conducted as cross-section with a professional squad, longitudinal data of youth academies were analysed (see Table 1). Over a period of five years (2011 to 2015) and five age classes (U14 to U19) were available. Whereas data for the youth academy only contain as

Table 1. Overview of the two data sets.

Set 1: Professional Squad (n = 22)				Set 2: Youth academy (n = 280)	
Year	2016			2011, 2012, 2013, 2014, 2015	
Age class	Adult			U14, U15, U16, U17, U19	
Extension	Left and right			Left and right	
Speed	60°/sec	180°/sec		60°/sec	240°/sec
Repetitions	6	20		N/A	N/A
Available data	Torque	Degree value	Sample rate	Torque	

summative information the maximum strength values (extension and flexion) left and right, the data set of the professional team contains the raw data of the isokinetic strength testing recorded in ProEMG.

Example For software modeling, the two data sets complement each other. Set 1 gives a lot more opportunities to analyze each trial deeper and set 2 allows to present longitudinal data. Based on this, different analyses have been developed.

Maximum strength diagnostics with hamstring-quadriceps ratio (HQR) are already standard. Figure 6 shows a first UI for these data. The presentation is still simple. All data can be selected dynamically, e.g. different years, periods, age classes or player positions. The colour coding (e.g. red presents the left quadriceps) goes through the entire software. Each selection affects every chart. Within the table, colored numbers show, which player is 10% above or below the team average. Even at this simple presentation can be seen, that there are great imbalances at the HQR. Only one value is below 10% (green) and another two are below 20% (orange) of the HQR.



Fig. 6. Simple UI for the maximum strength data with different charts.

The main way of presenting information within the UI of the CMIS is using graphs and KPIs. Figure 7 shows four different statuses of the representation of the transverse section of the thigh. Each colour is connected to a muscle group. KPIs and slider give a quick overview of the HQR of a player. These simple graphics are an example, how graphics should look like and how it could work at a CMIS.

The long-term analysis (Fig. 8) should help to monitor the development of different groups. Strongly aggregated information should help, to get a first impression. Afterward, the software provides more detailed information. It has to be found out which levels of aggregation is needed. At this point, the feedback from the users is indispensable.

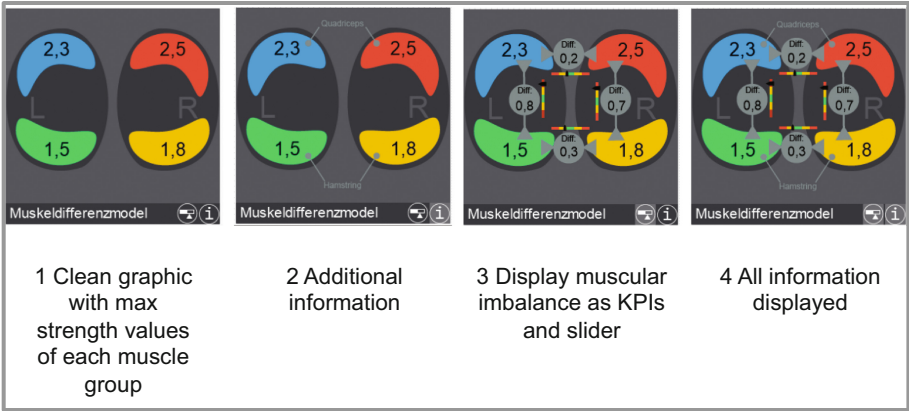


Fig. 7. Muscle differential model as part of the CSD.

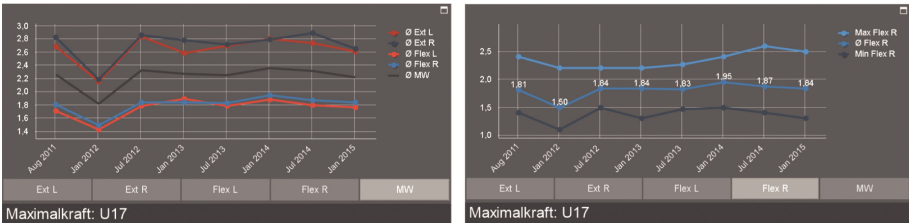


Fig. 8. Line charts of the maximum strength over several years with different aggregations.

The shown examples are just the beginning of analyzing maximum strength values. The next steps should develop deeper analysis to get better predictors of injuries. More data will be included and other views on this data should be done.

6 Outlook

The current system is a first step using management information systems within a football club, which is at present under-developed. Next steps after implementing the software will be that a comprehensive strategy of data collection together with the necessary organizational measures is realized. The ultimate criterion, though, will be the active use of the system and improvements in daily operations attributed to it.

Once established in this way, an increasing database accumulating historical data will allow for new types of analyses, e.g. data mining technologies for discovering even more systematic cross-connections between the data from different sources and machine learning technologies for describing typical developmental paths of athletes.

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Development of Real-Time Analysis System of Match Playing Time for Water Polo Player

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Abstract. According to the official rule of water polo, player substitution is allowed whenever the coach needs without restriction of times. Thus, information of total playing time for each player will be effective in real-time for the water polo coach as one of decisive information for player substitution. The purpose of present study is to develop a system which enables water polo coach to grasp the playing time of each player by using a tablet device during the match in real-time. A relational database software FileMaker Pro12 and mobile template FileMaker Go were used to develop playing time calculating system for water polo, and an iPad-mini was used as inputting apparatus. An operator with iPad-mini observe the match and input official playing time information and the cap number of player who participating on the water polo field at every substitution time. Total playing time for each player are showed on the same screen as inputting on iPad, which enables the coach to know objective information concerning the opportunity of player substitution.

Keywords: Water polo · Database application · Playing time · Tablet · Real-time analysis

1 Introduction

The official rule of water polo decided that the number of person playing in the field is seven, and the maximum number of substitution player is six, respectively [1]. The Player substitution is allowed at any time and without restriction of times. Generally, the water polo coach exchanges player at the beginning of each period, after each goals of both teams, end of time out of both teams, and in-play situation same as ice hockey match. Therefore, distribution of playing time and rest time which correspond to physical fitness of each player is very meaningful to maximize the total team performance of water polo match.

In the international major water polo tournament, such as Olympic, World Championships, and European Championships, total playing time of match for each player is presented after the end of the match officially. These water polo match data are handled and database was shared by Omega and Swiss Timing Ltd on their website since 2007 [2].

Total playing time information is beneficial for the coach to check their own team and opponent from physical fitness viewpoint. However, there is no information concerning playing time of each player during the match in real-time. Therefore, the purpose of this study is to develop a system which enables coach to grasp playing time of each player by using a tablet device during the match in real-time.

2 System and Apparatus

FileMaker Pro (FileMaker, Inc.), one of the popular relational database program, was used as a notational analysis tool for water polo match as for official match supporting or as tactical support for coaching [3, 4]. The advantage of FileMaker Pro is as follows; (1) high variability and original function system which enable users to customize a database for their object, (2) data compatibility which will be used on other application program by csv format, (3) cross-platform operating system such as Microsoft Windows, Apple Mac-OS. In addition, FileMaker Pro is directly linked with FileMaker Go (FileMaker, Inc.) mobile application template, which is operated by Apple iOS. As a result, the FileMaker Pro solution is useful on the iOS product such as iPad mini (Apple, Inc.), with the advantage of light weight (331 g) and long life battery (23.8Wh rechargeable lithium-polymer battery).

FileMaker Pro12 (FileMaker, Inc.) and FileMaker Go were used to develop a solution to analyze playing time of each player (Playing Time counter for Water Polo; PTWP) of one match in real time. One person, who has the knowledge concerning the rule of water polo, observes the water polo field from the point where he can watch the cap number of players, and operates PTWT through the match. The team bench will be ideal position for the operator, because he can communicate with coach concerning the playing time of player whenever he wants to know.

The size of tablet screen was 160 mm x 120 mm. The input screen is consisted of three categories of buttons (Fig. 1). The number of button for player category is thirteen for each team, for time category is sixty-eight, for setting category is eleven, respectively. Each button was colorized by their category and team to enhance sight perceptibility and visibility. The radio button is used for player category and time category to explicit which number was chosen. The size and distance for each button are appropriate to inputting by finger or stylus pen with ease.

With regards to preparation before the match and before each period, match number and period number are required to input, respectively.

With regards to operation during the match, operator input participating player on the field by using radio button which arranged on the screen for each substitution. Additionally, official match time information of substitution moment is needed to input, too. Generally, the coach of water polo tends to substitute their players at dead time, such as after score the goal, during time out, or after the exclusion of player. The operator has enough time to input all of participating player information during dead time without difficulty.

GAME5PERIOD2SERIAL2

JPNVS

KAZ

JPN'S MATCH

NEXT

BEFORE THE PERIOD

END OF THE PERIOD

TOTAL PLAYER7

TOTAL PLAYER7

OTHER MATCH

TABLE

SUM

DATA EXTRACT

JPN RESET

VS RESET

ALL RESET

MINUTES

SECONDS

110:43

210:00

32:43

42:43

52:43

68:00

78:00

810:43

98:00

108:00

112:43

122:43

1300:00

110:43

28:27

3OUT4:59

42:16

510:43

64:40

76:03

88:00

91:57

10IN5:44

118:46

122:43

1300:00

86420

7531

595857565554

535251504948

474645444342

414039383736

353433323130

292827262524

232221201918

171615141312

111009080706

050403020100

Fig. 1. Input screen and button layout of PTWT (5'17" of 2nd period, Japan vs Kazakhstan)

Total playing time information of each player of both teams is displayed on the screen at all times, which enables the coach to provide judging information for substituting the player in real-time.

As found in Fig. 1, Round radio buttons are used for input participating player and official playing time of substitution. Square buttons which four corner are rounded are operation button. Three or four digit numbers combined by colon are total playing time for each player. This screen means number two is substituted for number eight in Japan team, number ten is substituted for number three in Kazakhstan team, on 5'17" of 2nd period.

According to official rule of water polo, playing time clock is stop during dead time, such as between foul and free throw, between goal and restart, respectively [1]. The moving distance of player during dead time is a little to take account of playing time. Thus, playing time of each player is counted based on official playing time clock. Error message is utilized for miss operating which occurs frequently during tense atmosphere in the match.

After the end of match, all the information concerning total playing time of player is transmitted to laptop computer to summarize, viewing, printing, and storage.

3 Conclusion and Limitation

PTWT enables the coach of water polo to recognize information related to physical fitness information of each player based on playing time. Most of excellent water polo coach try to realize fitness condition for each player previous to the match, and estimate ideal playing time for them. Moreover, they try to grasp the fatigue state of each player

by using observation during the game to substitute them and give them rest. However, mental condition of coach is not only stable during the match but also in excitement or nerves, which leads them to some false detection. The playing time of each players which calculated by PTWT will show the objective information to the coach concerning the judgement of substitution during the match.

However, as one limitation of this system, one person (may be an assistant coach) need to concentrate for player substitution of both teams and operate tablet throughout the match time with keeping a cool head. The number of staff of the team is limited until three on the official regulation of water polo [1], and consequently PTWT will be beneficial in so far as request of head coach to specialize an assistant coach for operation of PTWT.

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Measurement

Reconstruction of 3D Ball/Shuttle Position by Two Image Points from a Single View

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Abstract. Monocular 3D reconstruction is an important problem. We solve the problem of reconstructing a 3D ball or shuttle position from a single-view television video. The contextual constraint is vital in this paper, which is implemented by a confirming point. The confirming point represents all the contextual cues, such as human pose, stroke technique, and shadow on the ground. The confirming point also tells us where the 3D point is. The confirming point decision is made by a human operator. Thus, the proposed method is a mixture of computer vision and human intelligence. Moreover, we propose a new air-ball friction model. This model provides a more accurate result because the aerodynamic drag force cannot be ignored in ball game.

Keywords: Computer vision · Monocular · 3D reconstruction · Badminton

1 Introduction

Monocular 3D reconstruction is an important problem and has received much interest both in the computer vision community and in the “computer science in sport” community. We consider the problem of estimating 3D ball or shuttle position from a single-view television video. Compared with multi-view methods, there are many advantages to use single-view (or monocular) method, including lower equipment cost, lower complexity, and higher portability of camera set-up. Moreover, most of the publicly available sport videos are single view, especially the data of competitors. Finally, the television sport video provides more realistic and more important information than the data from laboratory.

Monocular 3D reconstruction is a challenging area with many unsolved problems. In 2015, Shen et al. solved the model-drifting problem using a mixture of physical model and geometric model [8]. The hidden assumption of this solution is that the ball must hit the ground plane and the 3D position of ball bounce is known. However, this assumption is not valid in volleyball game or badminton game because the ball/shuttle is not allowed to bounce. We solve this problem using a mixture of computer vision and human intelligence.

Monocular 3D reconstruction is an ill-posed problem and is inherently ambiguous: a single RGB image on its own does not provide depth cue explicitly (i.e., given a single view image, there are infinite numbers of 3D scene structures explaining this image). However only one explanation is, in fact, correct. In this paper, we present a novel method for the reconstruction of 3D ball/shuttle position in sports television using a new constraint, namely, confirming point.

2 Method

2.1 The 3D Line Constraint

A camera is a mapping between the 3D world and a 2D image. Camera mapping is represented by a matrix P . According to the pinhole camera model, P is a 3×4 projective matrix defined by

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = P \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (1)$$

where $m_{11}, m_{12}, \dots, m_{34}$ are the elements of matrix P , (X, Y, Z) is the position in the 3D world and (u, v) is the position in the 2D image. Let $m_{34} = 1$ because the Eq. (1) uses homogeneous coordinates system.

Given N ($N \geq 6$) court-to-image point correspondences, the projective matrix P is computed by solving the following linear equations

$$\begin{bmatrix} X_1 & Y_1 & Z_1 & 1 & 0 & 0 & 0 & 0 & -u_1 X_1 & -u_1 Y_1 & -u_1 Z_1 \\ 0 & 0 & 0 & 0 & X_1 & Y_1 & Z_1 & 1 & -v_1 X_1 & -v_1 Y_1 & -v_1 Z_1 \\ & & & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ X_N & Y_N & Z_N & 1 & 0 & 0 & 0 & 0 & -u_N X_N & -u_N Y_N & -u_N Z_N \\ 0 & 0 & 0 & 0 & X_N & Y_N & Z_N & 1 & -v_N X_N & -v_N Y_N & -v_N Z_N \end{bmatrix} \begin{bmatrix} m_{11} \\ m_{12} \\ m_{13} \\ m_{14} \\ m_{21} \\ m_{22} \\ m_{23} \\ m_{24} \\ m_{31} \\ m_{32} \\ m_{33} \end{bmatrix} = \begin{bmatrix} u_1 \\ v_1 \\ \vdots \\ u_N \\ v_N \end{bmatrix} \quad (2)$$

Suppose P is known and a 2D image point (u, v) is given, a set of 3D points maps to this image point. This set constitutes a ray or line in 3D space passing through the camera center. Monocular 3D reconstruction is very difficult because one 2D image point back projects to one ray, which is defined by two planes:

$$\begin{aligned} (um_{31} - m_{11})X + (um_{32} - m_{12})Y + (um_{33} - m_{13})Z + (um_{34} - m_{14}) &= 0 \\ (vm_{31} - m_{21})X + (vm_{32} - m_{22})Y + (vm_{33} - m_{23})Z + (vm_{34} - m_{24}) &= 0 \end{aligned} \quad (3)$$

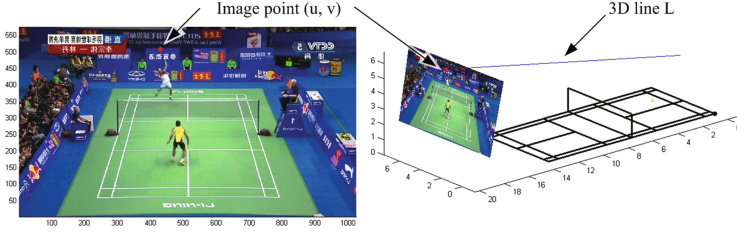


Fig. 1. Image point and its corresponding 3D line

As shown in Fig. 1, the image point (u, v) back project to a 3D line L . It should be emphasized that the aim of this paper is to find the 3D point (X, Y, Z) on this 3D line.

To find the 3D point, it will be necessary to add constraints on the solution, which at the beginning result from a physical model. A well-known physical constraint is gravity. Ohno et al. showed that the 3D flying ball trajectory can be estimated by fitting a physical model of 3D ball movements in the world to the observed 2D ball trajectory in the image [5]. Ribnick et al. proved the minimum condition for the existence of a unique solution [6]. The basic assumption behind these methods is that the ball motion is governed only by gravity. This assumption suffers from the model-drifting problem. Shen et al. added a bounce point constraint to fix this problem [8]. However, this constraint is NOT available in badminton or volleyball game because the shuttle or ball does not hit the ground plane until one player scores a point.

In this paper we will develop a new method that uses *confirming point* to reconstruct 3D shuttle position in a badminton game when the racket hits the shuttle.

2.2 Cylinder Constraint

As shown in Fig. 2, one image point back project to one ray. Suppose we have a new camera #2, then two matching image points back project to two rays (L and L' in Fig. 2). The 3D point (X, Y, Z) can be reconstructed by intersecting these two rays. This method is well-known as multi-view (or triangulation) method and camera #2 introduces a new constraint on the solution, which indicates the 3D point (X, Y, Z) on line L . Hawkeye system is a typical example.

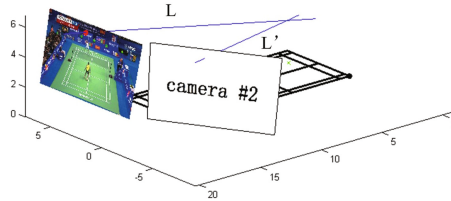


Fig. 2. Multi-view method

In monocular 3D reconstruction, however, camera #2 is not available. In computer vision literature, there are several other constraints, including camera motion [4], shading [9], structured lighting [7], motion blur [2], learned pattern [3], etc. However these constraints cannot be directly used in sport games. We found that contextual constraints are very useful in sport games. We give a simple example of player's positional cylinder constraint.

In a badminton game, a player hits a shuttlecock across the net using a racket. The strike-point cannot be far away from the player's body, which is modeled by a cylinder at the player's position $(4.29, 1.41, 0)$ in Fig. 3. Humans make rough inferences about the approximate 3D point using cylinder constraint. Empirically, the radius of the cylinder is 1.84 m, and the height of the cylinder is 3.2 m. In Fig. 3, the image point $(595, 68)$ back projects to a 3D line. After introduction of contextual constraint (player's position here), the 3D line L is reduced to a red colored line segment within the cylinder.

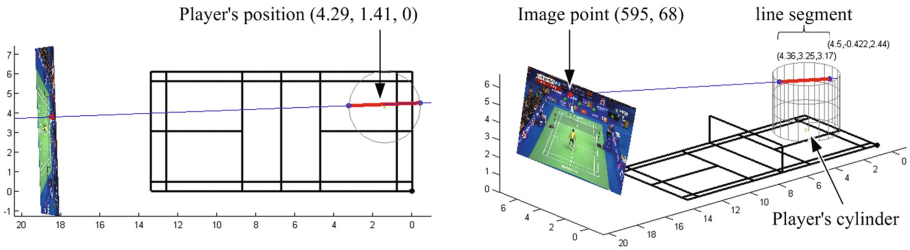


Fig. 3. Player's positional constraint

2.3 Confirming Point

The cylinder constraint has two problems: (1) it provides approximate range not accurate 3D position; (2) the player's position is ambiguous when he plays a vertical jump.

First, the more accurate 3D point can be inferred from 3D human pose when the player hits the shuttle. In other words, the contextual constraint is the human skeleton structure. Monocular 3D pose estimation has received much interest in computer vision community due to the wide range of applications. It is a difficult problem because there are large numbers of degrees of freedom (around 30) to be recovered. Especially, there are many difficulties in real sports data (e.g., motion blur, self-occlusions, and self-similarity). Therefore, fully automatic 3D pose estimation is currently not possible, and manual labeling is required [1].

The complete 3D skeleton requires a human operator to manually label all joint locations in image. These labeling is a heavy work in badminton game because the average length of a rally is more than 10 shots. Moreover, a full-body 3D human pose is not necessary if we want only the 3D position of the shuttle in sports television. In this paper, we use a confirming point rather than all joint points. This confirming point can be easily labeled as follows (Fig. 4).

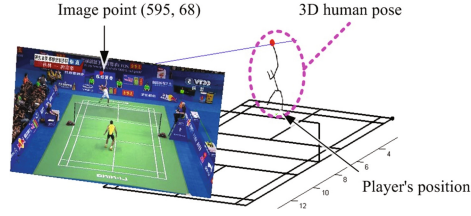


Fig. 4. The 3D point can be inferred by 3D human pose

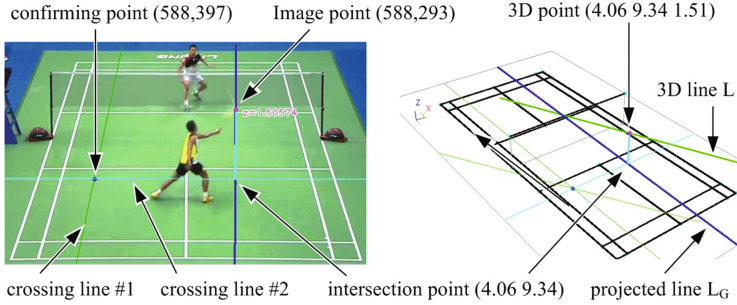


Fig. 5. The 3D point is inferred from image point and confirming point

In Fig. 5, for example, the player hits the shuttle on his backhand side in the midcourt. The operator clicks the shuttle position (588,293) in the image, and the corresponding 3D line L is computed according to (3). The operator's mouse moves on the computer screen, and the confirming point is shown both in the image (left) and in the 3D visualization window (right). The confirming point (588,397) is decided by the operator according to the contextual cue: human pose, stroke technique, shadow on the ground, and shuttle trajectory. The confirming point has two assistant crossing lines, and each line parallels to the X or Y axis. Crossing line #2 intersects with the projected line L_G of L on the ground plane. The intersection point (4.06 9.34) indicates that the 3D point is (4.06 9.34 1.51). In short, two 2D image points provide one 3D point.

Please note that the confirming point decision is made by human operator, because contextual cues are too complicated to be captured by mathematical model using current computer vision technology. For example, the player's position is ambiguous when he plays a vertical jump smash. The fact is that the shadow of the player indicates that he is jumping (left in Fig. 6), but the image itself gives the incorrect position (jump height is 0 m). However, the temporal cues show that the correct position of player is not on the ground, and the vertical jump height is 0.42 m (right in Fig. 6). Therefore, a human operator can easily tell the correct confirming point, but computer can not.

The confirming point plays a vital role in our monocular 3D reconstruction system. It represents all of the contextual information perceived by humans,

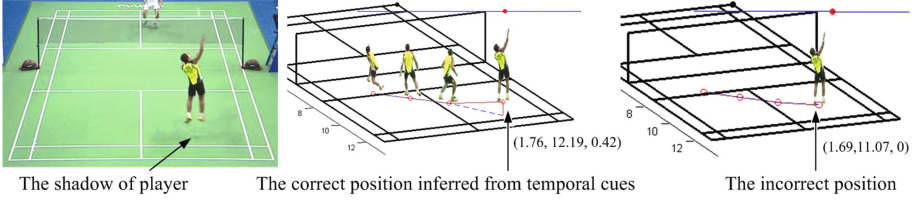


Fig. 6. Example of a vertical jump

and helps us to tell the 3D position from a single view. Hence, our method is a mixture of computer vision and human intelligence.

2.4 Air-Ball Friction Model

Given 3D position of ball/shuttle, complete 3D trajectory can be estimated using Shen’s method [8]. But a main shortcoming of [6, 8] is that it doesn’t model the air-ball friction. In this paper, we propose a novel air-ball friction model.

$$\begin{aligned} X(t) &= X(0) + tV_X - \frac{1}{2}\alpha V_X t^2 \\ Y(t) &= Y(0) + tV_Y - \frac{1}{2}\alpha V_Y t^2 \\ Z(t) &= Z(0) + tV_Z - \frac{1}{2}(g - \alpha V_Z)t^2 \end{aligned} \quad (4)$$

where $(X(t), Y(t), Z(t))$ denote the position of the ball at time t , and (V_X, V_Y, V_Z) denote the velocity of the ball at time 0, g denotes the acceleration of gravity ($g = 9.8 \text{ m/s}^2$) and α is *air-ball friction coefficient*. The aerodynamic drag force is a function of ball velocity (V_X, V_Y, V_Z) and air-ball friction coefficient (α). This 3D ball trajectory model is very simple and effective.

First, we can directly estimate α from a monocular view data.

$$\theta^* = \arg \min_{\theta} (F) \quad (5)$$

where $\theta = (X(0), Y(0), Z(0), V_X, V_Y, V_Z, \alpha)$ and F is a cost function [8]. But this α estimation method is troublesome. Evidence shows that the monocular view data itself cannot provide a correct result.

Moreover, we collected a Hawkeye ground truth dataset GT and $GT(i)$ is i^{th} smash speed ($i = 1, 2, \dots, G$) in badminton game from television. We estimate α by minimization:

$$\alpha^* = \arg \min_{\alpha} \sum_{i=1}^G (\sqrt{V_X^2 + V_Y^2 + V_Z^2} - GT(i))^2 \quad (6)$$

We found that the best air-ball friction coefficient α^* is 3.135 and the best mixture weight w is 780 in [8]. Given these two parameters, our α fixed method provides a reliable result. Please refer to Sect. 3 for more details.

3 Experiment

We implement the proposed method using C++ and run it on a 3.3GHz PC with 2G RAM. The videos are recorded from the Badminton - Singles Men of the Rio 2016 Olympic Games. Using the monocular 3D reconstruction method in [8] and two image points method in this paper, we obtain 1363 shuttle trajectories. Some final strikes are visualized in Fig. 7.

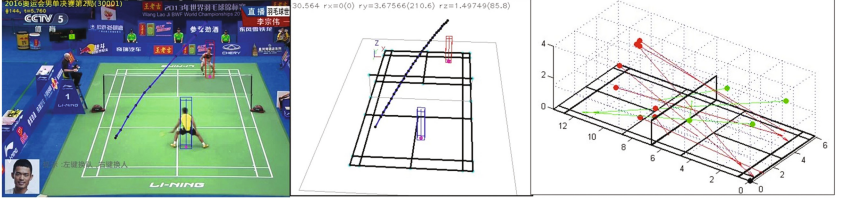


Fig. 7. Our monocular 3D reconstruction system and the final strikes

Table 1. The Comparison of four methods.

Method	Mean-Square error (m/s)	Back-project error (pixels)
Ribnick 2009 [6]	2.27e+018	23.212
Shen 2015 [8]	5989.14	209.828
Our α estimation method	5989.08	206.85
Our α fixed method	1559.37	67.0942

Second, we collected a Hawkeye ground truth dataset from the original badminton games in television broadcast, and compared four methods: Ribnick 2009 [6], Shen 2015 [8], our α estimation method and our α fixed method ($\alpha = 3.135$). As shown in Table 1, Ribnick 2009 is the best according to the back-project error. But it is the worst according to the MSE because it suffers from the model drifting problem [8]. Our α fixed method is the best. The experiment result shows that the aerodynamic drag force (α) influences the monocular 3D reconstruction significantly and cannot be ignored.

4 Conclusion

The proposed method uses two constraints to solve the monocular 3D reconstruction problem. The first constraint is the ball/shuttle image point recovering a 3D line from camera to ball. The second constraint is the confirming point telling us where the 3D point is on this line. The confirming point represents all the contextual cues, such as human pose, stroke technique, and shadow on the ground.

The confirming point decision is made by a human operator. Therefore, the proposed method is a mixture of computer vision and human intelligence.

The proposed method models air-ball friction to improve monocular 3D reconstruction. The aerodynamic drag force influences monocular 3D reconstruction significantly and cannot be ignored. The air-ball friction coefficient α is 3.135 in badminton game.

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A Comparison of Smoothing and Filtering Approaches Using Simulated Kinematic Data of Human Movements

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Abstract. Gathered kinematic data usually requires post-processing in order to handle noise. There are three different approaches frequently used: local regression & moving average algorithms, and Butterworth filters. In order to examine the most appropriate post-processing approach and its optimal settings to human upper limb movements, we examined how far the approaches were able to reproduce a simulated movement signal with overlaid noise. We overlaid a simulated movement signal (movement amplitude 80 cm) with normal distributed noise (standard deviation of 0.5 cm). The resulting signal was post-processed with local regression and moving average algorithms as well as Butterworth filters with different settings (spans/orders). The deviation from the original simulated signal in four kinematic parameters (path length, maximum velocity, relative activity, and spectral arc length) was calculated and checked for a minimum. The unprocessed noisy signal showed absolute mean deviations of $54.78\% \pm 12.16\%$ in the four kinematic parameters. The local regression algorithm revealed the best performance at a span of 420 ms with an absolute mean deviation of $2.00\% \pm 0.86\%$. For spans between 280–690 ms the local regression algorithm still revealed deviations below 5%. Based on our results we suggest a local regression algorithm with a span of 420 ms for smoothing noisy kinematic data in upper limb performance, e.g., activities of daily living. This suggestion applies to kinematic data of human movements.

Keywords: Smoothing · Filtering · Kinematics · Simulation · Human movement

1 Introduction

The use of kinematic data of human limbs or centers of pressure in science is broad and covers a wide range of research fields such as neurorehabilitation [1–3], orthopedics/prosthetics [4–6], human engineering [7–9], and (peak) performance in sports [10–12]. The validity of the analyses does not only rely on the appropriate set of metrics but also on the quality of the gathered signal. For many applications the velocity profile of a moving body part, which can belong to a player in sport or be the limb of a stroke patient, is derived from positional or acceleration data. These can be obtained using tracking technology (video-, magneto- or GPS-based) or accelerometers. Depending on the technology that is used, the signal can be prone to systematic and stochastic noise. Such noise is usually post-processed by smoothing and/or filtering the signal. Commonly used

tools are local regression algorithms [1, 13], moving averages [14, 15], Butterworth filters [16–18], or non-parametric regression approaches [19]. Using such tools, one has to maintain a balance between reduction of noise and loss of original information.

In this study, we created a velocity profile similar to one resulting from hand movements in everyday actions [13], incorporating two movements, one small smooth movement and one larger unsmooth movement (Fig. 1). The computation of the velocity profile was based on the work of Flash and Henis [19]. Similar approaches to simulate velocity profiles have been used by Balasubramanian [20], Hogan [21], or Rohrer [22]. After the addition of normal distributed noise, different smoothing approaches (two algorithms and one filter) were applied and the gathered kinematic parameters were compared with the known values of the original data. Based on this analysis, we propose an optimal configuration for the post-processing of noise-contaminated signals in the kinematic analysis of human upper-limb movements with more than one velocity peak.

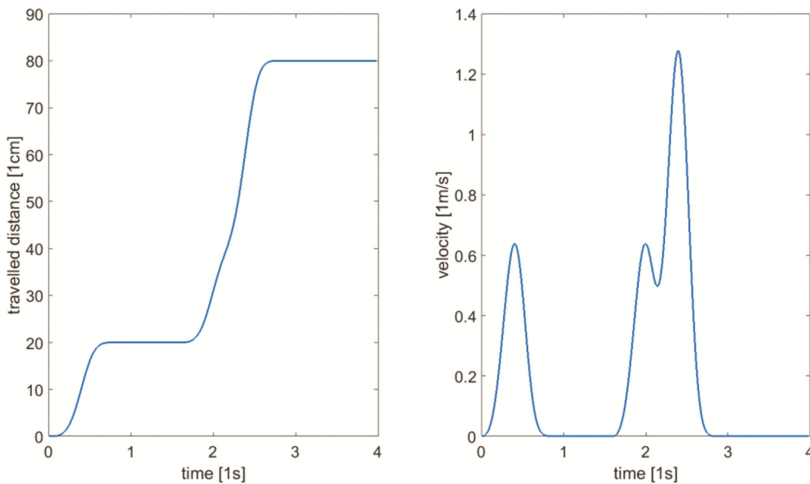


Fig. 1. The original signal and its velocity profile (100 Hz), which was used for the analysis of different smoothing approaches and their impact on kinematic parameters.

2 Methods

Using MatLab (MATLAB R2015a, Mathworks, MA), we created a velocity profile similar to hand movements in everyday actions [13] (100 Hz, Fig. 1). The movements covered 80 cm in path lengths and reached a maximum velocity of 1.28 m/s. The velocity profile consisted of one small smooth and one larger unsmooth movement. Unsmooth means that the movement contains sub-movements which are overlaid with latency, resulting in multiple velocity peaks (Fig. 1, right). The computation of the overlaid sub-movements was based on the work of Flash and Henis [19]. In a next step, the signal (positional data) was corrupted by the addition of normal distributed noise with a standard deviation of 0.5 cm (Fig. 2).

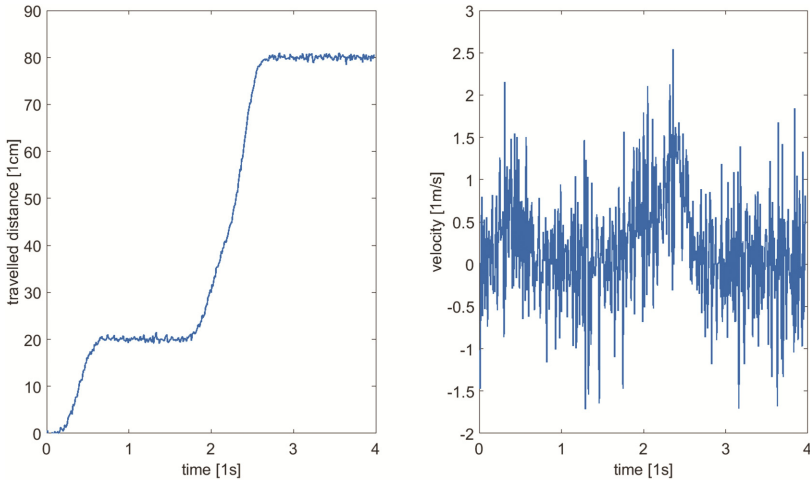


Fig. 2. Example of a corrupted signal and its velocity profile. The overlaid noise was normal distributed with a standard deviation of 0.5 cm.

For the comparison of the original signal with the post-processed noise-contaminated signals the following kinematic parameters were gathered:

- **Path length:** The distance covered by the movements.
- **Maximum velocity:** The fastest velocity peak.
- **Relative activity [13]:** The quotient of movements faster than 0.05 m/s and the signal length.
- **Spectral arc length [20]:** The smoothness of the movement, derived from the arc length of the signal's Fourier magnitude.

Their values were set in relation to the values of the original signal by the following formula:

$$relative\ deviation = \left| \frac{Value(i)}{Value(0)} - 1 \right|$$

Further, the mean of the four used parameters was calculated in order to have an estimate of the average deviation of information from the true values (original signal). The applied smoothing approaches were:

- a **local regression** algorithm ('loess')
- a **moving average** ('moving')
- a forward-reverse low-pass **Butterworth filter** ('butterworth') with a cut-off frequency of 10 Hz.

The cut-off frequency of 'butterworth' was set to 10 Hz in order to not exclude any potential human movement [23, 24], cut-off frequencies are usually chosen between 5 and 20 Hz [20, 25]. Over 200 iterations, the different approaches were applied to the corrupted signal with changes of the window size (span); for 'loess' and 'moving' of

10 ms per iteration (10 ms–2000 ms), and changes of the order of ‘butterworth’ with steps of 1 per every 10th iteration (1–20).

Since trial durations in kinematic analyses can vary between less than a second and several minutes (e.g. in neurorehabilitation [1, 24]), the spans of the applied algorithms can be at a large proportion of the signal. Since, in this exploration, the signal was rather short with a length of 4 s, the behavior of the algorithms and the filter at the signal’s boundaries are taken into account.

This procedure was repeated 500 times, in order to have a more stable estimate of the three approaches.

Note: The term ‘iteration’ is used according to the definition of informatics as a repeated, non-approximate block of commands. The terms ‘smooth’ and ‘smoothness’ are used according to kinematic analyses, e.g. in neurorehabilitation, and not the mathematical definitions.

3 Results

Over the course of iterations, all approaches revealed an expected outcome:

- Close to the noise-contaminated signal (small window sizes (spans)/low orders), the mean relative deviation lies between absolute values of 25% and 50% from the original values.

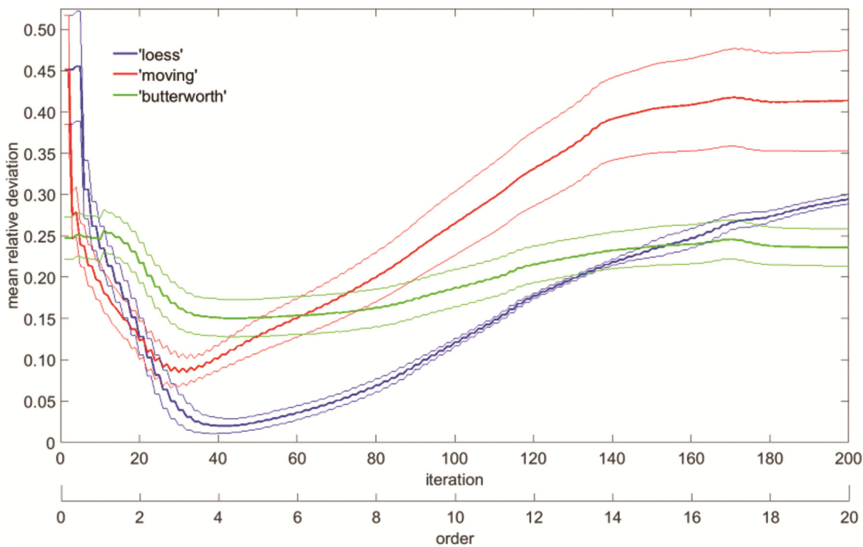


Fig. 3. Display of the mean relative deviations from the original signal’s values of ‘loess’ (blue), ‘moving’ (red), and ‘butterworth’ (green) over the course of 200 iterations and 20 orders, respectively. The standard deviations in positive and negative direction are displayed with thin lines. The iteration equals the window size (span) in 10 ms for ‘loess’ and ‘moving’ and order the order of ‘butterworth’.

- With growing window sizes (spans)/orders, valleys appear, where the approaches were able to reduce the noise and the mean relative deviations are decreasing.
- Towards larger window sizes (spans)/higher orders, the signal becomes more and more damped by the approaches, so that the deviations are growing similar to root functions.

Figure 3 illustrates the mean relative deviation of ‘loess’, ‘moving’, and ‘butterworth’ over the course of window sizes (spans)/orders.

A minimum value of mean relative deviation was reached by ‘loess’ at a window size (span) of 420 ms, with a mean deviation of $2.00\% \pm 0.86\%$ from the original values. The unprocessed corrupted data, in comparison, showed a deviation of $54.78\% \pm 12.16\%$ from the original values. The four kinematic parameters had deviations of 0.91% (path length), 1.91% (maximum velocity), 3.19% (relative activity), and 1.99% (spectral arc length) from the values of the original signal. For ‘loess’ window sizes (spans) between 280 ms and 690 ms, the mean deviation was still below 5% (Fig. 3).

4 Discussion

Resulting from our calculations, the local regression algorithm was able to reduce the noise and retain the original signal in the best way, with an information deviation of $2.00\% \pm 0.86\%$ at a window size (span) of 420 ms. Our proposal for an optimal smoothing configuration is therefore a 420 ms local regression algorithm. An acceptable mean deviation below 5% was reached in the range of 280 ms–690 ms. Of course, this has to be taken with caution, since it is derived from one purely computational signal with fixed noise characteristics, although the noise is strong with a standard deviation of 0.5 cm in a movement covering 80 cm. This can be taken as a first suggestion towards an optimal way of post-processing noisy signals, like in our case positional data of human upper-limbs. Future investigation should cover a broader range of signals to estimate if the suggested smoothing configuration is reliably best suited.

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How to Accurately Determine the Position on a Known Course in Road Cycling

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Abstract. With modern cycling computers it is possible to provide cyclists with complex feedback during rides. If the feedback is course-dependent, it is necessary to know the riders current position on the course. Different approaches to estimate the position on the course from common GPS and speed sensors were compared: the direct distance measure derived from the number of rotations of the wheel, GPS coordinates projected onto the course trajectory, and a Kalman filter incorporating speed as well as GPS measurements. To quantify the accuracy of the different methods, an experiment was conducted on a race track where a fixed point on the course was tagged during the ride. The Kalman filter approach was able to overcome certain shortcomings of the other two approaches and achieved a mean error of -0.13 m and a root mean square error of 0.97 m.

Keywords: Cycling · GPS · Kalman filter · Road cycling · Feedback

1 Introduction

With the increasing popularity of advanced cycling computers, new possibilities for feedback become available for the majority of road cyclists. This allows to guide cyclists during training and competition, for example to provide the rider in real-time with a strategy on how to ride. Garmin Ltd., one of the leading manufacturers of cycling computers, introduced a rudimentary real-time feedback feature with their ‘Virtual Rider’. It shows information about how far off the rider currently is, compared to a previous ride in terms of time and distance. A critical point in providing this kind of track specific feedback is to know the precise location of the cyclist on the track. Moreover, Garmin Ltd. recently provided an API (‘Connect IQ’) allowing to develop custom apps for their cycling computers that may provide real-time position dependent feedback for athletes.

Using commercially available sensors and devices is subject to various problems. Distances derived by an odometer (as included in commercially available

speed sensors) can lead to errors if the rider chooses a slightly different route than in the reference recording, e.g., by cutting curves. An additional source of error when using an odometer is, that the exact circumference of the wheel must be known. A small error in this parameter leads to a continuously increasing error in the calculated distance during the ride. Another option that can provide information about the position of the rider is the Global Positioning System (GPS), or similar but less common systems like GLONASS and Galileo. The accuracy of the GPS position strongly relies on satellite coverage and landscape settings and can be expected to range from 5 m to 10 m in a stationary setting [1, 2]. Studies indicate, that this range is also feasible for dynamic applications like cycling: In an experiment where cyclists were riding on cycling lanes in a city, 55% of the recorded data points were located in a range of less than 2.5 m outside of the actual lane and 86% in a range of less than 10 m outside of the actual lane [3]. From the experience of recreational road cyclists, the best self-reported accuracies of satellite based positioning on commercially available cycling computers are about 3 m.

In professional road cycling, the difference between the top athletes' performances has become very small and therefore the term 'aggregation of marginal gains' has become popular. It describes the concept that small performance improvements in different areas will add up and make a significant difference. One of these areas is the right choice of strategy during races. For individual time trials optimal strategies can be derived mathematically, based on modeling the cycling dynamics and the riders fatigue. First experiments in the laboratory indicate potential of this method to improve performance in real races [4]. In order to transfer the laboratory results to the field it is crucial to precisely determine the current position on the course, to provide the proper feedback. In this paper we experimentally investigated the accuracy of different methods to determine the riders current position on a course in order to get a clear view about what accuracy can be expected.

2 Methods

The position on the course is defined as the traveled distance during a reference ride along the course trajectory. Such a trajectory can be defined by georeferenced road data, or by one's own measurement. In this paper, we define the reference course trajectory by measuring the lateral and longitudinal GPS coordinates during a ride. These measurements are approximated by a cubic smoothing spline with the MATLAB function 'csaps' [5]. The arc length of the spline ideally corresponds to the recorded distance provided by the odometer. This approach defines a mapping between a two-dimensional GPS coordinate and a one-dimensional distance along the course. Therefore, the feedback can be defined corresponding to that distance, e.g., at an arbitrary position given by a distance of s [m], the athlete should ride applying a computed pedal power of $P(s)$ [W]. Thereby, the methods used for the mapping are not crucial. For this study, we chose the described approach since the spline approximation both

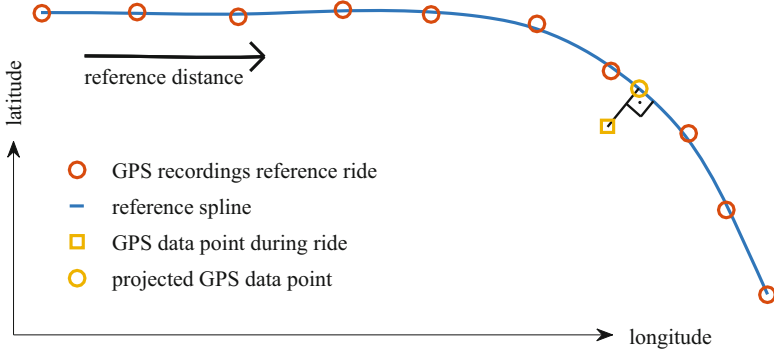


Fig. 1. Schematic representation of the reference trajectory. In a first step, the reference trajectory is calculated based on measured GPS coordinates and corresponding distance values. The red dots represent the recorded latitude and longitude GPS coordinates. Through those points a smoothing spline is placed, the arc length is provided by the recorded distance (blue). With this reference trajectory, in future measurements, GPS coordinates are projected onto the reference spline to get the arc length of the projected point on the spline (yellow), which corresponds to the reference distance of the recorded GPS point.

smooths the recorded data and interpolates between two recorded data points in a suitable manner.

We consider three methods of deriving the reference distance during a subsequent ride:

1. Associate the reference distance with the distance provided by the odometer of the external speed sensor.
2. Projection of GPS coordinates onto the reference trajectory. The arc length of the reference trajectory at the projected point provides the reference distance. See Fig. 1.
3. Kalman filter [6] based on the discrete dynamic system

$$\begin{aligned}x_{t+1} &= x_t + \Delta t v_t \\v_{t+1} &= v_t\end{aligned}$$

with speed v , reference distance x and sampling interval $\Delta t = 1$ s. The measurements used in the Kalman filter are the speed values provided by the external speed sensor and the reference distance as derived in method 2. The covariance matrix of the observation noise R was estimated by previous experiments and resulted in the following values:

$$R = \begin{bmatrix} 1.042 \text{ m} & 0 \\ 0 & 0.00176 \text{ m s}^{-1} \end{bmatrix}$$

Method 1 is based solely on measurements of the speed sensor, while method 2 incorporates only GPS measurements. To increase the accuracy and compensate



Fig. 2. Experimental setup: The left picture shows a screenshot of the application. The number in the top shows the current speed and the gray box is the button that was pressed each time the marker was passed. The picture in the middle shows the physical realization of the marker: A rounded bar, taped on the 50 m line of the track. The Garmin hub speed sensor is presented in the right picture, mounted on the hub of the front wheel of the test bike.

shortcomings of these two methods, the data of both sensors is combined in method 3 using a Kalman filter.

All data was collected with an Android application on a Sony Xperia Z1 Compact smartphone. The application was generated with the Pegasos [7] framework. GPS was recorded by the phone's internal GPS sensor while speed and distance were measured by an external Garmin hub speed sensor (Fig. 2, right). Additionally, a button was integrated to the Pegasos front end to allow the rider to tag specific positions on the road during the ride, which are used for the evaluation of the method. Figure 2 (left) shows a screenshot of the application. The smartphone was mounted on the handlebar in reach of the right hand, to allow to press the button with the thumb while still being able to hold the handlebar firmly. Speed data was recorded with a sampling frequency of 3.79 ± 0.71 Hz and GPS with a sampling frequency of 1.00 ± 0.02 Hz. For further use, all data was resampled uniformly at 1 Hz by applying a low pass filter in form of a Gaussian filter with $\sigma = 1$ s.

Two rides were conducted on the 4th lane of a 400 m oval running track. Both rides were performed on the same bike, by the same rider who was instructed to stay in the middle of the lane. Each ride started at the beginning of the 100 m track and a marker was placed at the 50 m line. At the marker position the rider pressed the button on the smart phone to tag the position. To improve the accuracy of the button press, the marker was a 2 cm high, rounded plastic bar as shown in Fig. 2 (center). With this construction, the rider did not only have a visual feedback of the position of the marker but also a haptic one. In the first ride, the reference trajectory was recorded. One lap on the course was performed, the first pass of the marker tagged the beginning of the lap and the

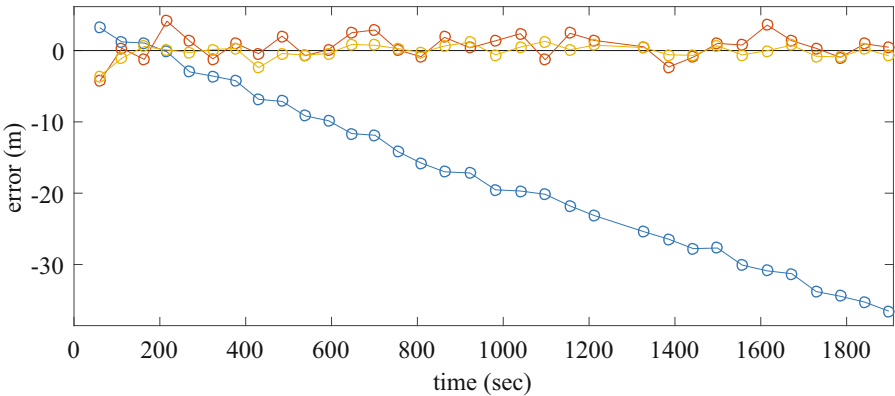


Fig. 3. Position error at the marker points relative to the marker tags in the reference ride. The blue curve shows the recorded distance of the speed sensor, the red curve the distance determined by projecting the GPS coordinate onto the reference trajectory and the yellow curve shows the result of the Kalman filtering.

Table 1. Mean error and accuracy (root mean square error) of the position derived by the different methods at the marker positions in meter.

Method	Mean error	Accuracy
Speed sensor	−17.27	20.93
GPS projection	0.58	1.80
Kalman filter	−0.13	0.97

second passing tagged the end of the lap. This lap was replicated to form a virtual reference ride of 35 laps. In the second ride, all 35 laps were completed in a row on the track. The button was not pressed correctly in lap 1 and 24, hence the number of valid marker tags totals to 33.

3 Results

At each marker position, the positioning error was determined as the difference of the distance in the reference ride and the corresponding distance derived by the different methods. Figure3 shows these errors of the different distance estimation methods at the marker positions. In the distance returned by the speed sensor, a clear drift over time can be detected while the errors of the methods considering GPS measurements stay in the same range throughout the whole test. Table 1 shows the resulting mean error and the root mean square error of the derived position at the markers. The root mean square error is provided as a measure of accuracy of the different methods. Projecting the GPS coordinates onto the reference trajectory led to a mean error of 0.58 m and a root

mean square error of 1.80 m. The extension using the Kalman filter improved these results with a mean error of -0.13 m and a root mean square error of 0.97 m. Additionally, the GPS sensor provided information about the accuracy of the measured horizontal coordinates for each measured point. The mean self-proclaimed accuracy at marker positions was 3.07 m.

4 Discussion

In order to determine a precise, absolute position on a course, using only the odometer is obviously not applicable, because the error increases over time. Three potential problems occur if using the odometer only: (1) A consistently different choice of line in the test ride compared to the reference ride introduces a drift as can be seen in Fig. 3. Since we chose a course with a high portion of uniform curvature this phenomenon most likely was amplified. (2) The total distance is calculated over the number of rotations of the wheel. So, if the wheel diameter is estimated wrongly, an error sums up with each rotation. (3) The distance at the starting point has to be estimated correctly, since the error that is done there is passed on to all consecutive measurements. Altogether, the distance provided directly by the odometer is not suitable to precisely estimate the riders position on a given course.

Due to the shortcomings described in the previous paragraph, GPS measurements have to be taken into account. They provide a global measure, independent of previous measurements, which avoids that the error constantly increases during the ride. For the pure GPS measurements, a self-proclaimed accuracy of about 3.07 m in the two-dimensional horizontal plane was provided by the sensor. Since the proposed application is to estimate the position on a given course, the two-dimensional GPS coordinates are projected into one dimension: the distance along the course. This projection already improved the accuracy of the determined position. One serious issue with using GPS only is that under certain circumstances the GPS signal can vanish, for example in a dense forest or in a tunnel. In this case, no or only very poor information about the current position is available. Therefore, a Kalman filter was employed, incorporating GPS and speed measurements. It can handle the loss of sensor information by using model predictions and by including both, speed and GPS sensors. The accuracy of the determined position is further improved by this method. The achieved accuracy of 0.97 m is about twice as good as with the method projecting the GPS coordinates onto the spline and about three times better than the self-proclaimed accuracy of the original GPS measurements.

5 Conclusion

To provide cyclists with feedback during training sessions or races, the precise position on the course may have to be known. The naive approach, taking the distance provided by the bike computer is unstable and cannot be recommended. A Kalman filter incorporating the GPS position along the course as well as speed

measurements provides stable and accurate distance values. The achieved accuracy of 0.97 m should be sufficiently good for most practical applications and is better than using only GPS coordinates. To validate the accuracy of the experimental approach, it is planned to include measurements of the accelerometer of the smart phone. With this additional data, it should be possible to detect the marker position more precisely than with the virtual button operated by the rider. Future work will apply the proposed Kalman filter method to provide real-time feedback to finish a course in the shortest possible time. A strategy for the whole course is precalculated and during the ride, the proposed speed for the current position is presented to the rider.

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Virtual Reality in Sports

Missing Depth Cues in Virtual Reality Decrease Performance of Three-Dimensional Reaching Movements

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Abstract. Goal-directed reaching movements in three-dimensions are important for our interaction with the environment. Instrumented setups displaying virtual targets for rehabilitation training of reaching movements often provide limited depth cues, which may affect movements. This work aims to quantify effects of limited depth cues on reaching movements. We developed a virtual environment for assessing three-dimensional reaching movements that allows different depth cues to be enabled or disabled. By imposing a fixed spatial tolerance around targets for speed-accuracy trade-off, completion time normalized with the straight-line distance to the target was used to measure reaching performance. In the present study, 8 (control) subjects using a typical monitor setup applied in rehabilitation were compared to 7 subjects using a head-mounted display and receiving additional depth cues, namely, hard-referenced objects of known size, motion parallax due to tracked head-movements, and stereopsis. Control subjects required on average 9.88 s/m straight-line distance, while subjects using the head-mounted display required only 3.15 s/m straight-line distance. Additionally, movement trajectories of control subjects showed a different pattern, indicating a lack of reliable depth information. Thus, state-of-the-art rehabilitation setups are challenging already for healthy subjects. This challenge can be reduced by improving the provided visual depth cues.

Keywords: Head-Mounted display · Perception · Reaching assessments · Upper-Arm rehabilitation

1 Introduction

Patients suffering from spinal cord injury or stroke often have a reduced muscle function or muscular control, impairing the ability to perform goal-directed reaching movements [3]. Goal directed reaching is an essential movement component to enable grasping, moving, or manipulating objects [2]. Therefore, reaching movements are of great interest during rehabilitation for restoring natural function, learning compensatory strategies, or to quantify impairments [3].

Augmenting reaching movements with instrumented setups and virtual reality is beneficial, since the labor-intensive work of placing or indicating targets can be moved away from a human therapist into virtual reality. Additionally, virtual reality can increase motivation [6], which may well be necessary due to the repetitive and boring nature of such reaching movement trainings.

To plan and perform natural reaching movements, human subjects must have clearly perceived the target's position in three-dimensional space. For perceiving depth, i.e. the distance from an object to the human, humans rely on a variety of different depth cues [8]. However, when indicating reaching movements within virtual reality, some of these depth cues are typically not available. For example, oculomotor cues that provide depth information due to eye muscular activity are not used, since the distance to the focused virtual reality display is not the distance to the object. Recent studies in instrumented arm rehabilitation with three-dimensional movements involve setups using typical two-dimensional computer monitors (e.g. see [4, 5, 11]). These setups also do not provide stereopsis by separate images for each eye, or motion parallax by rendering dependent on the viewer's perspective. Other depth cues, such as linear perspective, occlusion, and texture gradients are easy to recreate with computer graphics, and were typically provided. The relative importance of different depth cues has been shown to depend on the task [1], but for rehabilitation-typical reaching movements we have not found investigations of the effects of the different depth cues. Knowledge on the effect of such missing depth cues is crucial for designing virtual reality environments for reaching therapy, and for understanding impairment quantifications obtained with such virtual reality environments.

Recent development and releases of commercially available consumer grade virtual reality systems with head-mounted displays (HMDs, e.g. HTC¹ Vive, Oculus² Rift, Sony³ PlayStation VR) provide affordable solutions to provide both stereopsis and motion parallax. Besides the provided depth cues by HMDs, existing setups do typically not fully exploit the retinal image size depth cue. Often, only an approximate object size is known contextually by the subject (e.g. approximate size of a fruit to be grasped), but no hard-reference of exact object size is provided at both hand avatar and reaching target. Therefore, this work aims to quantify effects of typically missing depth cues on three-dimensional reaching movements. In other words, we want to demonstrate the open potential between current state-of-the-art rehabilitation setups and what we can technically realize with a HMD for performing three-dimensional reaching movements.

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³ Sony Corporation, Minato, Tokyo, Japan.

2 Methods

2.1 Apparatus

The experiment setup consisted of a desktop computer (Windows 10⁴, i5-6600 K⁵ CPU @ 3.5 GHz, GTX⁶ 1080 GPU), a desktop monitor for the control group (HP LP2065, 20", 1600 × 1200 @ 85 Hz), and the HTC¹ Vive virtual reality system (see Fig. 1). The HTC¹ Vive consists of a head-mounted display (1080 × 1200 per eye @ 90 Hz, 110° horizontal field of view, 0.6 kg), a handheld controller (0.2 kg), and the optical tracking system Lighthouse⁷ offering “sub-millimeter accuracy” and between “250 Hz to 1 kHz” update rates [10]. Virtual reality and data logging (at approximately 1 kHz) were realized in a custom-made Unity⁸ game using the OpenVR [9] library.



Fig. 1. User wearing the HTC Vive head-mounted display, holding a handheld controller, and a “Lighthouse” tracking device on a tripod in the back.

2.2 Task and Protocol

The subjects had to move a cylindrical pointer (0.05 m length, 0.01 m diameter) from a resting position on the table towards spherical targets and keep a part of the pointer inside the sphere for a two second dwell time (see Fig. 2). The target sphere was displayed as a semitransparent yellow table tennis ball (0.04 m diameter). If any part of the pointer was within the target sphere, the pointer color changed from pink to green.

The quantitative outcome of the individual reaching movements was time to completion. Subjects were instructed to perform the task as fast as possible. The controlled spatial tolerance given by the 0.04 m sphere was considered to control speed-accuracy trade-off and justify time to completion as a reasonable performance metric.

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⁶ Nvidia Corporation, Santa Clara, California, USA.

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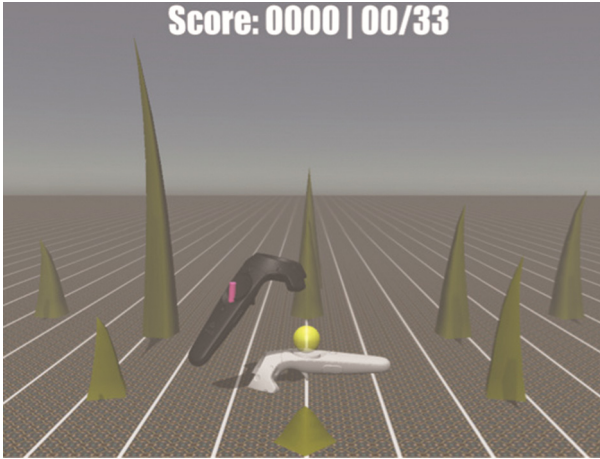


Fig. 2. Virtual reality rendering of the task with all realized monocular depth cues, the cylindrical pointer in pink, and the starting position indicated by the yellow target sphere.

In total, 33 targets were presented to each subject. Targets were distributed throughout a quarter-sphere of 0.32 m radius. The first 4 targets were placed at the extremes of the workspace and served to test the participants' range of motion and familiarize them with the task procedure. All subjects received the same targets in the same order.

At the start of the trial, the target sphere was displayed at the starting position. Once the subject reached the starting position, a 15 s countdown was started. The first 10 s of this countdown were provided as a resting period before presenting the next target to avoid fatigue. During this first 10 s, the countdown from 15 to 6 was visually displayed. The next target was presented 5 s before the countdown ran out, to allow the subject to locate the target in space and plan the movement. To avoid occlusion of the scenery, the start countdown from 5 to 0 was not displayed visually. Instead the last 3 s (3, 2, 1) were indicated with a 440 Hz sounds of 0.2 s duration each, and the start sound at the end of the countdown with a frequency of 880 Hz and a duration of 0.2 s duration, mimicking the start sounds commonly used for car races. The start sound was accompanied by a small vibration impulse realized with the handheld. False starts, i.e. moving out of the 0.04 m diameter sphere around the starting position before the start signal, were indicated by removal of the target sphere and excluded from analysis. After false start or task completion, the start position was indicated with the target sphere. Once the subject returned to the start position, the next 15 s countdown was started.

To keep subjects motivated, they received performance feedback about their last reaching movement during resting time. The score S_i of movement i was calculated as

$$S_i = 300 \cdot \frac{d_i}{\Delta t_i},$$

where d_i was the straight-line distance between start and target position in meter, and Δt_i the time to task completion in seconds (without the last two seconds of dwell time). The gain was chosen, such that typically a two to three-digit integer score could be displayed.

2.3 Virtual Environment and Depth Cues

The virtual environment for the reaching tests was designed to provide minimal rendered objects and distraction, while allowing to toggle as many different visual depth cues as possible (see Fig. 2). Visual depth cues (see e.g. [8]) that were realized with our setup, their implementation in the virtual scenery, and the group definitions are summarized in Table 1. The control group did not receive the visually rendered handheld and ghost handheld as hard-referenced objects of known size. Their only object of known size was the table tennis target sphere, to emulate the condition that is commonly used in virtual reality enhanced rehabilitation.

Table 1. Depth cues, their implementation and group definition.

Depth cue	Implementation	HMD group	Control group
Aerial Perspectives	Virtual fog	✓	✓
Linear Perspective	White parallel lines on the scenery ground	✓	✓
Motion Parallax (Scenery Motion)	Constant virtual wind moving blades of grass, closer blades of grass appear to be moving more	✓	✓
Occlusion	Blades of grass, that may partially occlude each other, target, or pointer	✓	✓
Shadows	Shadows of grasses, pointer, target, handhelds, and ghost handheld	✓	✓
Texture Gradients	Grey texture at scenery ground, where a more detailed rendering is visible closer to the subject	✓	✓
Motion Parallax (Subject Motion)	Virtual scene is rendered from the current viewing position and angle, which can be tracked by the HMD, close objects seem to move more	✓	
Retinal Image Size	Object of known size at both, pointer position by showing virtual black handheld, and at target position by white ghost handheld	✓	
Stereopsis	Individual screens for each eye in the HMD providing different images for both eyes	✓	

2.4 Subjects

In total, 17 healthy volunteers were recruited and pseudo-randomly assigned to the groups. The pseudo-randomization was performed based on a randomly permuted assignment list, ensuring approximately equal group sizes. The experimental protocol and subject recruitment was approved by the ETH Zurich Ethics Commission (EK 2014-N-21).

Subjects did not have any experience performing the test task prior to the experiment. However, three subjects from the lab correctly guessed the experimental hypothesis during the trial. All subjects had a range of motion compatible with the experiment, were 18 to 40 years old, participated in sports at least half an hour per week, and reported no non-corrected sight or hearing defects.

All subjects had to perform the task with their right hand. Subject handedness was determined with the Edinburgh Handedness Inventory [7]. One left-handed and one ambidextrous participant were excluded from data analysis. From the remaining 15 subjects, 8 were assigned to the control group.

2.5 Statistical Evaluation

The primary outcome variable time to task completion was normalized with the straight-line distance between start and target position. For each subject, one average time over distance was calculated. The 7 data points from HMD were compared against the 8 from the control group. First, a Levene test for the assumption of equal variance was performed. In case variance between the groups was significantly different, a non-parametric Kruskal-Wallis test was performed instead of one-way ANOVA. The significance level was fixed to $\alpha = 0.05$.

3 Results

The HMD group completed the reaching tasks on average in $\mu_{hmd} = 3.15$ s/m straight-line distance, the control group took in average $\mu_{ctrl} = 9.88$ s/m (Fig. 3). The Levene test for the assumption of equal variance failed ($F_{1,13}^{levene} = 15.9$, $p = 0.0015$). The Kruskal-Wallis test showed significant group differences ($F_{1,13}^{chisq} = 10.5$, $p = 0.0012$).

The 30th target position served as an illustrative example of differences in movement trajectories. While subjects in the HMD group showed relatively direct handheld trajectories, the control subjects failed to do so (Fig. 4).

A total of 16 false starts occurred that were excluded from analysis. The HMD group showed 3 false starts out of 203 movements (7 subjects, 29 evaluated targets) and the control group showed 13 false starts out of 232 evaluated movements (8 subjects, 29 evaluated targets).

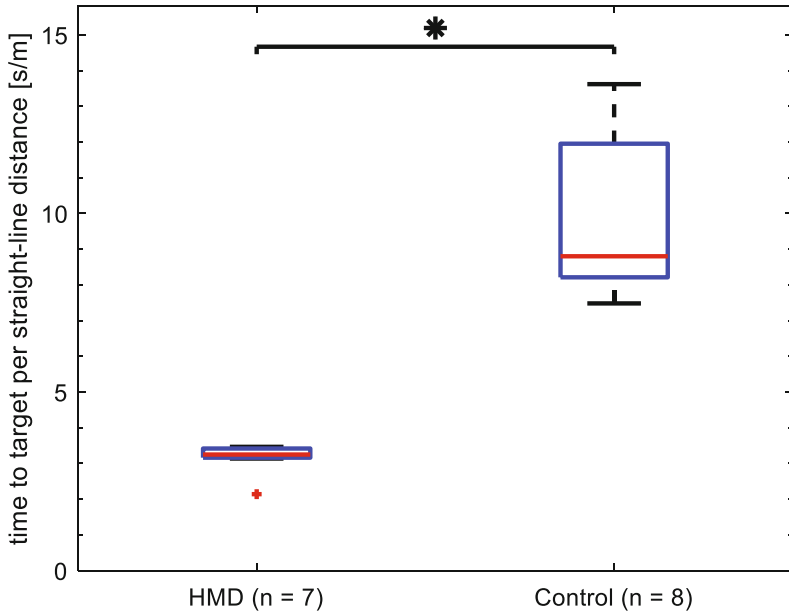


Fig. 3. Between group comparison of subject average time to target normalized by the straight-line distance, with group median as horizontal red lines, 50% coverage intervals as blue boxes, $\pm 2.7\sigma$ or 99.3% coverage intervals with black whiskers, one outlier as a red cross, and the significant group difference with a starred black horizontal bar.

4 Discussion

The observed differences in three-dimensional reaching movement performance (Fig. 3) between the group with the Head-Mounted Display (HMD) and the control group with the standard screen exceeded our expectations. The three-dimensional reaching-task was designed to resemble reaching movement training or impairment assessments from rehabilitation. Therefore, the task was supposed to be very simple for healthy subjects. However, we could show that already healthy subjects perform very unnatural movement trajectories (Fig. 4) in three-dimensional reaching if a rehabilitation typical two-dimensional screen setup is used. These unnatural and longer movements partly explain the observed increased time to completion, however, do not mitigate the problem itself. In rehabilitation and impairment assessments, the goal is to train or measure natural reaching, but for a typical screen setup already healthy subjects show difficulties to perform natural movement trajectories. This finding may be less critical for rehabilitation training, where high-intensity training and a high number of repetitions are of primary concern. However, this finding might be critical for impairment assessments where these additional difficulties arising from location perception could mask impairment effects.

Possible limitations that could be partially responsible for the observed performance difference are motivation bias and incomplete subject blinding. The subjects were

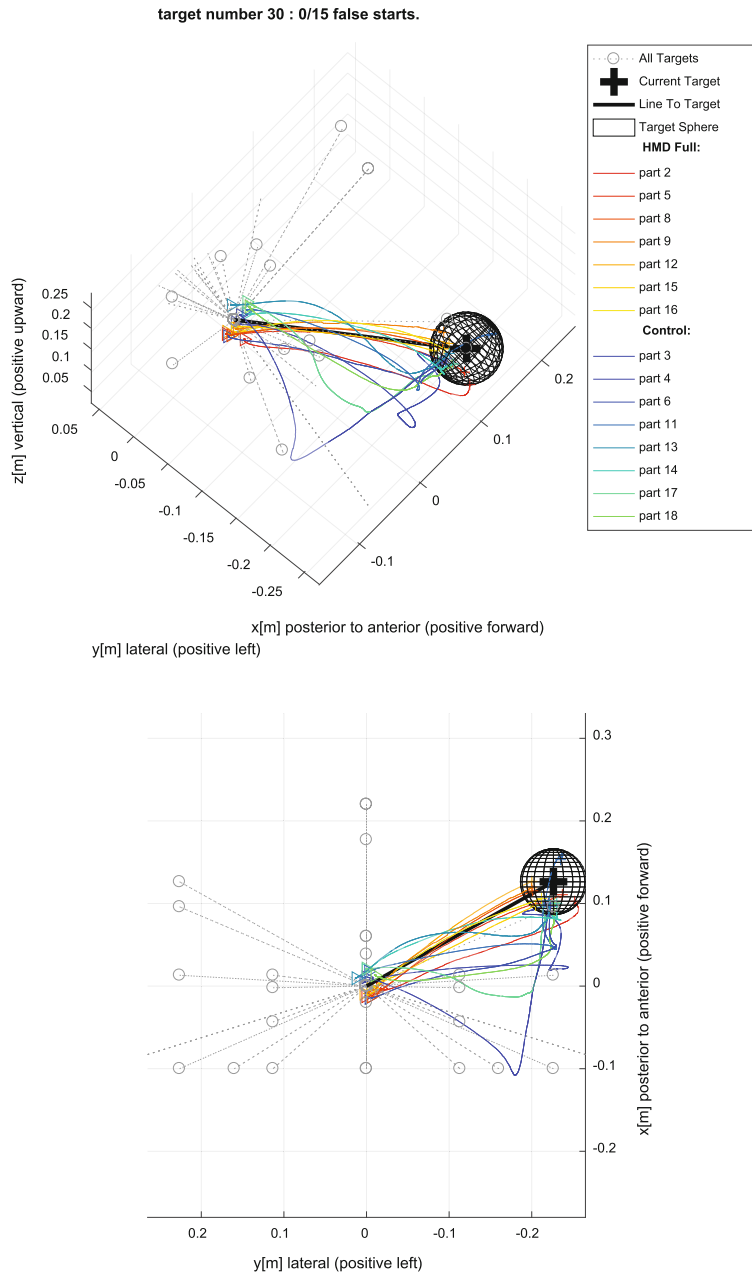


Fig. 4. Illustration of movement trajectories in 3D view (top) and top-down view (bottom) for target number 30. HMD subjects are illustrated in colors from red to yellow and control subjects from blue to green.

recruited for a study testing a new virtual reality system. Control subjects were not wearing the HMD but using a two-dimensional screen, which may be frustrating when expecting to test a new virtual reality system. In contrast, using immersive virtual reality is assumed to be motivating, especially for subjects in the HMD group that had no or only little experience with head-mounted displays before. Additionally, subjects may have correctly guessed the research question, and subjects not using the HMD may have easily guessed their assignment to control. However, we consider it unlikely that the movement trajectory differences shown by the control group (Fig. 4) are purely based on these limitations.

Another potential contributor for our big group difference is that the HMD setup was very immersive and blocked the real-world view completely. Eventually, subjects in the HMD group could purely rely on subconscious motor control, meaning they only had to actively think about the goal of the movement but not about movement execution. However, subjects in the control group eventually used visuomotor control consciously to control and synchronize both their hand movements in real world and the displayed movements on the screen. Thus, this additional mental load of actively using visuomotor control and the need of learning a visuomotor mapping from real to virtual world could be partially responsible for slower movements and therefore for their higher times to complete the reaching movements.

However, the trajectories performed (Fig. 4) seem to indicate that a major reason for the observed group differences is a behavioral difference in reaching strategy. Compared to HMD subjects that move close to a straight-line, subjects of the control group seem to first move to a correct two-dimensional position and then move back- or forwards in depth until they hit the target. Since hitting the target was visualized with the pointer changing its color, this movement strategy can be considered failsafe, meaning even without any depth perception this strategy would allow completing the task. Subjects in the control group seemed to have limited trust or benefit of the provided depth cues and therefore fell back on this less efficient but failsafe strategy. Thus, rehabilitation typical screen setups seem to provide serious challenges for depth perception, which can heavily affect subjects' reaching movements and impede their performance.

Eventually, subjects could learn to use the provided depth cues by practice and would then move away from this failsafe strategy. This means that impairment assessments based on three-dimensional reaching movements with such a setup would then at least partially measure virtual reality specific perception instead of the ability to perform reaching movements.

The rather high variability in reaching performance of control participants was partly expected (Fig. 3). When performing challenging movement experiments with humans, between participant variability in motivation, dexterity, cognitive abilities to find successful strategies, and different previous experience levels of perceiving virtual worlds may accumulate to a high variability in the measured performance. For participants performing under less challenging conditions, these participant preconditions may lose their impact on the performance metrics. E.g. if the depth perception in the HMD condition was not requiring any conscious mental load, variability in motivation or cognitive abilities have less effect on the measured performance. Participants using the HMD showed very low between subject variability in performance and significantly less

than control. This might be an indication that the HMD group shows flooring effects, where depth perception requires no conscious mental demand beyond what all subjects are all very skilled to do from real life.

Based on our observations in this study, we cannot attribute which of the additional depth cues (Table 1) was responsible for, or mainly contributed to, the observed group differences. Since we have not provided the same retinal image size cues to both groups, we do not want to conclude on the benefit of HMDs in general. Our study was designed to show the open potential for improved depth cues between current state-of-the-art setups and what we could technically realize. Since state-of-the-art rehabilitation environments do typically not contain objects with a hard-referenced known size, i.e. the handheld in our case, we have decided to avoid this cue for control. Nevertheless, our virtual environment included the other realizable depth cues and was designed with the goal to maximize clarity of each cue. Therefore, we are confident that our control group received better or at least equal depth information than subjects using typical state-of-the-art rehabilitation setups.

5 Conclusion

Missing depth cues in state-of-the-art monitor setups may change strategy and heavily impede performance when performing three-dimensional reaching movements with virtual targets. Impairment assessments obtained by reaching movements using such conventional monitor setups should be interpreted with care. Thus, we conclude that state-of-the-art rehabilitation setups leave a great open potential to make three-dimensional reaching movements in virtual reality more natural by providing better depth cues.

If head-mounted displays are necessary, or if optimizing or adding additional depth cues to the virtual reality environments is sufficient to exploit this potential stays an open question. Ranking the different depth cues' importance for three-dimensional reaching movements is subject of further research.

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Development of an Autonomous Character in Karate Kumite

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Abstract. Virtual Reality (VR) has become common practice in the field of sports, but autonomous virtual environment (VE) systems, especially in fast reacting sports, are rare. The current study demonstrates the development of an autonomous character (AC) in karate kumite, which performs attacks against a freely moving, real athlete. The development of the AC consists of four steps: selection of relevant karate techniques, development of a decision system, creation of an animated model of the AC, and the evaluation. A Cave Automatic Virtual Environment (CAVE) and a Head Mounted Display (HMD) were chosen for the VE. The evaluation of the AC in the VEs was conducted by expert interviews ($n = 6$). The results reveal a feeling of comfort for all athletes in VR which underpins a high degree of realism in the VEs. Moreover, the HMDs are seen as more suitable than CAVEs for presenting a karate specific environment. Based on these results the developed AC seems applicable for anticipation research and training in karate kumite. The discussion includes further possible improvements for the AC as well as future directions for further investigations and training programs using the AC. Moreover, the procedure of the AC's creation can be transferred to other sports.

Keywords: Virtual reality · Autonomous character · Karate kumite

1 Introduction

Virtual Reality (VR) is an often-used instrument in sports science and sports exercise [48]. It offers many advantages such as standardized experimental conditions and immersion due to spatial information and realistic visualization. Authentic sport settings can be repeated and manipulated, which is not possible in real settings [1, 38, 43, 49]. Further advantages of VR are the possibility of a standalone training [5, 44], the safe training environment, and the individual design [13, 14, 49]. Other authors [8, 46] highlight the importance of a realistic visualization of VR environments and their virtual characters to ensure a natural behavior of real athletes. [33] demand the development of

sport specific settings that are close to normal training and competition conditions. It is a known fact, that expert athletes can only use their expertise in familiar and natural settings [29]. Nevertheless, further demands exist to develop realistic virtual environment (VE) systems in sports including high immersion: the reduction of latencies, and cybersickness (symptoms of illness, nausea and dizziness) [2, 23, 36, 47].

Several VE systems already exist in different sports: e.g. rowing [38, 42], rugby [26], basketball [13, 14], handball [6, 7], baseball [22, 35, 49], soccer [15], dancing [11] and martial arts [4, 34]. Despite of these VE sports systems, only few inherit autonomous behavior of the virtual characters. [27, 36] emphasize the importance of interactivity in VR applications, which reduces cybersickness. Through interactivity athletes have more influence in the VE sports system and hence have a greater feeling of immersion. This is also described through the model of “Interlocking” [45], where the user’s intentions and possibilities and the affordances of the environment should fit together. Presence is a subjective sensation of being and interacting in a virtual environment. The more realistic the VE is and the greater are the affordances and possibilities to act, the higher is the feeling of presence and thus of immersion. Therefore, it is extremely important that users feel comfortable within and accept VR as a new world [41]. In this context it is also important that users have a natural perception in VR [37].

First approaches of autonomous, adaptive VE systems in sports already exist. [3] used an autonomous system to analyse anxiety and pressure in pistol shooting. [17] developed an interactive cricket simulator for training. Further, there are few autonomous virtual coaches in sports: [20] evolved a first approach for an adaptive immersive virtual golf training system. [11] presented a virtual dance tool for autonomous learning. [16, 19] created a multimodal coaching system for squats with a virtual coach giving real-time feedback. [9] developed an interactive virtual rugby environment to examine deceptive body movements.

In conclusion, autonomous systems can also be used in training as well as in research. However, there is still a research gap in the development and use of autonomous VE sports systems, which adapt independently to the athlete’s movements to ensure realistic sports specific behavior. Especially, autonomous VE systems in fast reacting sports and studies in karate-kumite are rare [32]. Therefore, our research is focused on the development of an autonomous character (AC) in karate kumite, which is able to execute a karate attack autonomously against a real athlete. The development is described from a sports science point of view and can also be used for the creation of ACs in other sports. In the discussion we present future investigations and training programs, which are possible with the ACs. Furthermore, insights regarding the preferable display mode (CAVE or HMD) are given.

2 Methods

2.1 Development of an AC from a Sports Science’s Perspective of View

General Approach

Autonomous characters (AC) for karate kumite were created by the following steps: competition analysis of several high-class international karate kumite matches, design

of the AC based on 3D motion capture data, and development of a decision system for autonomy of the ACs. The aim of the current work is that an AC can choose and conduct an appropriate attack towards a real athlete, who is moving karate specifically (stepping with varying distances), but does not perform an attack himself. The athlete is aware that he is going to be attacked and has to react as in a real competition. The AC recognizes the position of the real athlete permanently through the real-time tracking of the athlete's head and two hand targets.

Competition analysis: selection of relevant karate techniques

A male karate expert with long-term international success as athlete and experience as coach carried out the competition analysis of 45 high-ranking international karate competitions (including world championships, European championships and high-class international cups in the years 2012–2013. The results show that the arm techniques Gyaku-Zuki (GZ), Gyaku-Zuki overrun (GZo) and Kizami-Zuki (KZ) are the most often and most successfully applied attacks with 87% of all attacks. The remaining 13% of attacks are foot techniques, which are not included in the AC. Based on these results, which are in line with [28], the above mentioned arm attacking techniques were chosen for the AC. GZ, a punch with the back arm, is performed from a stable position, while GZo is executed during a forward motion. KZ is a punch with the front arm. GZ can be performed jodan (attack towards the head/chin) and chudan (attack towards the chest/solarplexus), while KZ and GZo are jodan techniques only.

Creation of the AC

Three ($n = 3$) healthy karate athletes (one woman and two men, age: 24–31 years, long-term and successful international experiences in karate kumite, with coaching experiences, shotokan style) were chosen for animating three ACs. All athletes were informed about the procedure, risks and the aim of the study. They took part on voluntary basis and gave their written consents. The study was conducted under the ethical guidelines of the academic institutions. All employed attacking techniques, sports specific stepping and turning movements were recorded using an infrared motion capturing system (VICON Tracker by using in A.R.T., Weilheim, Germany, 120 Hz) with twelve cameras (MX-13, VICON, Oxford, UK). The targets were placed according to the guidelines of A.R.T, with the exception, that the abdomen target and both foot targets were replaced

Table 1. List of movement recordings for each of the three ACs. GZ: Gyaku-Zuki, GZo, Gyaku-Zuki overrun, KZ: Kizami-Zuki.

Kind of variation	Quantity/movement description
Distance to the target	3-4 (GZ), 2 (KZ), 2 (GZo)
Vertical variation to the target area	4 (GZ), 2 (KZ), 2 (GZo)
Horizontal variations to the target area	3 (GZ), 3 (GZo), 3 (KZ)
Stepping in place	Regular, irregular
Stepping in all directions	Forward, backward, to the left, to the right
Turning in place	To the right and left, both 45° and 90°
Turning around an opponent	To the right and left, both 45° and 90°

by self-created targets using 3D-print and CAD models, because the original targets were not suitable for karate specific movements (either they hurt or they were too loose). All attacks started from a pre-defined position and were executed towards a target, replacing an imaginary opponent. The distance and height of the target varied according to the skill of the athlete and the feasibility for a correct attack execution due to the karate regime. For further detail see Table 1. All conditions were repeated three times. All movements were monitored by a karate expert.

Development of a decision system for an AC

The autonomy of the ACs is based on a decision system developed in a previous work [31], and aims to ensure a natural (not predictable) behavior of the AC. Based on the competition analysis it was defined that the ACs execute the GZ and GZo with the right hand, and the KZ with the left. In normal stepping position (kamae position) the left foot and left hand are ahead, the right foot is behind, and the right hand positioned near the hip.

The decision system runs through four processing stages: Firstly, the AC has to be aligned towards the real athlete. The alignment is achieved when the AC's front hand points towards the athlete's head and front hand being tracked continuously. Oriental adaptations are attained by use of one or several recorded turning techniques. Secondly, the AC follows the athlete's movements to get in a distance which is appropriate for an attack. The ACs' positional adaptations are achieved by using recorded forward, backward and sideways steps. All karate athletes executed their attacks within distances of 1.80 m to 2.40 m (distance between rear foot of the attacker and the target), the AC measures the distances (from its rear foot to the plain of the athlete's head) continuously. Thirdly, the AC is in a possible attacking distance and executes an offensive. The distance between the real athlete and the AC is divided into 20 cm steps resulting in six ranges (under 1.60 m, 1.60 m–1.80 m, 1.80 m–2.00 m, 2.00 m–2.20 m, 2.20 m–2.40 m, over 2.40 m). For each range we used cumulative probabilities (given in percent and based on the competition analysis) for executing attacks. Fourthly, the AC selects the appropriate target position on the opponent's body. Depending on the position of the hands and thus the defensive position of the real athlete, this can either be the chin (jodan) or the chest (chudan). For more technical details see [50].

Technical implementations

The motion data of each real athlete was captured with the described motion capturing system and a frequency of 120 Hz. The captured data represent target positions of each body limb and are saved as raw data files for each athlete and type of attack. Target information that was lost during the capturing process was interpolated linearly using valid target positions before and after the gaps. Since the gaps were not bigger than a few frames in most cases, no relevant features of the movement were lost during this process. All tracking raw data were filtered using different operations such as LULU smoothing, Savitzky-Golay-Filter and Gaussian Smoothing. To gain best results the parameters of the filtering process were determined separately for each data file. The raw tracking data were then converted to a standard motion representation which describes the orientations of the joints of a skeleton. The 3D model of the AC was

designed analogously to a real athlete who perceives the opponent's movements, decides on an appropriate reaction and performs it. The perception of the AC was realized by an online tracking system that yields the positions of markers mounted to the athlete's glasses and attached to his hands. This data was provided to the decision system which then chose an appropriate movement for the AC (attack or a step movement). The decision system worked at a frequency of 30 Hz which enables the AC to adapt to changes in the environment. The motion generator produced a motion defined by the decision system (using probability distribution related to empirical findings for hand techniques: 46% are GZ, 23% are GZo and 31% are KZ) and ensured that given constraints as e.g. distances, positions and orientations are met. For the decision on target height (jodan or chudan) the positions of the athlete's hands were taken into account by computing of protecting area and comparison with the width of two fists. Finally, the created motion data was used to animate the 3D model which was visualized in a stereoscopic displaying system, a CAVE or a HMD. The system was implemented using C++ with Qt library published under the LGPL license. The response time (time between the action of the real athlete and reaction of the AC) was around 150 ms. This includes the processing time of the tracking system, network delay, latency of the displaying hardware, and the time needed for all computations involved. For further technical details see [50].

2.2 Evaluation

Participants

Six ($n = 6$) experts took part in the interviews on voluntary basis. Four experts are also coaches in karate kumite and thus can give a reliable feedback if the ACs could be used as a tool for karate training in the future.

Procedures

The first aim of the interviews was to analyze whether the ACs and the virtual environment are realistic enough and if the experts feel comfortable in VR. Parts of the interview are based on the questionnaires IPQ [40] and SSQ [21]. The evaluation method is based on to the one described in [17]. It is also possible to evaluate such systems by questionnaires or different scales (e.g. MEC-SPQ or SPES), but we chose expert interviews to get a specific evaluation for our ACs to assess, inter alia, the movement behavior and interactivity. The second aim was to analyze whether a CAVE or a HMD would be more appropriate for future investigations. The experts examined the ACs in two immersive VE systems: in a CAVE (Cave Automatic, Virtual Environment with four screens, size $2.30\text{ m} \times 2.30\text{ m}$, resolution of 1400×1050) and by using a HMD (Head Mounted Display (Oculus Rift, DK2). They observed the ACs in each setting for 15 min (five minutes per AC in both virtual environments). To reach a higher feeling of immersion within the virtual environment of the HMD, the user's hands were visualized. Unfortunately, our HMD was not wireless, so the cable was held away to reduce the risk of tripping. The participants were free to move and response karate specifically towards the offences.

Subsequently, the interview (open questions) was made and the answers were both journalized and recorded. The interview consisted of seven complexes, had a total of 35

questions (three to seven items per complex) and lasted 30 min. Complex 1 included seven questions concerning realistic and natural attack behavior and movement behavior of the ACs. Furthermore it was asked, if movement velocities and behavior of different parts of the ACs' body (hands, posture of head and trunk, face and hip rotation) were displayed realistically. Complex 2 included five questions. It was questioned, if the ACs followed the athletes correctly and if they attacked from realistic positions and distances and hit adequate body regions of the athletes. Complex 3 contained four questions regarding the realistic appearance of the ACs and the virtual environment. The participants were also asked if they felt immersed in the VE system and if they had still perceived the real environment. In complex 4 five questions were used to determine, if further feedback or implementations are needed for future sports applications. Furthermore, the participants were asked, if they need an acoustical or a tactile feedback, natural facial expressions in the ACs or if the visual appearance is sufficient enough. The athletes could give further ideas for improvements, additionally. Complex 5 consisted of four questions. The participants were asked if they took the VE system seriously, if they had fun, and if they felt comfortable in VR. Furthermore, they were asked for symptoms of cybersickness. Complex 6 was composed of three questions. It was assessed if the attacks were recognizable and correctly performed, and if one could respond adequately. Furthermore, the participants should indicate if they were able to behave in the VE system as they would have done in reality. Complex 7 (six questions) inquired if the participants recommended HMD or CAVE for future investigations.

Analysis of data

The participant's answers were summarized and analyzed accordingly to the methods of [18, 25]. All answers were scaled into three categories (good, moderate and poor), due to certain keywords or phrases. These are shown as descriptive statistics in Table 2. Answers to complex 7 were scaled into seven categories (−3 to +3).

Table 2. Results of the expert interviews in [%]

Complex	Output device	Good	Moderate	Poor
1: Realistic behavior of the AC	CAVE/HMD	31	66	3
2: Realistic autonomous setting	CAVE/HMD	13	87	0
3: Realistic visualization of the AC and the VE/immersion	CAVE	18.75	81.25	0
	HMD	93.75	6.25	0
4: Integration of further feedback and implementations	CAVE/HMD	64.17	25.83	0
5: Experts feelings and experiences in VR	CAVE	90	7.5	2.5
	HMD	92.5	5	2.5
6: Experts movements, reactions and attack recognition	CAVE	30.84	45	24.16
	HMD	55	30.84	14.16

3 Results

Complex 1 (realistic behavior of the ACs) is sufficient, but still developable. Complex 2 (realistic interaction) is only moderate. The results of complex 3 (realistic visualization of VR and immersion) indicate that CAVE and HMD have a different level of realistic visualization and immersion. While visualization and immersion are only moderate in the CAVE, they are satisfactory using the HMD. The visual presentation of the ACs (karate Gi), the environment (a gym and mats), the presented movements, and the attack distances were rated realistic. Moreover, the decision system works correctly and fulfills the demands for VE sports systems, allowing a sport specific behavior. Complex 4 (integration of further feedback and implementations) confirms that the visual implementation is sufficient. Ideas for further feedback and implementations are described in the discussion. Complex 5 shows that the participant's feelings, experiences and cognition are good in both virtual environments, CAVE and HMD. All participants felt great motivation and no one reported symptoms of cybersickness, although the latency of the AC's movements were around 150 ms. However, the latency did not lead to changes in the reaction behavior or seem to disturb the athletes. Nevertheless, all participants recommend further decrease of latencies to ensure greater immersion and more natural training feeling. Complex 6 (participant's movements, reactions and attack recognition) reveals that they can move and respond better with the HMD. In the discussion we show possible improvements for the ACs and possible future usage in regards of anticipation research and as a tool for competition and training.

In complex 7 the participants were asked if they prefer CAVE or HMD for future investigations in karate kumite. All athletes rate the HMD as preferable tool. In a seven-scale (-3 to +3) the CAVE was valued with 0.25 ± 0.5 (mean $\pm$ SD), while the HMD was valued with 1.875 ± 0.25 (mean $\pm$ SD). The interviews showed that all users prefer the use of a HMD due to greater immersion and more freedom of movements, which allow sports specific motions and responses. The CAVE used in the current study was too small, which resulted in the participant's feeling of being constricted. The HMD was very light and compact and thus did not disturb the users. While the HMD encloses both eyes and thus completely excluded the real world, it was easier to accept the VR here as a real environment. In regard to the question, if the experts would recommend using the current autonomous VE system for future sports application, such as anticipation research and training, all stated that after some improvements the system would be appropriate for further utilization.

4 Discussion and Outlook

The present study provides sports scientific foundations for the development of an autonomous character (AC) in karate kumite with regard to future anticipation research and training. Autonomous characters have the advantage, that they can be used without the need of further training partners or coaches. Nowadays, new visualization technologies are more accessible and there is a need to understand if and how VR can be used to train athletes in sport. This work makes a contribution in showing how to create an

autonomous VE sports system and how this system could be used to analyze and train motor response and decision making in karate. It allows both, to study and to routinely test athletic responses to realistic and controllable stimuli. The current work also shows that karate athletes prefer a HMD over a CAVE, so for future investigations and training in VR, we recommend the use of compact and small HMDs in a room which is big enough for sports specific movements. The presented procedure can also be transferred to other sports (e.g. martial arts and racquet sports). Finally, our study provides essential steps for the development of an autonomous character (AC), which can be used in a great number of sports to perform interactions with real athletes. These include the selection of relevant techniques, the creation of an animated model of the AC, the development of a decision system, and its evaluation. In the following sections we present possible improvements for the ACs and potential uses.

Possible improvements for the ACs

The study shows the development and evaluation of an AC in karate kumite from a sports scientific point of view. The results underpin the development of a realistic autonomous setting and hence achieve the demands of [4, 8, 32, 33, 46].

There has been a rapid improvement of techniques in the last years, so future virtual sports systems with reduced latencies should be available soon [17, 36]. The usage of a prediction system would also be helpful to decrease latencies of the decision system [27]. Already [10] present a prediction system in table tennis, what could also be used in other fast reacting sports. We intend to increase the interaction, what is also demanded by our athletes, so the ACs would be able to react to a real athlete's attack. Therefore, a movement analysis and a cue detection, which are relevant to anticipate an upcoming attack, are necessary.

Our experts recommend further feedback in the autonomous VE system like acoustics (e.g. stepping noises) and haptics, but especially haptic feedback is still rare in VE sports systems [43, 49]. A possible setup for haptic feedback would be to attach e.g. vibration sensors at the body of the real athlete (for a correct AC's hit detection), and at the athlete's hand for a correct hit detection of the athlete's attack. The approach of [12] would therefore be a possible solution, where piezoelectric sensors were attached to detect upcoming forces (correct hits). However, feedback should be implemented cautiously. Firstly, it could evoke a dependency of users to it, and secondly the athletes could be disturbed by further technical devices [43]. In regards to the importance of natural facial expressions of the ACs different opinions exist. Currently, the ACs have a stiff face with a neutral expression. While some athletes state that facial expressions are very important, others declare that they do not look into the opponent's face. Using eye tracking, [39] showed that expert karate athletes focus on the opponent's head, what could be a cue that the face and especially facial expressions may play a role. However, the use of eye tracking alone inherits the problem that actual attention as well as information in peripheral vision cannot be captured and measured directly. Although athletes focus on the head, which is a central point, they can detect small peripheral changes in the opponent's kinematics and thus have an overall view. As a result the influence of facial expressions and the detection of relevant anticipatory cues need further examinations in future studies.

Another possibility for improving the ACs for karate training in VR is to capture more athletes. The athletes embodying the ACs in the study at hand are all from the same sports club which is organized in the German Judo-Karate Association (DJKB) e.V. Athletes from different national and international clubs and different associations could be included and be the basis for several ACs with different fight behaviors. Moreover, further attack techniques (e.g. foot techniques) should be integrated in the ACs. [30] already demonstrated that high-class experts also have little individual variations within their movement patterns. A further potential approach could be to mix all the movements of all current ACs. Specifically, the same character would attack, but with all recorded movement variations (all distances and all heights for attacks), which would not be possible in reality.

Potential uses of the ACs

One of the aims of VE sports systems are to implement realistic sports scenarios for practical training (e.g. [13, 14, 38]), and to analyze relevant anticipatory cues (e.g. [4, 9]). All experts of the present study have the opinion that an AC is a useful tool for anticipation and reaction training and further investigations in anticipation research. Anticipation investigations can be made using the ACs in combination with the methods from [32]. Here, the real athlete's first response to virtual attacks is being detected.

Another possibility to analyze anticipation is the usage of eye tracking in VR in combination with temporal and spatial occlusion techniques. The reaction to spatial and/or temporal occluded attacks might give insights to relevant anticipatory cues. Further it is possible to use the ACs for practical training, e.g. the improvement of response times and type of response. Virtual opponents have the advantage that they do not get tired and thus can be a suitable tool for speed and reaction training. Additionally, the speed of the virtual attacks could be manipulated to create different levels of difficulties. ACs can also be used for competition preparation. The AC can be manipulated to act and attack like an actual future opponent. Different fighting types (offensive, defensive) can be embodied, too. However, when using HMDs in VR training, the athlete's body should be virtualized [24]. It would also be desirable to use a wireless HMD to create even more freedom of movement.

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Miscellaneous

Students' Use of and Attitudes Towards Information and Communication Technologies in Sport Education Cross-Sectional Surveys Over the Past 15 Years

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Abstract. In this paper, cross-sectional surveys of students' access to, use of and attitude towards computer, the Internet, mobile technologies and eLearning are described covering a period of 15 years. A sample of 1,106 students (age: $M = 21.5$ years, $SD = 2.9$) of sport science participated in the study. They completed a questionnaire addressing the topics mentioned above. Main findings are that almost every student has access to the computer, mobile ICT, and the Internet. There are differences between study programs favoring students attending study programs that combine sport and computer science. Furthermore, attitudes have changed over the past 15 years. Finally, gender differences concerning the use of and attitudes to the computer and the Internet but not eLearning still exist. Important limitations of the studies are the sequential cross-sectional design and sample distortions.

Keywords: ICT · eLearning · Equipment · Attitude · Gender · Sport education

1 Introduction

Innovative information and communication technologies (ICT) always offer new options for education. This is particularly the case with the Internet. In the year 2000, the web-based education commission of the US government published a report on the power of the Internet for learning [6]. This report identifies new and fascinating promises of web-based education (WBE) and possible barriers for successful implementation such as access to the Internet.

However, the success of web-based and ICT-supported education is not only a matter of technical equipment, it is also depending upon the attitude of all the persons being involved in the complex process of teaching and learning [7, 9, 10]. In this regard, the attitudes of students and teachers are ambivalent: On the one hand, the added values of eLearning are clearly recognized and acknowledged, but on the other hand, numerous critical factors influencing implementation are mentioned. For example, Almaberah et al. [1] found that students (sample: 536 undergraduate students, 382 females, 154 males; different study programs) used the Internet mainly for communication (social networks, chats) and information search. In a sample of 300 students

(165 females, 135 males; different faculties and study programs), Bashir et al. [2] found that students used the Internet mainly at the university, at home, and at the department. About 90% used the Internet daily or at least twice a week. The dominant reasons for Internet use were the preparation of class assignments (63%), research (45%), and knowledge update (33%), followed by communication (32%) and entertainment (24%). In three case studies, Fayer [5] found that students deemed four aspects important for successful online courses: strong course organization, appropriate time-flexible feedback by the teachers, confidence in the instructor's competencies (content, support), and relevance of feedback and coursework. In a sample of 158 students (65 females, 93 males; different study programs: sport & computer science, sport science, Physical Education), Wiemeyer [9] found positive attitudes towards eLearning; furthermore, attitudes differed significantly regarding gender and study program. In a sample of 434 students of social sciences (156 males, 278 females; undergraduates and postgraduates), Dumčienė et al. [4] found that traditional learning was favored over both pure eLearning and blended learning. Furthermore, the authors found significant differences according to study program (undergraduates: more positive towards eLearning than postgraduates) and gender (males were more satisfied with eLearning than females; see also [8]). Furthermore, a recent review [3] confirmed significant gender differences concerning attitudes towards technology in general; however, effect sizes were low.

In summary, there is evidence that students' attitude towards eLearning is heterogeneous and depends on gender and study program as well as quality of content and learning material. Furthermore, due to fast developments in ICT, quality of technology seems to have decreased in importance.

Therefore, the purpose of this contribution is to analyze the attitudes, behavior, and ICT equipment of students involved in sport education regarding eLearning. In particular, we were interested in answering the following questions:

- Q1: How do students of sport science use computers, the Internet and mobile ICT?
- Q2: What is the attitude of students towards using computer and the Internet for learning?
- Q3: Are there characteristic profiles for general usage of computers versus mobile ICT?
- Q4: Are there any systematic changes over the past 15 years from 2002 to 2016?
- Q5: Do the attitudes of students differ depending on the particular study program they are part of?
- Q6: Do students of sport science also show the often-reported gender differences?

2 Methods

2.1 Sample

A total sample of 1,106 students (372 females, 733 males; age: $M = 21.5$ years, $SD = 2.9$) of sport science (three types of program: sport & computer science, sport science, or Physical Education) at Technische Universität Darmstadt, mainly in the first term (90%), participated in the study.

Table 1 shows the distribution of students concerning term and study program.

Table 1. Distribution of students in the sample. Note: 15 students did not answer.

Term	Sport science & computer science		Sport science		Physical education		Total
	<i>N</i>	%	<i>N</i>	%	<i>N</i>	%	
Winter 2002/03	26	36%	36	50%	10	14%	72
Winter 2003/04	14	16%	63	73%	9	10%	86
Winter 2004/05	20	27%	50	68%	3	4%	73
Winter 2005/06	16	16%	49	51%	32	33%	97
Winter 2006/07	22	22%	29	30%	47	48%	98
Winter 2007/08	21	27%	3	4%	54	69%	78
Winter 2008/09	21	27%	3	4%	55	70%	79
Winter 2009/10	1	1%	2	3%	70	96%	73
Winter 2010/11	2	6%	10	29%	22	65%	34
Winter 2011/12	33	40%	29	35%	21	25%	83
Winter 2012/13	23	27%	26	31%	35	42%	84
Winter 2013/14	18	24%	34	46%	22	30%	74
Winter 2014/15	13	18%	39	55%	19	27%	71
Winter 2015/16	7	11%	44	71%	11	18%	62
Winter 2016/17	0	0%	13	48%	14	52%	27
	237		430		424		1091

The gender distribution in the different study programs is illustrated in Table 2. Proportion of females and males is significantly different in the study programs ($\chi^2(df = 2) = 13.60; p < 0.001$). In general, more males than females attended the three programs. Furthermore, the unequal distribution of females and males is most pronounced in the computer science programs.

Table 2. Gender distribution in the different study programs

Gender	Sport & computer science	Sport science	Physical education	Total
Female	56	153	158	367
Male	180	277	266	723

2.2 Questionnaire

The questionnaire consisted of six parts (see Table 3): First, personal data were assessed. In the second part, experiences with the subjects ‘mathematics’, ‘physics’, and ‘informatics’ were reported (scale: 6 grades from “very good” to “very bad”). In the third part, questions concerning access and usage of computers and Internet were asked. In the fourth and fifth part, attitudes towards the use of computers and eLearning was assessed, respectively (scale: 6 grades from “fully applies” to “does not apply at all”).

Table 3. Structure of the questionnaire.

Part	Items
Personal data	5 items: Age, gender, study program, terms, subjects
School experiences	6 items: Experiences with mathematics, physics, and informatics
Computer access & usage	5 items
Attitudes towards computers	13 items (9 items added since term 2006/07)
Attitudes towards eLearning	16 items
Usage of mobile technologies	5 items (added since term 2014/15)

Finally, access to and use of mobile ICT was reported; this part has been added in term 2014/15.

2.3 Research Questions and Hypotheses

Regarding Q4 to Q6 mentioned in the introduction, the following hypotheses were tested:

1. H1: Considering the period of 15 years from 2002/3 to 2016/17, significant term-related differences were expected concerning access to and usage of computers, the Internet, and mobile technologies.
2. H2: Differences related to study program (with an advantage for students in the computer science programs) were expected.
3. H3: According to extensive existing evidence, gender differences were expected.

2.4 Data Processing and Statistical Analysis

All statistical analyses were performed using the software package SPSS 24.0.0.0. Frequencies, Mean and Standard Deviation were used as descriptive measures. Due to the fact that normal distribution was violated in 10 variables (Kolmogorov-Smirnov test: $p < 0.20$), group, gender and term differences were analyzed by non-parametric tests (Kruskal-Wallis test). Significance levels were set to 0.05. In multiple follow-up analyses, non-parametric tests (Wilcoxon and Mann-Whitney U test) were applied including Bonferroni correction of significance level.

3 Results

3.1 Q1: Access to and Usage of ICT – State and Changes

Almost every student has access to a computer (99.2%), smartphones (99.3%), and the Internet (computer or notebook: 97%; smartphone: 100%). There was a significant increase of computer access and Internet access via computer from 2002/3 to 2016/17 (computer access: $\chi^2(df = 28) = 52.5$; $p = .04$; Internet access: $\chi^2(df = 28) = 80.1$; $p < 0.001$).

Within the last three terms, there has been no significant change in smartphone access ($\chi^2(df = 2) = 1.33; p = .51$).

Digital access to the Internet is clearly dominating for computers (84.8%; analogue: 5.1%; unknown: 7.6%). Mobile Internet access (smartphones) is dominated by W-LAN (72.5%; HSDPA, HSPA+, LTE: 13.1%; GSM, GPRS, EDGE, UMTS: 11.1%; unknown: 3.1%).

Total duration of computer use is 14.4 ± 10.5 h per week (Range: 0–90 h/wk). Communication (3.4 ± 4.3 h/wk) and surfing (3.3 ± 3.6 h/wk) dominate, followed by information search (2.4 ± 2.4 h/wk) and writing texts (2.5 ± 2.9 h/wk). Gaming (1.4 ± 3.5 h/wk) and learning (1.0 ± 2.0 h/wk) play a minor role.

3.2 Q2: Attitude Towards Computer and eLearning

Concerning attitude towards computer (Maximum = 6), students are rather confident (4.8 ± 1.3) with the computer. The item “I like spending time with the computer” (liking) is moderately agreed (3.6 ± 1.3), as well as the item “I use computers exclusively for study and work” (study only; 3.1 ± 1.7). The item “I use computers only if it is inevitable” (reluctance) shows low agreement (1.5 ± 1.1).

In general, students show a moderately positive attitude towards eLearning. Students put high demands on eLearning (see Table 4). They expect interactive and activating learning elements comprising animations, tasks, exercises, and simulations. Furthermore, exchange with classmates and teachers is preferred. Fixed and self-determined order of eLearning show equal approval. Greater interest, demands and engagement as well as technology-mediated contact to teachers are much less approved.

3.3 Q3: Characteristic Profiles for General Usage of Computers Versus Mobile ICT

Since the winter term 2014/15, students were asked to report separately on computer and smartphone use. Students spend about 14.4 ± 10.9 h per week with the computer compared to 22.0 ± 23.2 h per week with their smartphones. This difference is significant (Wilcoxon test: $N = 153; z = -4.33, p < .001$).

In Fig. 1 durations of use for computers versus smartphones are compared. In the questionnaire, five application purposes were addressed separately for computer and mobile ICT: communication, surfing, information search, gaming, and learning. Wilcoxon tests (with Bonferroni correction) confirm significant differences between computers and smartphones concerning communication ($N = 153; z = -9.13, p < .001$), information search ($N = 154; z = -3.43, p = .001$), and learning ($N = 149; z = -5.08, p < .001$), but not for surfing ($N = 157; z = -0.73, p = .47$) and gaming ($N = 150; z = -0.72, p = .47$).

Table 4. Attitudes of sport students towards eLearning (total sample). Scale (6 grades): “1 – does not apply at all” to “6 – fully applies”

Statement	<i>N</i>	<i>M</i>	<i>SD</i>
1. The learning program should contain tasks and exercises in order to apply the learning content.	1098	5.0	1.1
2. The learning program should contain animations and visual illustrations.	1098	5.0	1.1
3. The learning program should contain simulations in order to go throw different options for solution.	1101	4.9	1.1
4. The learning program should motivate for active engagement with the learning content.	1097	4.6	1.2
5. I would like to exchange with my classmates about the learning matter.	1099	4.5	1.2
6. The learning program should allow me a free working procedure.	1097	4.5	1.2
7. The learning program should afford more than a lecture.	1101	4.4	1.2
8. I would like to have regular contact to the teacher.	1097	4.2	1.2
9. I would prefer to work alone with the learning program.	1099	3.6	1.3
10. The learning program should prescribe a fixed order for working through it.	1101	3.6	1.3
11. With a computer at home I would structure my learning time better.	1098	3.4	1.4
12. My personal effort with computers would be equal or greater compared to a lecture.	1098	3.4	1.3
13. I would be satisfied if the lecture-related material (e.g., scripts and slides) was available online.	1102	3.4	1.4
14. A learning program would put more demands on me than a lecture.	1095	3.3	1.4
15. A technology-mediated contact to the teacher would be sufficient for me.	1081	3.2	1.3
16. I suppose that learning with a computer is more interesting than a lecture.	1094	3.0	1.4

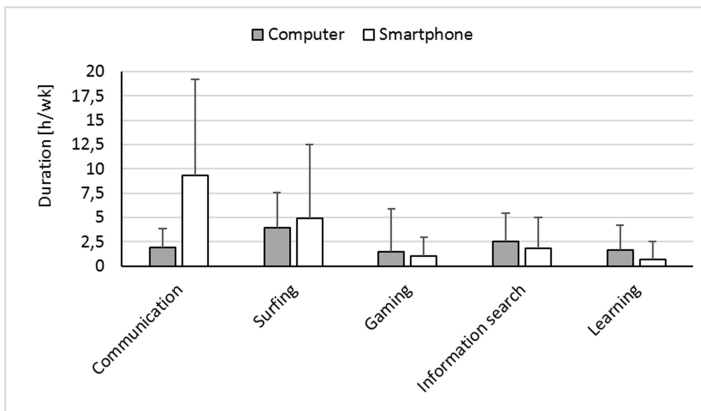


Fig. 1. Comparison of activities using computers versus smartphones (Columns: means; error bars: standard deviation). Smartphones are used predominantly for communication and surfing, whereas surfing and information search dominate for computers.

3.4 H1: Differences Related to Term

Concerning duration of computer usage, seven different application purposes have been addressed in the questionnaire (see Subsect. 3.1). Kruskal-Wallis tests reveal that there are significant term-related differences concerning all seven application purposes.

Follow-up analyses (U tests; with Bonferroni correction) show the following significant differences: Use of computers for communication was more extensive from the first three terms (terms 2002/03 to 2004/05) compared to term 2009/10, followed by increasingly lower values until term 2014/15 (see Fig. 2A). Application of computers for surfing was significantly lower in the first six terms compared to the following terms (see Fig. 2B). Use of computers for information search show fluctuations between 1.5 and 4.2 h per week (see Fig. 2C). Application of computers for learning was relatively low in the first seven terms, followed by an increase from term 2009/10 on (see Fig. 2D).

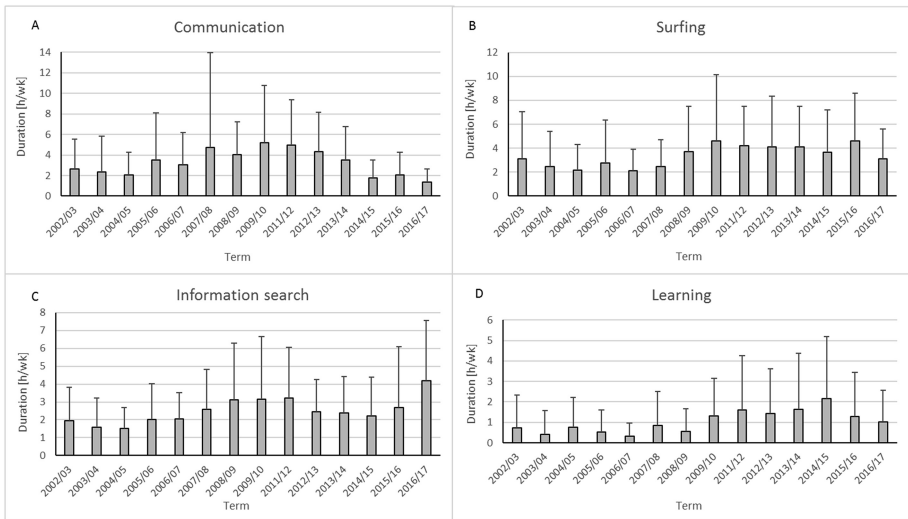


Fig. 2. Computer use for communication (A), surfing (B), information search (C), and learning (D) from term 2002/03 to term 2016/17. (Columns: means; error bars: standard deviation).

Concerning the attitude towards computers, Kruskal-Wallis tests reveal significant term differences for confidence ($\chi^2(df = 14) = 33.65$; $p = .002$), liking ($\chi^2(df = 14) = 103.04$; $p < .001$), and reluctance ($\chi^2(df = 14) = 27.01$; $p = .02$), but not for study only ($\chi^2(df = 14) = 16.77$; $p = .27$). Follow-up analyses (U tests with Bonferroni corrections) do not show a clear picture. There is just a tendency for confidence to increase a little bit from term 2002/03 to 2016/17. The other variables show more or less unsystematic fluctuations.

Concerning attitude towards eLearning, Kruskal-Wallis tests confirm significant term differences for exchange with peers ($\chi^2(df = 14) = 57.11$; $p < .001$), technology-mediated contact to the teacher ($\chi^2(df = 14) = 41.54$; $p < .001$), allocation of learning

time ($\chi^2(df = 14) = 27.57$; $p = .016$), more interesting learning ($\chi^2(df = 14) = 34.94$; $p = .001$), added value compared to lecture ($\chi^2(df = 14) = 23.93$; $p < .05$), tasks and exercises ($\chi^2(df = 14) = 25.71$; $p = .038$), and simulations ($\chi^2(df = 14) = 33.58$; $p = .002$).

Again, U tests (with Bonferroni correction) do not show any systematic developments. Rather unsystematic fluctuations are found around the item-specific basic level (i.e., overall mean). One exception is the item “I suppose that learning with a computer is more interesting than a lecture” (added value compared to a lecture); this item shows a significant decrease from the terms before 2011/12 to the following terms.

3.5 H2: Differences Related to Study Programs

Concerning usage of computers, Kruskal-Wallis tests reveal significant differences concerning communication ($\chi^2(df = 2) = 35.32$; $p < .001$), surfing ($\chi^2(df = 14) = 15.19$; $p = .001$), gaming ($\chi^2(df = 14) = 13.12$; $p = .001$), information search ($\chi^2(df = 14) = 26.12$; $p = .001$), and writing texts ($\chi^2(df = 14) = 30.10$; $p < .001$). Follow-up analyses (U tests with Bonferroni correction) confirm that students in sport & computer science use computers significantly longer for communication, surfing, gaming, and information search than students of sport science. Compared to PE students, students in sport & computer science surf significantly longer and use the computer significantly shorter for writing texts. Furthermore, PE students use computers significantly longer for communication, information search, and writing texts than students of sport science.

Concerning attitudes towards computers, Kruskal-Wallis tests confirm significant differences for confidence ($\chi^2(df = 2) = 28.63$; $p < .001$), liking ($\chi^2(df = 2) = 32.39$; $p < .001$), study only ($\chi^2(df = 2) = 11.57$; $p = .003$), and reluctance ($\chi^2(df = 2) = 16.24$; $p < .001$). Follow-up analyses (U tests with Bonferroni corrections) confirm significantly more positive attitudes of students attending the computer & sport science programs compared to students of sport science and PE, respectively.

Concerning attitude towards eLearning, Kruskal-Wallis tests show significant differences for learning alone ($\chi^2(df = 2) = 7.75$; $p = .021$), technology-mediated contact to the teacher ($\chi^2(df = 2) = 16.67$; $p < .001$), more interesting learning ($\chi^2(df = 2) = 31.26$; $p < .001$), effort ($\chi^2(df = 2) = 13.59$; $p = .001$), added value compared to lecture ($\chi^2(df = 2) = 9.83$; $p = .007$), tasks and exercises ($\chi^2(df = 2) = 6.55$; $p = .038$), animations and illustrations ($\chi^2(df = 2) = 7.84$; $p = .02$), and simulations ($\chi^2(df = 2) = 7.80$; $p = .02$). Follow-up analyses (U tests with Bonferroni corrections) confirm the following differences (see Fig. 3):

- Computer & sport science versus Sport science: Students of computer and sport science significantly prefer technology-mediated contact to the teacher, estimate eLearning more interesting, and would engage more for eLearning than students of sport science.
- Computer & sport science versus Physical Education (PE): Students of computer and sport science expect significantly greater added value of eLearning than students of PE.

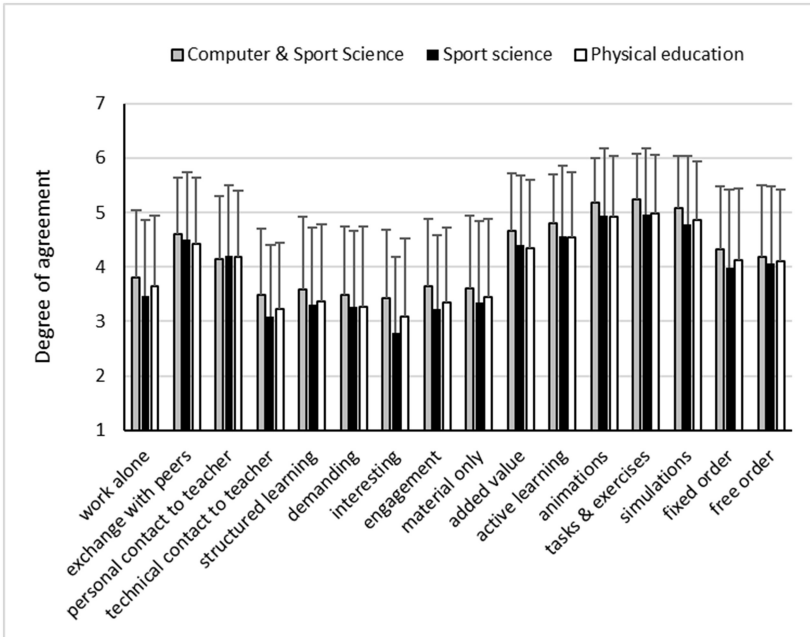


Fig. 3. Differences in attitude towards eLearning related to study program. (Columns: means; error bars: standard deviation).

- Sport science versus Physical Education: Students of PE estimate eLearning significantly more interesting than students of sport science.

3.6 H3: Gender-Related Differences

Concerning usage of computers (see Fig. 4), U tests with Bonferroni correction confirm that females spend significantly less total time at the computer ($N = 1087$; $z = -6.50$, $p < .001$) as well as for surfing ($N = 1047$; $z = -8.59$, $p < .001$), gaming ($N = 1041$; $z = -9.53$, $p < .001$), information search ($N = 1084$; $z = -2.68$, $p = .007$) and learning ($N = 1082$; $z = -3.07$, $p = .002$) than males. No differences are found for communication ($N = 1081$; $z = -0.54$, $p = .59$) and writing texts ($N = 1077$; $z = -1.47$, $p = .14$).

Concerning attitude to computers, U tests with Bonferroni correction confirm that females report significantly less confidence and liking as well as more reluctance towards the usage of the computer (see Fig. 5).

Since the term 2007/2008, nine additional gender-sensitive items concerning attitude towards computers have been added to the questionnaire. U tests with Bonferroni correction confirm that females report significantly more difficulties and reluctance using computers. In addition, solution strategies differ significantly: Whereas females prefer requesting for (external) help, males tend to solve the problem themselves by trial and error.

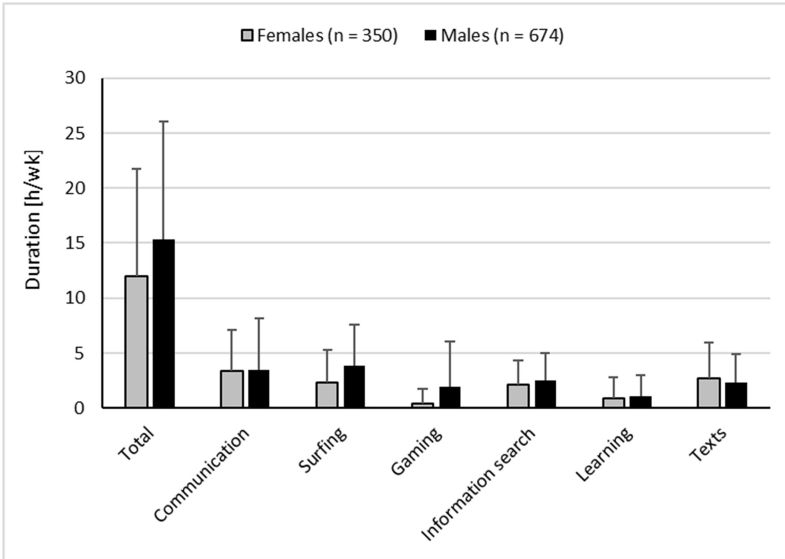


Fig. 4. Gender-related differences in computer use. (Columns: means; error bars: standard deviation).

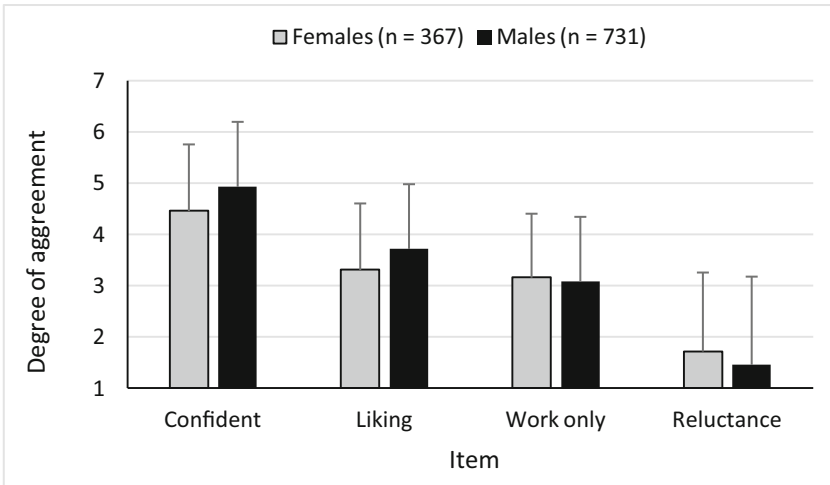


Fig. 5. Gender-related differences in attitudes towards computers. (Columns: means; error bars: standard deviation).

Concerning attitude towards eLearning, U tests with Bonferroni correction confirm that females agree less with the statement that eLearning is more interesting than a conventional lecture (2.8 ± 1.3 versus 3.1 ± 1.4 ; $N = 1093$, $z = -3.35$, $p = .001$). All other differences are not significant.

4 Discussion and Conclusions

The purpose of the study was to assess access to, use of computers and ICT as well as attitudes towards ICT and eLearning over a period of 15 years.

We found the following results:

- Students have full access to computers, smartphones and the Internet (Q1). Therefore, technology seems to be no issue any more.
- In general, attitudes to eLearning are moderately positive (Q2). Students show high expectations concerning the quality of content and communication. The reason for this result may be, that, compared to other studies [1, 2, 4, 5], all students are attending study programs at a technical university. Furthermore, students are predominantly at an early stage of their academic career [4].
- Comparing the extent of computer versus smartphone use (Q3), there is a characteristic profile: computers are used mainly for information search and learning, whereas smartphone use dominates in communication. Both computers and smartphones are used for surfing.

The dominance of communication, surfing and information search partly corresponds to the findings reported in [1], but not to [2] where specific study-related activities dominated. The reason may be that students in our sample were at the beginning of their study program. Therefore, activities specifically related to attending the study program were less pronounced.

- There is a significant influence of terms (H1/Q4). However, systematic changes can only be found for usage (i.e., transient or permanent increase of communication, surfing, information search, and learning).
- Concerning study program (H2/Q5), students of computer & sport science show a significantly more extensive usage of ICT and more positive attitude towards eLearning than the other students. These differences may be partly explained by the greater affinity of the computer & sport science students to the subjects of math and informatics compared to the students attending other sport and PE programs.
- Significant differences were also found for gender regarding the use of and attitude towards computers (H3/Q6), confirming previous research regarding technology and computers [3, 8]. However, attitudes towards eLearning do not show significant gender effects (with one exception). Nevertheless, the significant gender differences concerning problem solving must be considered. Females prefer immediate external support and are less motivated to use computers.

However, the study has two important limitations. On the one hand, the sequential cross-sectional design does not warrant conclusions for truly longitudinal developments over 15 years. Rather, effects of epoch and term are confounded. A second limitation of the current study is the fact that students are not equally distributed to all

cells of study program and term (see Table 1). Therefore, there is a distortion in the subsamples, for in different terms study programs contributed differently to the results. Sport science students dominate in the first four (i.e., 2002/2003 to 2005/06) and last terms (i.e., 2013/14 to 2015/16), whereas PE students dominate in terms 2006/07 to 2010/11. One possible reason is that there has been a change in the study programs offered by the university (i.e. from Diplom to B.Sc. and M.Sc. in computer science and from Magister to B.A. and M.A. in sport science). Consequently, there was a transient break-down concerning the computer science and sport science programs. However, reconsidering the inconsistent term-related differences, there is no relation between different proportions of students and changes of attitudes.

The study revealed important information concerning overall as well as differential attitudes to computers and eLearning. Particularly the program- and gender-related differences should be taken into account when offering eLearning to students.

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BIMROB – Bidirectional Interaction Between Human and Robot for the Learning of Movements

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Abstract. The overlap between the workspaces of humans and robots has been increasing in the last decades. Situations in which humans interact with robots are becoming more frequent. The uni- and bidirectional interactions of humans and robots when learning movements have so far not been adequately investigated. The presented studies investigate the unidirectional interaction of humans and robots in different settings. Results present first indications for an efficient and effective interaction configuration of a bidirectional interaction between human and robot for the learning of movements and to combine their advantages of humans and robots.

Keywords: Human-robot-interaction · Dyad learning · Motor learning

1 Introduction

Over the last 50 years, robots have become indispensable in many application fields, e.g., in the car industry, rehabilitation and training. Currently, a paradigm shift is taking place from “classical” robots, which have been rigidly programmed towards a new, more adaptive type that is capable of learning. This new paradigm of common learning of humans and robots has huge potential for changing sensorimotor training in sports and rehabilitation. However, learning robots are still at an early stage of development. Their interaction with humans has been poorly understood and the dynamics resulting from humans learning tasks jointly with robots have not yet been deeply explored.

The core idea of the BIMROB project¹ is to combine the insights from the fields of human and robotic learning to lay the scientific foundations for improved training in human-robot dyads in various application fields, with a particular focus on sport movements.

¹ Funded by the Forum for Interdisciplinary Research (FiF) at the Technische Universität Darmstadt (www.fif.tu-darmstadt.de).

The purpose of this paper is to describe the project's objectives in general and to present selected experiments addressing important issues of human-robot learning.

2 Project Description

The main objective of the BIMROB project is to study the bidirectional interaction of humans and robots when learning movements aiming for finding their best configurations for both effective and efficient dyadic sensorimotor learning. Therefore, we address the unidirectional transfer from humans to robots or from robots to humans first, followed by their bidirectional interaction in dyad learning. This procedure results in four settings (see Fig. 1).

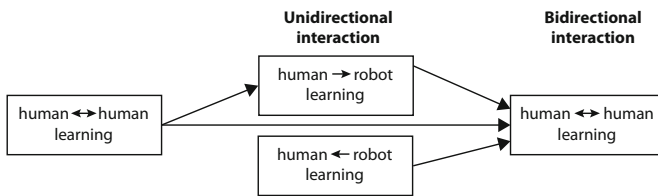


Fig. 1. Structure of the BIMROB project – four basic settings

The first task in the BIMROB-project was to establish the technical prerequisites. These are described in the following sections. Following this description, three exemplary studies² are presented for the settings of human-human learning and unidirectional human-robot learning.

2.1 BioRob System

The technical platform for the BIMROB project is a BioRob robotic arm (see Fig. 2).

This robotic system has been developed for the direct physical interaction between humans and robots. The system has four elastically actuated joints. Each joint is controlled by a motor located in the base. Each motor is coupled with four built-in elastic springs. Because of its lightweight design, the BioRob, which only generates little kinetic energy, is safe to use without collision detection even in high speed physical human-robot-interactions [1, 2].

In order to position the BioRob arm freely and/or to adapt it to individual anthropometric parameters human participants, a special lightweight frame has been developed (see Fig. 2). This allows for simple height adjustments as well as free positioning of the robot arm.

² For all experiments with humans, the approval by the ethic commission was obtained.

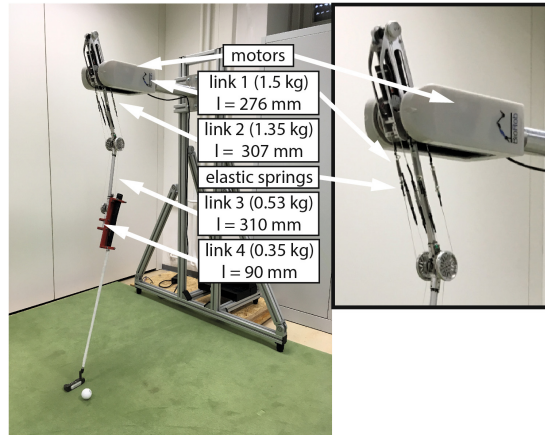


Fig. 2. BioRob with 4 DOF on a movable lightweight frame construction for experiments.

2.2 Artificial Putting Green

To study the interaction of robot and humans, we have constructed an artificial putting green. It is two meters wide and six meters long. Twelve aluminum profiles ($1 \times 1 \text{ m}$) were combined and covered by a carpet, whose properties are suitable to resemble putting (see Fig. 3). This design allows for the BioRob to be positioned freely on its base and human participants to move all over the green.

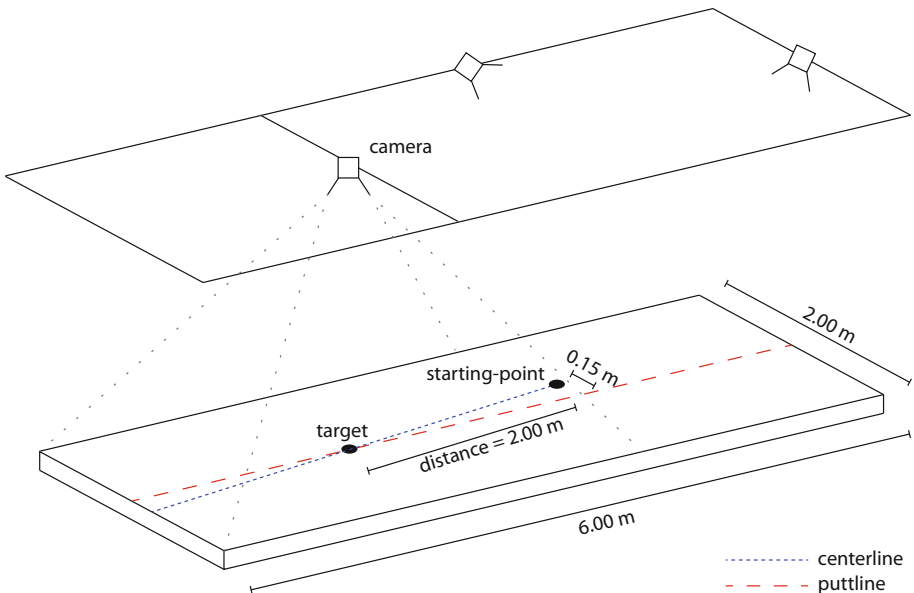


Fig. 3. Schematic illustration of the artificial putting green with cameras for the detection of final ball position, target and starting point, centerline as well as putt line.

Video cameras located above the putting green (see Fig. 3), are used to detect the final position of the ball after each trial. Prospectively, an automatic real-time ball detection will be developed allowing predictions about the ball's trajectory and its final position.

3 Human-Human-Interaction – A Dyad Learning Protocol

The first setting of the project considers the human-human interaction in motor learning. Therefore, the aim of this exploratory experiment was to develop an experimental design for dyadic learning and to identify topics addressed in human-human dyads (for a detailed description, see [3]).

Sample. 24 Students (12f, 12m) participated in the experiment. Inclusion criteria were that they had no previous experience in golf. Participants were assigned to three different types of dyads based on their gender: female-female, female-male and male-male.

Task and procedure. The Task was to putt towards a target located at a distance of two meters. In the dyads, participants practiced the golf putt (60 trials) in alternating order (120 putts per dyad). For example, while participant A performed his trials, participant B observed and vice versa. After every 20 trials per dyad, the pairs of participants were engaged in a dialog about their experiences [4].

Data analysis and processing. The radial distance to the target was calculated based on video recordings. All 60 dialogues (five per dyad) were also recorded with a video camera. Recordings were transcribed and analyzed using qualitative content analysis. Performance data were analyzed by two-factor ANOVA with repeated measures. In case of sphericity violation ϵ corrections were applied. The level of significance level was set a priori to 0.05.

Results. Putt performance, i.e., radial distance to the target, shows significant improvements over six blocks ($F_{3,72}$; $p = .028$; $\eta_p^2 = .121$). In the verbal reports, 867 statements were identified which could be classified into four main categories: concrete movement-related aspects (e.g., direction, distance, posture, grip, and oscillate), meta-statements (e.g., partner, exercise strategies and movement control), general movement-related statements (e.g., performance and movement sensations) and cause-effect relationships (e.g., if-then rules). In all dyads, references to the learning material were made, particularly to handle, posture and head position.

Discussion. On the one hand, the developed experimental setup was verified. On the other hand, important topics relevant for human-robot interactions were identified.

4 Human-Robot Learning – Observation Perspective

The second setting (see Fig. 1) considers the unidirectional human-robot interaction in motor learning: How can humans learn from robots?

When learning movements, the perspective of observation plays an important role in the perception of different spatial movement information [5]. In addition, mental rotation is required to transform the movement observed from an external perspective into an ego perspective. The study presented here examines the influence of the observational perspective on human movement learning in human robot dyads. According to Ishikura and Inomata [5], we had expected advantages for the frontal over the sagittal perspective.

4.1 Methods

Participants. 32 persons (14f, 18m) participated in the experiment. Inclusion criteria were no previous experiences in golf, field hockey or ice hockey. All participants gave informed written consent prior to the study. The age ranged from 15–35 years (mean 25.7 ± 4.1 y). Participants were assigned ad-hoc to three different groups based on their putt performance in the pre-test (see Table 1).

Table 1. Anthropometric data of participants and experimental groups with putt performance of the pre-test.

Condition/Perspective	n	Gender [m/f]	Age [years]	Putt performance (pre-Test) [cm]
Sagittal perspective	11	6/5	26.0 ± 2.7	46.5 ± 39.0
Frontal perspective	11	7/4	27.9 ± 3.7	47.4 ± 44.4
Control	10	5/5	22.9 ± 4.3	46.3 ± 36.0
Total	32	18/14	25.4 ± 4.7	46.8 ± 38.1

Apparatus and Task. The experiment was performed on an artificial putting green (see Fig. 3). Participants’ task was to hit standard golf balls (BEAST, Lady) with a

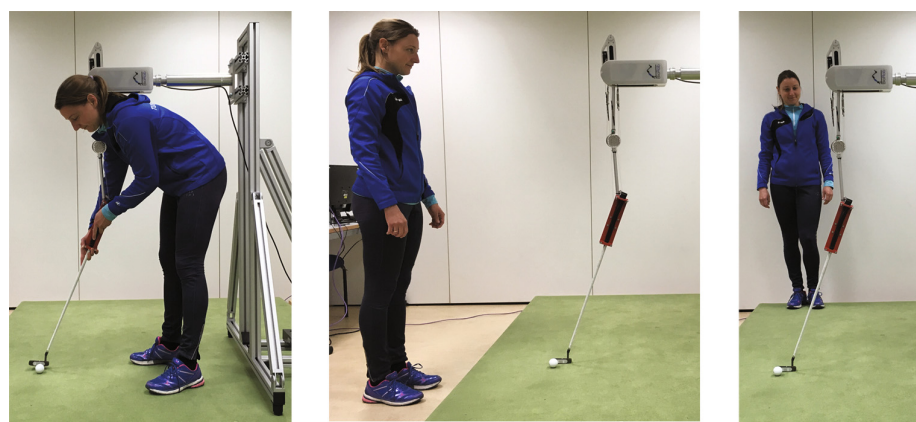


Fig. 4. Putting participant with a putter attached at the end-effector of BioRob (left) and observation perspectives (center: frontal; right: sagittal).

standard golf putter (MacGregor DX) attached at the end-effector of the BioRob (see Fig. 4), towards a target located two meters from the starting point. The starting point was located 15 cm aside from the center line to the right, while the target was located on the center line of the putting green [6] (see Fig. 3).

Procedure. All participants received written information about the procedure and completed a pretest consisting of seven unconstrained putt trials, i.e., putter was not attached at the BioRob, two different Mental Rotation Tests (MRT-Peters and & MRT-Bio [7]) and two different picture selection tests (PST). Based on their putt performance in the pretest, participants were randomly assigned to the three experimental groups. After the pretest, the frontal and sagittal groups completed a training phase of seven trials. The putter was attached to the end-effector of the robot arm (see Fig. 4). During the initial trial the BioRob was not actuated and just recorded the movement trajectory. After this first trial, participants took the assigned observation perspective (see Fig. 4) and the robot replicated the putt performed by the participant. In the next trial, the putt was again replicated by the robot, and the participants had the possibility to actively correct the putting movement of the robot and to change the trajectory. This sequence of observation and correction was repeated six times. The no-treatment control group did not perform putts with the robot; instead, the participants were shown an information video about the BioRob. After the training phase, all participants completed a post-test (putt performance and PST). The final position of the ball was recorded with a video camera located above the target point. Based on the video, the radial distance to the target was calculated for each trial in the pre- and post-test and also in the training phase.

Data analysis. Putting performance and PST data were analyzed by a two-factor MANOVA with repeated measures using SPSS (v24). The MRTs were examined for group differences with a univariate ANOVA. The level of significance level was set a priori to 0.05.

4.2 Results

All participants improved their putt performance from the pretest to the posttest, however, the difference was just above the level of significance ($F_{1,19} = 4.12$, $p = .052$, $\eta_p^2 = .12$). Interaction of group and test was not significant indicating no learning advantages for any group ($F_{2,29} = 0.45$; $p = .64$; $\eta_p^2 = .03$). However, the training groups achieved a much stronger mean improvement (frontal group +17.1% and sagittal group +22.8%) than the control group (+5.5%; see Fig. 5). No effects of perspective were found for PST ($F_{2,29} = .022$, $p = .98$) and MRT-Peters ($F_{2,29} = 2.03$, $p = .150$) but for MRT-Bio ($F_{2,29} = 3.551$, $p = .042$, $\eta_p^2 = .197$). Follow-up analyses (U-test) revealed a significant difference between the frontal group and the control group.

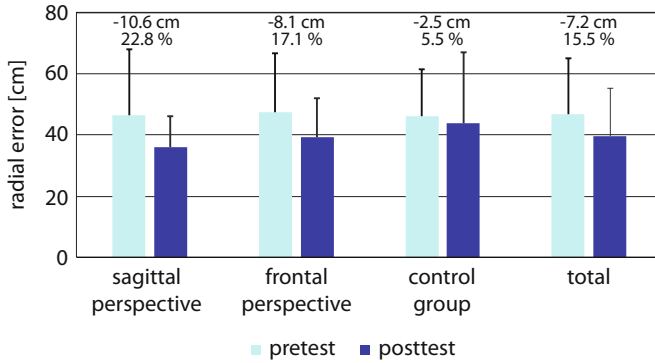


Fig. 5. Putt performance as radial distance from the target in the pretest and posttest.

4.3 Discussion

This study does not confirm the above-mentioned superiority of the frontal perspective. There are several possible reasons for this result: First, the study was underpowered ($1 - \beta = 0.38$). For detecting an effect size of 0.03, a total sample of 81 would have been required. Second, the study included switching between free putting in the pre- and posttests and constrained putting in the training phase in the treatment groups. This switching may have caused confusion in the treatment groups. Third, the training phase comprising seven trials may have been too short to find a clear training effect. Fourth, a more detailed analysis provided some hints that the participants adopted different strategies depending on the observation perspective. The participants of the frontal group seem to have tried to correct the movements of the robot based on the observed robot putt, whereas the participants of the lateral group seem to have focused on their own movements when correcting the robot movements.

In future studies, we plan to circumvent these issues by including a larger sample, more acquisition trials, and to avoid switching between free and constrained putting. Furthermore, the inclusion of a further training condition without observation and feedback is promising.

5 Human-Robot Learning - Iterative Robot Feedback

Part of our previous work focused on finding ways to transfer motor skills from humans to robots. For instance, through demonstrations of interactions between a human and a robot, it was possible to teach the robot how to assist a human partner in a box assembly task [8]. Further developments to this algorithm enabled the robot to learn motor skills from partially observed demonstrations executed at different speeds [9]. Moreover, the robot was able to optimize the speed of execution of movements taught by demonstrations [10] and to learn from incremental human feedback [11]. In this section, an approach is presented that explores transferring motor skills in the opposite direction, i.e. from robots to humans, with possible implications for sports and rehabilitation.

5.1 Introduction

Emken et al. [12] developed an iterative method that helps humans to achieve specific goals such as a pre-defined maximal foot height during the swing phase of walking. This method optimizes only a single parameter, rather than entire trajectories. In our work, we try to overcome this limitation by investigating how robots can help humans to learn movement trajectories through iterative feedback-based strategies. The basic idea is to adjust the correction strategy of the robot according to the estimated sensitivity of the human in order to optimize human motor skill learning. In the following, our proposed correction strategy of the robot, which allows for adjustments in the human's reaction, is presented. Afterwards, experiments in which humans practice a motor skill with the assistance of the BioRob (see Sect. 2.1) are presented. Finally, the experimental results and ideas for future work are discussed.

5.2 Feedback Strategy

A method was developed to teach arbitrary trajectories to humans with the help of a robot (see Fig. 7): First the robot guides the human along a pre-defined reference trajectory τ_H^{des} . During this phase, the force F_k applied by the robot increases in proportion to the deviation from the reference trajectory $(\tau_H^{des} - \tau_H^k)$ (iteration k, see Fig. 6):

$$F_k \propto \beta \cdot (\tau_H^{des} - \tau_H^k)$$

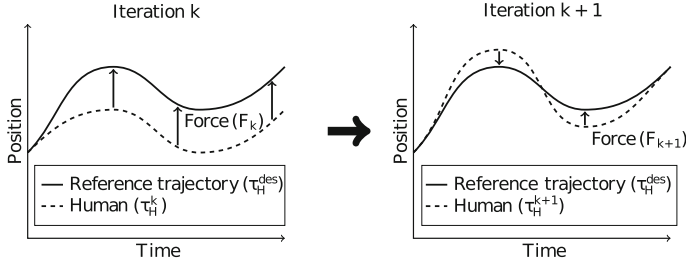


Fig. 6. Iterative method for motor skill learning in human-robot dyads. The force applied by the robot depends on the difference between the executed trajectory and the reference trajectory as well as on the estimated sensitivity of the human to the robot's feedback. It is assumed that the human's sensitivity can be estimated from a change in the executed trajectory from iteration k to iteration k + 1 and that this change is proportional to the robot's forces in iteration k.

The parameter β determines how strong the robot's feedback is, given a deviation from the reference trajectory. For positive or negative β values, the robot applies a force in the direction of $(\tau_H^{des} - \tau_H^k)$ or in the opposite direction, respectively. For example, if $\beta < 1$, the robot tends to apply less forces in proportion to the actual deviation (in case of high sensitivity of the human). On the other hand, if $\beta > 1$, the robot tends to apply higher forces, compensating for the low sensitivity of the human.

Through the robot's guidance, the human receives a haptic information about the trajectory to be learned. In the next iteration (iteration $k + 1$, see Fig. 6), the task for the human is to move the robot along the reference trajectory τ_H^{des} , to feel as little guiding forces by the robot as possible, probably resulting in a low deviation from the reference trajectory. If the human follows exactly the reference trajectory, he or she will not feel any guiding force by the robot. The trajectory τ_H^{k+1} executed by the human is recorded by the robot and is compared to the trajectory τ_H^k executed in the previous iteration in order to estimate the human's sensitivity. As a result, the robot's haptic feedback in the next iteration (iteration $k + 2$) depends on the estimated human sensitivity. A high estimated sensitivity leads the robot to apply less force and vice versa.

This iterative method starts from the assumption that the human uses the robot's correction forces, possibly with a damping or amplification factor, directly for the correction of the next movement:

$$\tau_H^{k+1} \propto \tau_H^k + \alpha \cdot F_k = \tau_H^k + \alpha \cdot \beta \cdot (\tau_H^{des} - \tau_H^k)$$

This equation is compatible with the model of Müller and Blischke [13]. The parameter α determines the human's reaction to the robot's feedback. For positive or negative α values, the human changes the trajectory according to or opposite to the robot's feedback, respectively. In addition, low or high α values indicate low or high sensitivity of the human to the robot's feedback. By choosing $\alpha = \beta = 1$, the robot's feedback exclusively depends on the deviation from the desired trajectory, i.e., without considering the sensitivity of the human (baseline strategy; BL). If the robot estimate the sensitivity parameter α and β is computed such that $\alpha \cdot \beta = \text{constant}$, the robot adapts its feedback to the sensitivity of the human (adaptive strategy; adap.). The parameters α and β can be computed with linear regression.

5.3 Experimental Procedure

Exploratory experiments with four participants (two males, two females, age ranging from 24–34 y) were conducted. The trajectory to be learned in these experiments corresponds to an arch-shaped movement from the middle to the left, to the right and back to

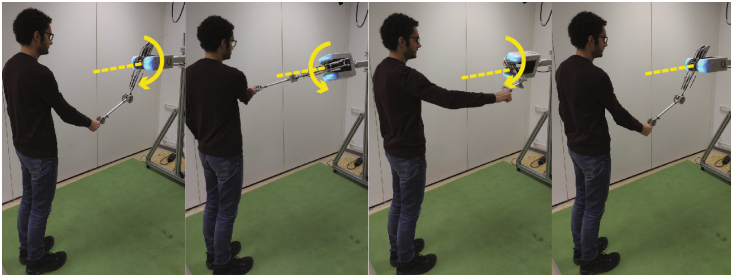


Fig. 7. Movement to be learned by the human with the assistance of the robot: arch-shaped movement from the middle to the left, to the right and back to the middle.

the middle (see Fig. 7.). Two participants trained using the baseline strategy, while the two other participants trained using the adaptive strategy.

In all iterations during training, the robot's controller compensates for its weight. The controller detects the robot's joint angles with a resolution of less than 0.5° and computes the motor commands with a frequency of 500 Hz.

For each participant, the workflow of the experiment (see Fig. 8) was as follows: ten training iterations with the baseline or with the adaptive method, five iterations with no feedback (early retention), ten minutes playing computer games and finally five iterations with no feedback (delayed retention). For the adaptive condition, $\alpha \cdot \beta = 0.5$.

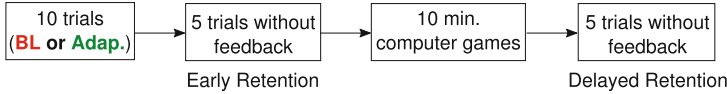


Fig. 8. Workflow of experiments.

5.4 Results and Discussion

Figures 9 and 10 show the root-mean-square error (RMSE) between actual and desired trajectory of joint one (most relevant joint for the movement in this experiment; see Fig. 7.) of the robot being manipulated by the human without and with time-alignment, respectively. The time-alignment was performed with dynamic time warping [14]. During training, the feedback given by the robot helps maintaining a low RMSE. In the early retention (ER) and delayed retention (DR) phases, the RMSE is higher due to the absence of feedback. In addition, most of the error is due to the misalignment in time between the trajectories the participants executed and the desired trajectory. The plots show no conclusive difference between the errors made by participants after training with the baseline and with the adaptive method.

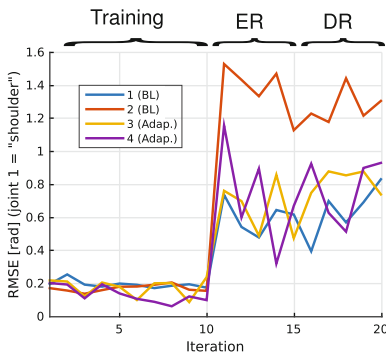


Fig. 9. Root-mean-square error between actual and desired trajectory of joint 1 of the robot being manipulated by the human (without time-alignment).

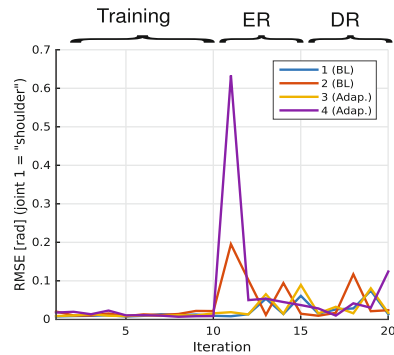


Fig. 10. Root-mean-square error between actual and desired trajectory of joint 1 of the robot being manipulated by the human (with time-alignment).

Figure 11 shows the computed values of the parameter α for the two participants who trained with the adaptive method. In these experiments, the values of the parameter α oscillate considerably from one iteration to the next, which results in oscillating values of the parameter β leading to inconsistent feedback by the robot. This difference in the intensity of the feedback from one iteration to the next might have caused confusion in the participants. Thus, it may be worth updating β not for every iteration, but only a certain number n of iterations.

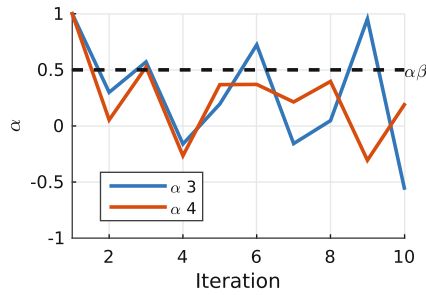


Fig. 11. Computed values of parameters α for participants 3 and 4, who trained with the adaptive method. These values show a large oscillation.

5.5 Conclusion

We proposed an iterative feedback-based correction strategy to teach motor skills to humans with the assistance of a robot. In our method, the robot is able to adapt its feedback to the estimated sensitivity of the human. Moreover, a baseline approach with no adaptation can be considered as a special case of our method.

The exploratory study could not confirm an advantage of the adaptive strategy.

Future work will explore variations of this strategy in which the robot does not adapt its feedback at every iteration, but after a number of iterations, potentially computing a better estimate of the sensitivity of the human.

6 Conclusion

First, the feasibility of the developed platform (BioRob system and artificial putting green) was confirmed. The robot system was able to perform and imitate the required putting movement with reasonable accuracy. The light-weight positioning system allows for adequate location of the robot.

Furthermore, the studies performed so far provide important insights into both human-human interaction and unidirectional human-robot interaction when learning putting movement. Observation conditions (e.g., video length and perspective) as well as robot learning from demonstrations and incremental human feedback and algorithms for robot support of human learning revealed results that will be included in further studies of bidirectional interaction of robots and humans. For example, several strategies

of learning (e.g., focus of attention, error correction, and feedback processing) were uncovered in the studies.

In future studies, the study of bidirectional human-robot interaction will focus on combining the advantages of both human and robot learning. The aim is to find adequate procedures including phases of self-optimization of humans and robots as well as phases of humans teaching robots and robots teaching humans.

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A Novel Multilocus Genetic Model Can Predict Muscle Fibers Composition

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Abstract. Muscle fibers composition is determined by the specific genetic characteristics to a large extent and it is connected with various physical traits of athletes. Establishment of genetic markers for sport prediction is an important task of great scientific and practical importance. We aimed to create a relevant model for prediction of the athlete muscle fibers structure based on a complex analysis of known genetic factors. The model included 14 single nucleotide polymorphisms and was tested for 55 subjects. Based on the performed ROC analysis, the model accuracy measured as area under the receiver operating characteristic curve (AUC) was 81% for professional athletes and 73% for non-athletes. The obtained results demonstrate that the proposed models can be used for sport testing.

Keywords: Sport testing · Muscle fibers composition · Genetic models

1 Introduction

Muscle fibers composition is connected with a set of physical traits of athletes such as endurance or strength and is significantly determined by various genetic characteristics. Its knowledge therefore can help in sport testing and prediction of athletic professional qualities. However, this area remains understudied and it is largely due to the complexity of the analysis of environmental factors, as well as the need to perform invasive procedures for histology samples of muscle tissue. A few data available to date show a significant role of genetic factors in athletic muscle fibers composition [1, 2]. Many genetics variants are also associated not only with the physiological characteristics of the muscles, but also with susceptibility to a variety of metabolic and cardiovascular diseases. Thus, the structural and functional phenotypes of muscles and predispositions to certain diseases have the same genetic background. Besides, predictive genetic sport testing of professional athletes can become an alternative to biopsy. Attempts to develop multilocus models for assessing muscle fibers composition have been undertaken since the high-throughput genetic studies became more accessible for personal use. Despite the ambiguity of the approaches, many of these models show acceptable accuracy [3, 4]. For applying the results of the conducted associative studies to the structure of muscles, we aimed to create a

relevant model for prediction of the athlete muscle fibers structure based on a complex analysis of known polymorphisms.

2 Materials and Method

Our model included a number of genetic variants previously described in a publication by Ahmetov et al. [5]. These variants are known to be associated with muscle structure and therefore can influence professional athletic characteristics. The list of these variants is provided at Table 1.

Table 1. Genetic polymorphisms associated with muscle structure.

Genetic variant	Gene	Genetic variant	Gene
rs1815739	ACTN3	rs1937	TFAM
rs11549465	HIF1A	rs660339	UCP2
rs2229309	NFATC4	rs1800849	UCP3
rs4253778	PPARA	rs2010963	VEGFA
rs2016520	PPARD	rs1870377	KDR
rs8192678	PPARGC1A	rs4646994	ACE
rs7732671	PPARGC1B	rs11091046	AGTR2

So far no quantitative parameters or degree of joint inheritance of the polymorphisms from the list have been known. The authors suggested that the degree of influence of all polymorphisms from the list is approximately the same, and the total contribution of genetics to the muscular structure is about 50%. On the basis of this statement we assumed that the best result in muscle structure assessment is achieved when a complete additive model is used [6].

Each of the described polymorphisms is the result of associative research and has three different states: with the predominance of the 1st type of muscles, intermediate and with the predominance of the 2nd type of muscles. Let each of the alleles leading to an increase in the number of muscle fibers of the 1st type be assigned a weight index equal to 1. The total risk index of the polymorphisms, R_i , will be equal to the sum of the weight indices for both alleles (0, 1, 2). Based on this, the maximal contribution of genetics to the development of muscles might be equal to 28 (which is the maximum weight of all 14 polymorphisms), while a result of summation equal to 0 will mean the development of the 2nd type muscles rather than the 1st one. The total risk index named as R_{tot} will be equal to the simple sum of risk indices for all model polymorphisms.

The risk index (R_i) is equal to the number of associated alleles with type 1.

$$R_i = n \quad (1)$$

The total risk index is equal to the sum of the risk indices.

$$R_{tot} = \sum_i R_i \quad (2)$$

Since the result of risk index calculating is only estimated, a transition was made from the numerical model to the personal status determining the overall ratio of fibers or predominance of a certain fiber type.

3 Results

The model was tested for 55 subjects who were divided into two group. The first group consisted of 30 amateur athletes – males aged 24 ± 4 years. Another group was formed of 25 physically active individuals who didn't achieved any ranks according to the Russian Unified Sports Classification System. They were also males of the same age as the first group and were regarded as controls in the present study. For all of the study participants there were results of biopsy of the thigh muscle reflecting the percentage of the 1st and the 2nd types of muscles. In addition, they were genotyped for the genetic variants listed above.

To assess the workability of the described genetic model, we performed a comparison between the result of the genetic test and the determined muscle structure on the basis of muscle biopsy. In case the result was the same, it was considered that the test was performed correctly, otherwise it was erroneous. We constructed a plot showing various genetic test indices corresponding to various cut-offs between muscle types in X-axis and percentage of individuals assigned to the 1st type muscle structure in Y-axis (see Fig. 1).

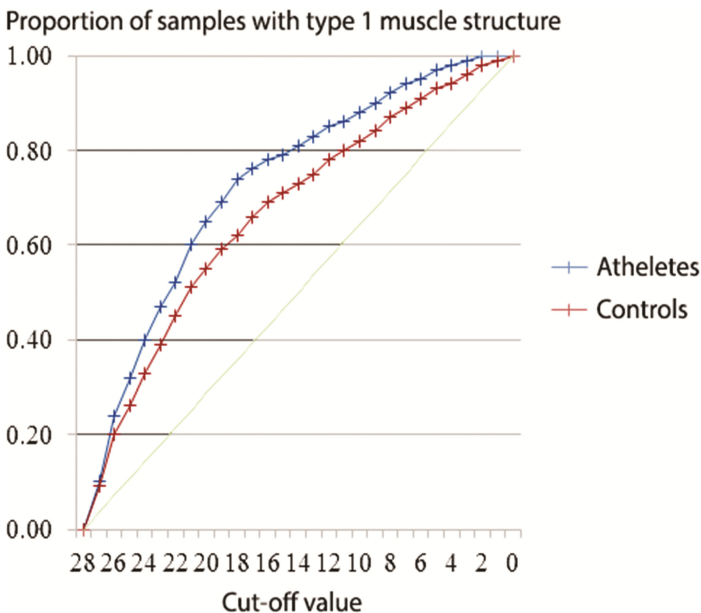


Fig. 1. Dependence of the number of samples assigned by the test to the group with muscle structure type 1 predominance from the numerical value of the muscle type detection cut-off for the model of muscle fibers structure prediction in athletes and controls.

Based on the performed ROC analysis, the model accuracy measured as area under the receiver operating characteristic curve (AUC) was 81% for professional athletes and 73% for non-athletes. The highest accuracy was achieved using the following criteria: 0–4 points indicated significant predominance of type 2 fibers, 5–10 points - predominance of type 2 fibers, 11–17 points - uniform muscle structure, 18–24 points - predominance of fibers of the 1st type, more than 25 points - a significant predominance of fibers of the 1st type.

We assume that a small difference between the groups arises from a significant influence on the development of the muscular structure due to professional sports activity, which directly confirms the statement about the 50% contribution of genetic factors to the muscle structure.

4 Discussion

As well as many studies of recent years, this investigation provides evidence that modeling of sport characteristics based on individual genetic data is feasible as well as the modeling of inherited disorders risk. Various mathematical approaches have been proposed for genetic modeling. The most common ones include multiplicative and additive models. The choice between them is determined by several factors. The multiplicative model is most often used when the measures of genetic variants effects are determined and vary from one variant to another. In case the measures are assumed to be similar or some of variants are likely to be linked, the additive approach is recommended to use. Concerning the present study, there is evidence on partial linkage disequilibrium between the genetic variants investigated and the average impact is suggested to be similar [5, 7].

Genetic markers are being successfully integrated for prediction of a broad range of medical and physiological traits. Genotype-phenotype associations related to physical characteristics, such as strength, power, performance, and elite athletic status are established by means of various associative studies [8, 9]. The reliability of such approach is also confirmed by the results obtained from a three-year pilot project on the analysis of the genetic predisposition to breast cancer and prostate cancer: CGEMS (Cancer Genetics Markers of Susceptibility) [10]. It is also supported by the data from studies funded by the Wellcome Trust Case Control Consortium on Coronary Artery Disease, Crohn's Disease, Rheumatoid Arthritis, and Type 2 Diabetes [11].

The values of area under the receiver operating characteristic curve (AUC) obtained in the present study are consistent with AUC of the currently used clinical non-genetic tests, such as PSA analysis, Treadmill test, and Mammography that have AUC values in the range 0.62–0.85. Moreover, the values were obtained without considering the information on the family history, without taking into account the influence of external factors and other clinically significant characteristics. In this connection, we can state the prospect of the proposed method of calculating individual sport characteristics on the basis of genetic variants for practical use.

5 Conclusion

The obtained final results demonstrate that additive genetic models such as proposed in this study show good diagnostic accuracy comparable with widely used diagnostic tests and can be used for sport testing particularly. The output of such models is acceptable for the formation of individual recommendations both to athletes (when choosing the optimal distance in the competitions) and to ordinary patients (for optimizing sport activity).

Acknowledgments. This work was supported by the Russian Science Foundation, Grant No. 17-15-01436: “Comprehensive analysis of the contribution of genetic, epigenetic and environmental factors in the individual variability of the composition of human muscle fibers”.

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Erratum to: Information Systems for Top-Level Football with Focus on Performance Analysis and Healthy Reference Patterns

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Erratum to:

**Chapter “Information Systems for Top-Level Football
with Focus on Performance Analysis and Healthy Reference
Patterns” in: M. Lames et al. (eds.), *Proceedings of the 11th
International Symposium on Computer Science in Sport (IACSS
2017)*, Advances in Intelligent Systems and Computing 663,
https://doi.org/10.1007/978-3-319-67846-7_7**

In Page 78, “quadriceps to hamstring ratio (QHR)” has to be changed as “hamstring-quadriceps ratio (HQR)”.

In Page 79, “quadriceps to hamstring ratio (QHR)” in one occurrence and “QHR” in two occurrences have to be changed as “hamstring-quadriceps ratio (HQR)” and “HQR”, respectively.

In Page 79, “QHR” has to be changed as “HQR”.

The erratum chapter and the book have been updated with the changes.

The updated online version of this chapter can be found at
https://doi.org/10.1007/978-3-319-67846-7_7

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